ECA31 NANOELECTRONICS MASTER RECORD

ELECTRONICS AND COMMUNICATION ENGINEERING

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TEST CASE QUESTIONS

Exp. 1: Simulation of scattering and absorption of Multi-layer Sphere and Core-Shell Nanoparticle using Mie-Plot tool

1. Scattering and absorption spectra for a single-layer sphere (core only).
2. Simulate the optical response of a gold (Au) nanoparticle of 50 nm diameter in water medium.
3. Simulate a core–shell nanoparticle with a gold core (30 nm) and silica shell (10 nm).

Exp. 2: Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.

1. Simulate the density of states for a quantum well with a thickness of 5 nm. How does it differ from a 2 nm well?
2. Compare the density of states for materials GaAs and InAs. What influences the differences?
3. Investigate the impact of temperature changes on the density of states for a 3 nm quantum well.

Exp. 3: Simulation of Energy States of Quantum Dot using NanoHub

1. Simulate how the lowest few confined energy levels change with quantum dot radius [Spherical dot, material = GaAs, effective mass $m^* = 0.067\ m_0$, potential well depth = 0.6 eV, radii: 2 nm, 5 nm, 10 nm]
2. Simulate a nanowire with defects. How do the band structure and electronic properties change?
3. Simulate the energy states of a spherical quantum dot with a radius of 5 nm. How do the states shift with a radius of 2 nm?

Exp. 4: Kronig Penney Model

1. Simulate the band structure for a periodic potential with a well depth of 0.5 eV. How does it change for a well depth of 1 eV?
2. Investigate the effect of varying the lattice constant on the width of the bandgap.
3. Compare the band structures for square and sinusoidal potential profiles.
4. Study the impact of increasing the barrier height on the effective mass of electrons.

5. Simulate the density of states for a one-dimensional lattice. How does it compare to a three-dimensional system?

Exp. 5: Band Structure Simulation for Graphene and CNTs using NanoHub

1. Simulate band structure of graphene and CNT using Si Material with Bulk Bands with 10
2. Simulate band structure of graphene and CNT using Si Material with Central Bands with 0.5
3. Simulate band structure of graphene and CNT using Ge Material with Bulk Bands with 30

Exp. 6: Demonstration of Quantum Tunneling

1. Simulate a particle tunneling through a barrier of height 1 eV and thickness 1 nm. How does the tunneling probability change with a thickness of 2 nm?
2. Investigate the effect of increasing the particle energy on the tunneling probability for a fixed barrier.
3. How does the tunneling probability differ for electrons and protons through the same barrier?
4. Simulate the effect of barrier shape (e.g., rectangular vs. triangular) on tunneling probability.
5. Study the tunneling dynamics for a multi-barrier system. How do the resonance effects influence the outcome?

Exp. 7: MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

1. Simulate the IV characteristics of a single-gate MOSFET with a gate length of 20 nm. How does reducing the gate length to 10 nm affect the performance?
2. Investigate the effect of increasing the oxide thickness on threshold voltage.
3. Compare the subthreshold slope for different materials used in the channel.
4. Simulate the impact of increasing the drain voltage on the saturation region.
5. Study how channel doping affects the drain current in the MOSFET.

Exp. 8: Simulation of VI Characteristics of FINFET, CNTFET and GFET using NanoHub

1. Simulate the VI characteristics of a FinFET with a gate length of 10 nm. How does increasing the gate length to 20 nm affect the performance?
2. Investigate the effect of varying the oxide thickness on the threshold voltage.
3. Compare the subthreshold slope for single-gate and double-gate FinFETs.
4. Simulate the drain-induced barrier lowering (DIBL) effect for different channel lengths.
5. Study the impact of increasing the fin height on the on-current of the FinFET.
6. Investigate the effect of increasing the channel length on the VI characteristics.
7. Study the effect of varying the chirality of the CNT on device characteristics.
8. Compare the performance of CNTFETs with semiconducting and metallic nanotubes.
9. Study the effect of varying the chirality of the CNT on device characteristics.
10. Simulate the VI characteristics of a graphene FET with a channel length of 10
11. nm. How does reducing the length to 5 nm affect performance?
12. Investigate the effect of increasing the doping concentration on the threshold voltage.
13. Simulate the impact of gate dielectric material on device performance.
14. Study how temperature affects the VI characteristics of graphene FETs.

Exp. 9: Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

1. Simulate the TMR ratio for a barrier height of 0.3 eV. How does increasing the barrier height to 0.6 eV affect the ratio?
2. Investigate the effect of varying the thickness of the tunneling barrier on the TMR ratio.
3. Study how temperature variations influence the TMR characteristics.
4. Compare the TMR ratio for devices using CoFeB and MgO materials. What factors contribute to the differences?
5. Simulate the impact of spin polarization on the TMR characteristics.

Exp. 10: Biosensor and Bio-FET characteristics using NanoHub

1. How does the threshold voltage (Vt) vary with pH when the electrolyte pH is increased from 4 to 8? Verify if the sensitivity aligns with the theoretical value of 59mV/pH.
2. How do the transfer characteristics (Ids vs. Vfg) shift when the pH of the electrolyte changes from 4 to 8? Analyze the trend and determine if the threshold voltage shifts consistently with pH.
3. Compare the BioFET sensitivity for low pH (e.g., pH 4) and high pH (e.g., pH 8). Are there any deviations in sensitivity, and if so, what could be the possible reasons?

4. How does the threshold voltage (Vt) of the BioFET respond to pH variation (4 to 8) when protein is used as the analyte instead of DNA? Compare the results to those obtained using DNA.
5. What is the effect of reducing receptor density (e.g., from 1×10^12 to 5×10^11) on the sensitivity and threshold voltage (Vt) of the BioFET for pH values
6. ranging from 4 to 8?

Exp. 11: Calculate the Potential of a pH SENSOR using NanoHub

1. Simulate the output potential of a pH sensor at pH 7. How does it change for pH 5 and pH 9?
2. Investigate the effect of electrode material (e.g., Pt vs. Ag/AgCl) on the sensitivity of the pH sensor.
3. Study the impact of temperature changes on the sensor's potential.
4. Simulate the performance of the pH sensor in solutions with varying ionic strengths.
5. How does the sensor's potential vary when using a buffer solution vs. a non- buffer solution?

Exp. 12: Rietveld refinement & lattice parameter determination (single-phase material) using FullProf software

1. Perform Rietveld refinement for a compound semiconductor (AlSb) and determine its lattice parameter. Compare with standard JCPDS values.
2. Perform Rietveld refinement for a metal oxide material (CeO_2) using the pseudo-Voigt peak profile. Analyze the Rwp and χ^2 values
3. Perform Rietveld refinement for a ZnO using the pseudo-Voigt peak profile. Analyze the Rwp and χ^2 values
4. Determination of lattice parameters for doped metal oxide material (Na0.5Bi0.5TiO3).

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Exp. 1. Simulation of scattering and absorption of Multi-layer Sphere and Core-Shell Nanoparticle using Mie-Plot tool

Aim:

To simulate and analyze the scattering, absorption, and extinction characteristics of multi-layer spheres and core–shell nanoparticles using the Mie-Plot tool.

Software:

Mie-plot tool (https://nanocomposix.com/pages/mie-theory-calculator)

Procedure:

- Open the Mie-Plot tool using the given link.
- Select a Core Material (e.g., Au, Ag, Si, TiO_2) from the dropdown menu.
- Enter the core diameter in nanometers.
- If the core type is set to Other, manually input the real part of the refractive index (n) in the box.
- To include a shell, select a Shell Material (e.g., SiO_2, Al_2O_3, ZnO) and enter the shell thickness in nanometers.
- If no shell is needed, select None for the shell option.
- Set the starting and ending wavelengths for the spectral range to be simulated (e.g., 300–800 nm).
- Enter the refractive index of the surrounding medium (e.g., 1.33 for water).
- Click "Plot!" to generate the extinction, absorption, and scattering cross-section plots.

Theory:

When light interacts with nanoparticles whose size is comparable to the wavelength of light, scattering and absorption phenomena occur. The **Mie Theory**, derived from Maxwell's equations, provides an exact analytical solution for the scattering of electromagnetic radiation by spherical particles.

For a **core–shell nanoparticle**, the optical response depends on:

- **Core and shell refractive indices**
- **Core diameter and shell thickness**
- **Surrounding medium refractive index**
- **Incident light wavelength**

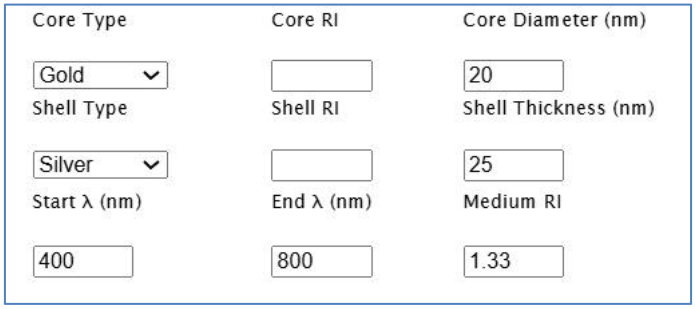
The **extinction cross-section (C_{ext})** represents the total light removed from the incident beam due to both scattering (C_{sca}) and absorption (C_{abs}):

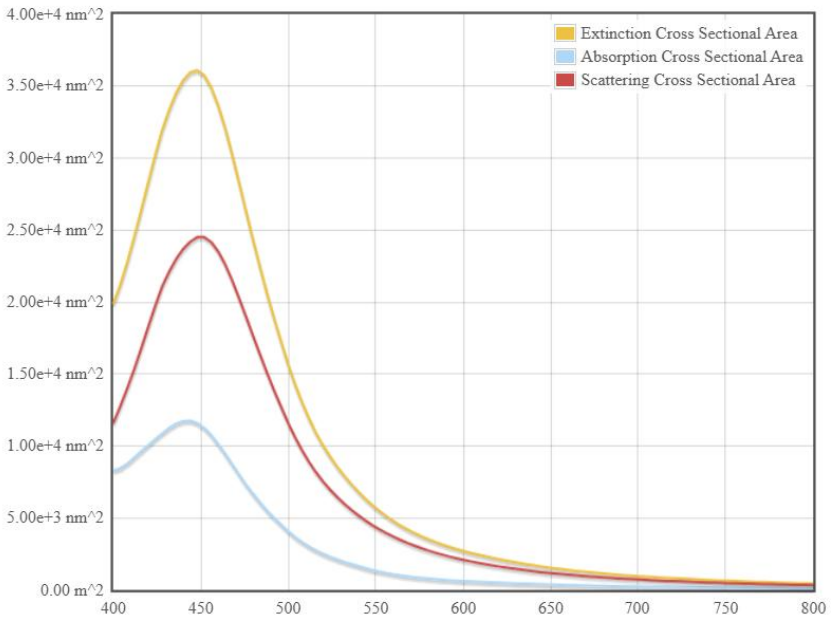
$$C_{ext}=C_{sca}+C_{abs}$$

Localized Surface Plasmon Resonance (LSPR) occurs when the conduction electrons in metallic nanoparticles oscillate in resonance with the incident light, producing strong absorption or scattering peaks. These optical properties can be tuned by modifying the particle size, composition, and surrounding medium.

Output:

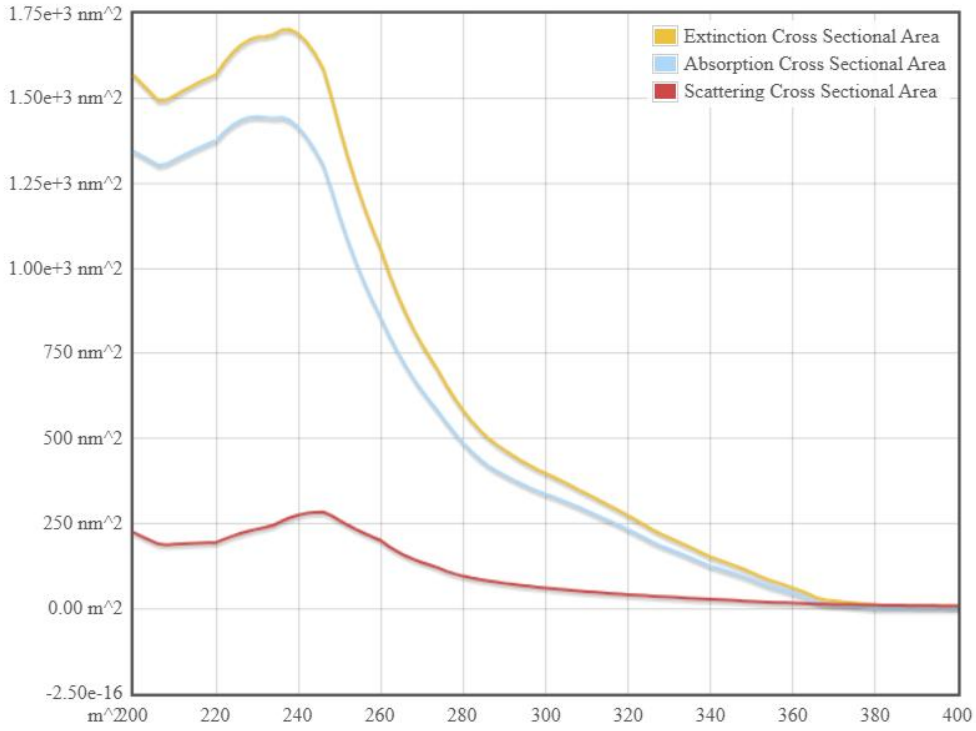
Scattering and absorption spectra for a core–shell nanoparticle.





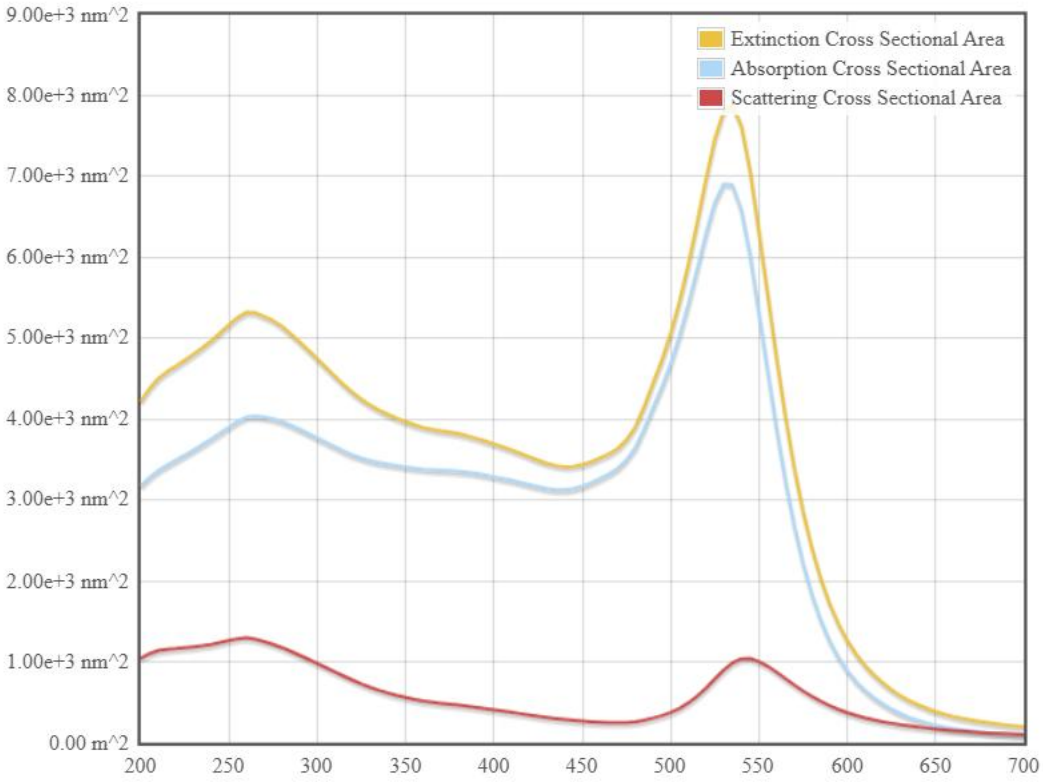
Case Study 1: Scattering and absorption spectra for a single-layer sphere (core only).

Core Type	Core RI	Core Diameter (nm)
TiO_2		30
Shell Type	Shell RI	Shell Thickness (nm)
None		0
Start λ (nm)	End λ (nm)	Medium RI
200	400	1.33

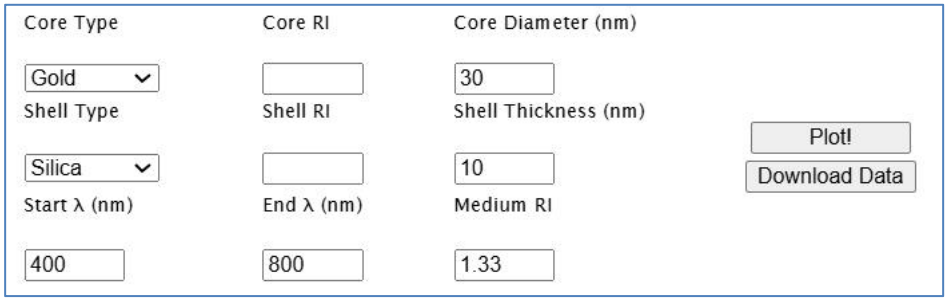


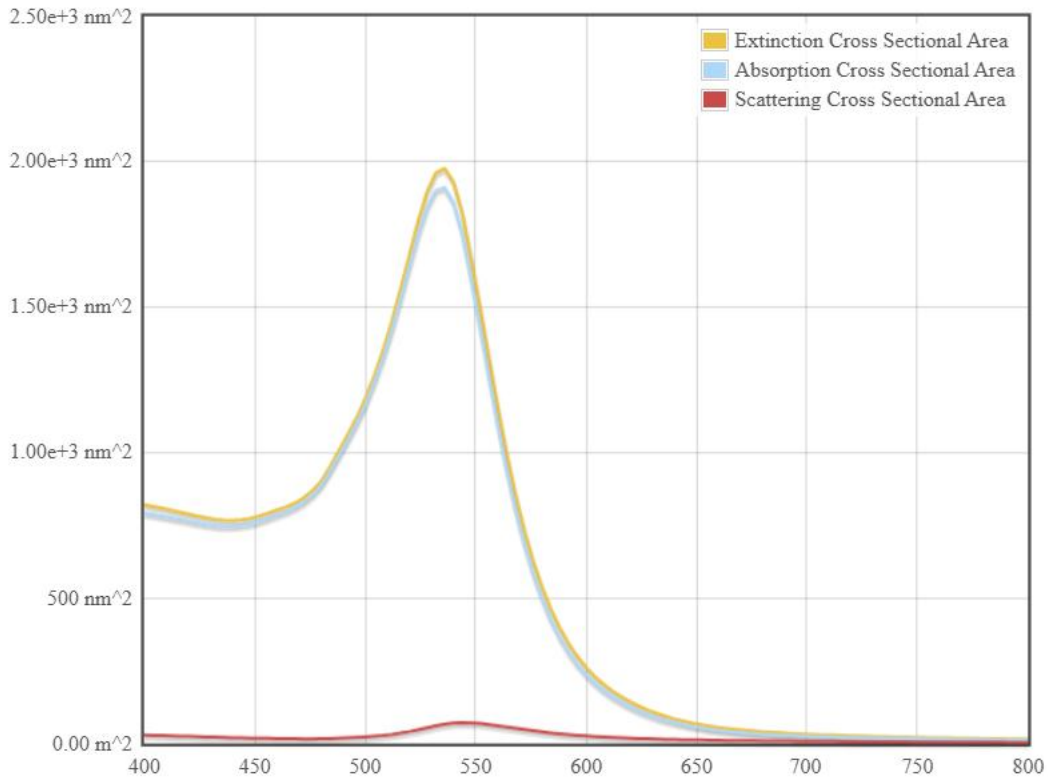
Case Study 2: Simulate the optical response of a gold (Au) nanoparticle of 50 nm diameter in water medium.

Core Type	Core RI	Core Diameter (nm)	
Gold		50	
Shell Type	Shell RI	Shell Thickness (nm)	Plot!
None		0	Download Data
Start λ (nm)	End λ (nm)	Medium RI	
200	700	1.33	



Case Study 3: Simulate a core–shell nanoparticle with a gold core (30 nm) and silica shell (10 nm).





Analytical:

Consider a spherical particle of radius r= 25nm and refractive index n_p=0.2+3.0i (gold) in a surrounding medium of refractive index n_m=1.33. Calculate the **size parameter (x)** using:

$$x = \frac{2\pi r n_m}{\lambda}$$

for an incident wavelength λ = 520nm. Interpret how this parameter influences scattering efficiency.

Result:

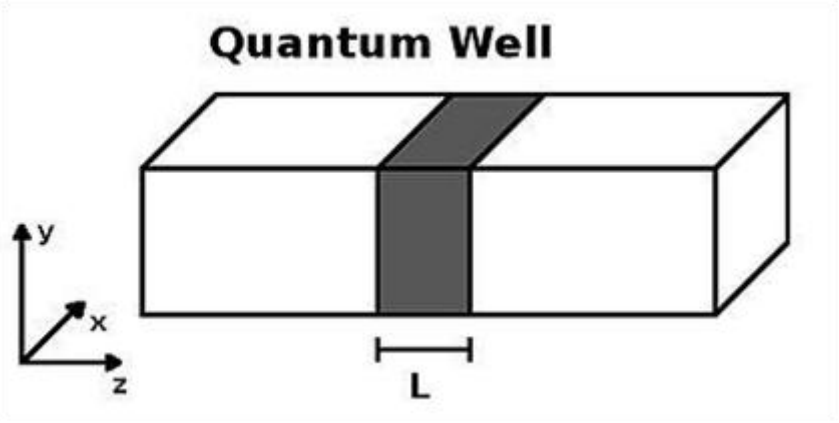
The scattering, absorption, and extinction spectra for multi-layer spheres and core–shell nanoparticles are obtained using the Mie-Plot tool. The results demonstrate the effect of particle size, shell thickness, and refractive index on the optical resonance behavior of nanoparticles.

Exp. 2. Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.

Aim: To simulate density of states of quantum well structure with various nano meter thickness.

Software: NanoHub Online Tool

Theory:



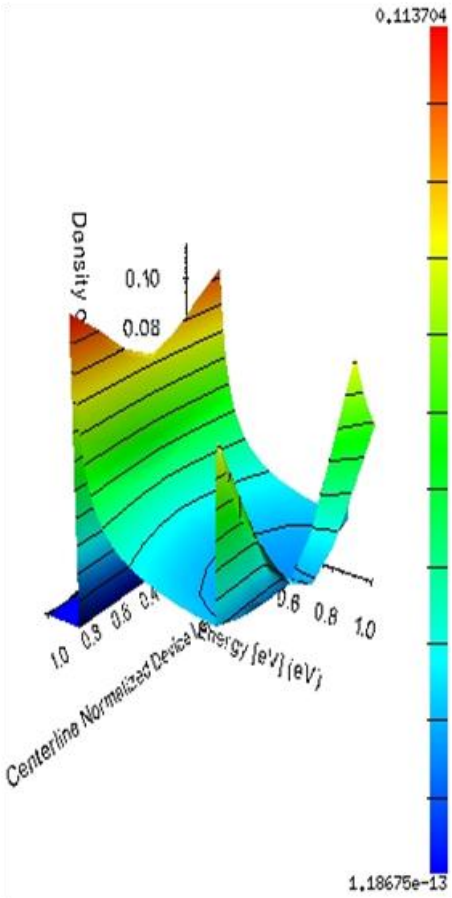
A quantum well is a potential well with only discrete energy values. The classic model used to demonstrate a quantum well is to confine particles, which were originally free to move in three dimensions, to two dimensions, by forcing them to occupy a planar region. The effects of quantum confinement take place when the quantum well thickness becomes comparable to the de Broglie wavelength of the carriers (generally electrons and holes), leading to energy levels called "energy subbands", i.e., the carriers can only have discrete energy values.

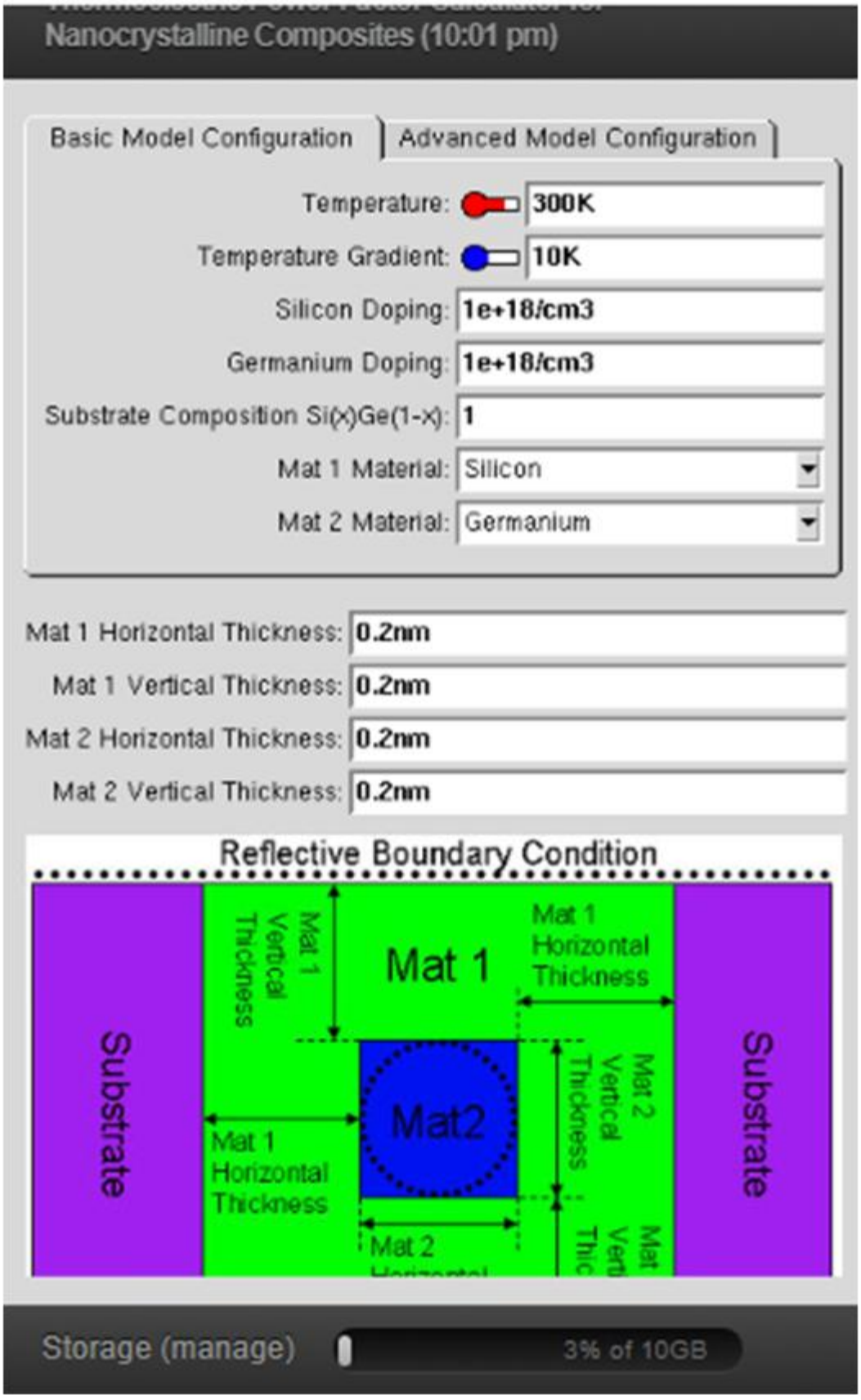
Procedure:

Open nanoHUB.
Goto Resources -> Tools.
Select Quantum wells in Tags.
Select Thermolelectric Factor calculator for nanocrystalline Composites in Resources.
Press Launch tool in info section.
Enter appropriate values in required section.
Select the Materials (Germanium/ Silicon).
Enter the material thickness required.
Press simulate to view results.
Select the Materials (Germanium/ Silicon).
Enter the material thickness required.
Press simulate to view results.

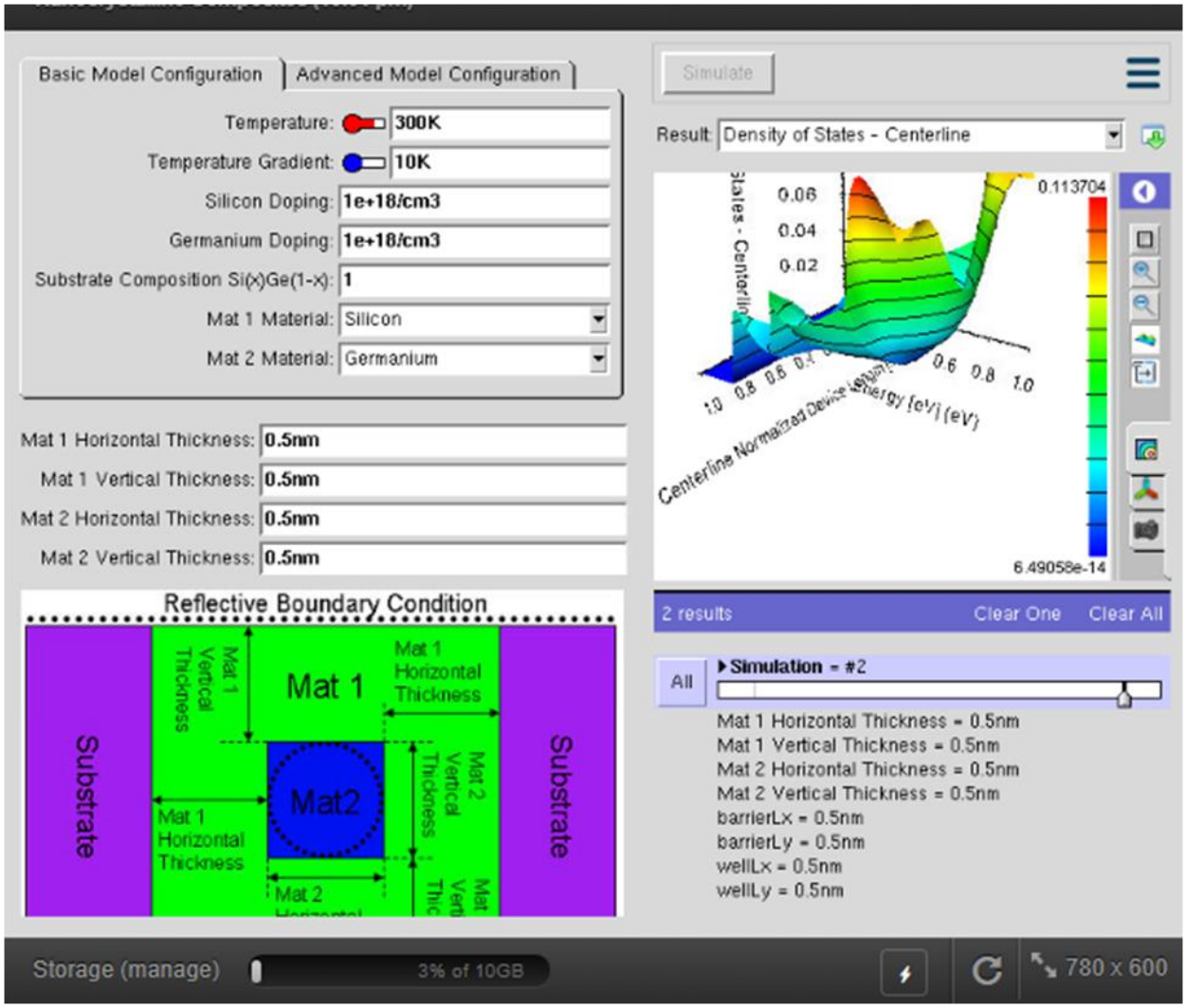
Compare both the results.

Output:

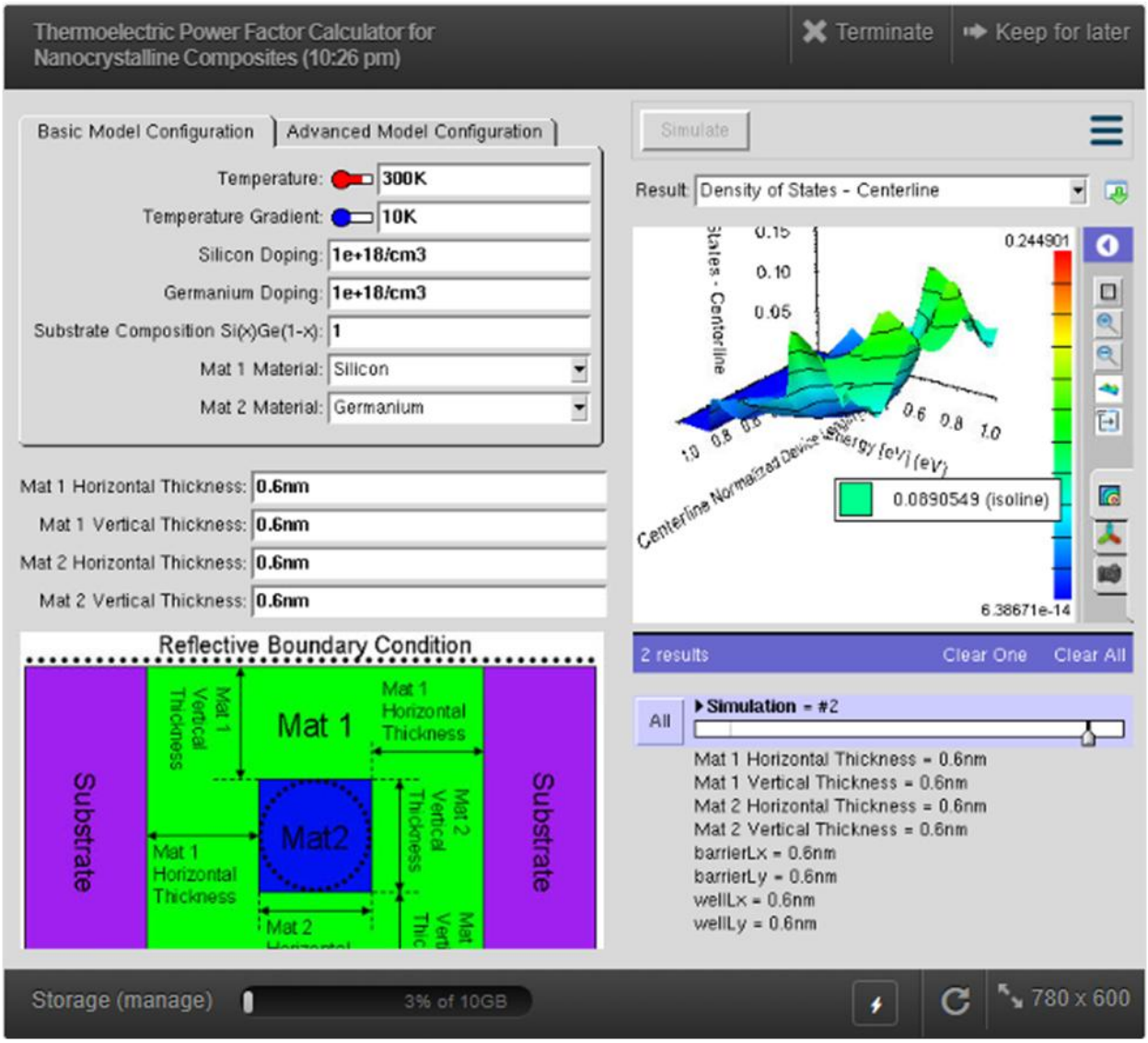



Nanocrystalline Composites (10:01 pm)
Basic Model Configuration
Advanced Model Configuration
Temperature:
300K
Temperature Gradient:
10K
Silicon Doping:
1e+18/cm3
Germanium Doping:
1e+18/cm3
Substrate Composition Si(x)Ge(1-x):
1
Mat 1 Material:
Silicon
Mat 2 Material:
Germanium
Mat 1 Horizontal Thickness:
0.2nm
Mat 1 Vertical Thickness:
0.2nm
Mat 2 Horizontal Thickness:
0.2nm
Mat 2 Vertical Thickness:
0.2nm
Reflective Boundary Condition
Substrate
Mat 1 Vertical Thickness
Mat 1
Mat 1 Horizontal Thickness
Mat2
Mat 1 Horizontal Thickness
Mat 2 Vertical Thickness
Substrate
Mat 2
Storage (manage)
3% of 10GB

Case Study 1: Simulate the density of states of quantum well of thickness 0.5 nm



Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm



Result:

Simulation of density of states of quantum well structure with various nanometer thickness is accomplished.

Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub

Aim:

To obtain the energy states of Quantum dots.

Software: NanoHub Online Tool

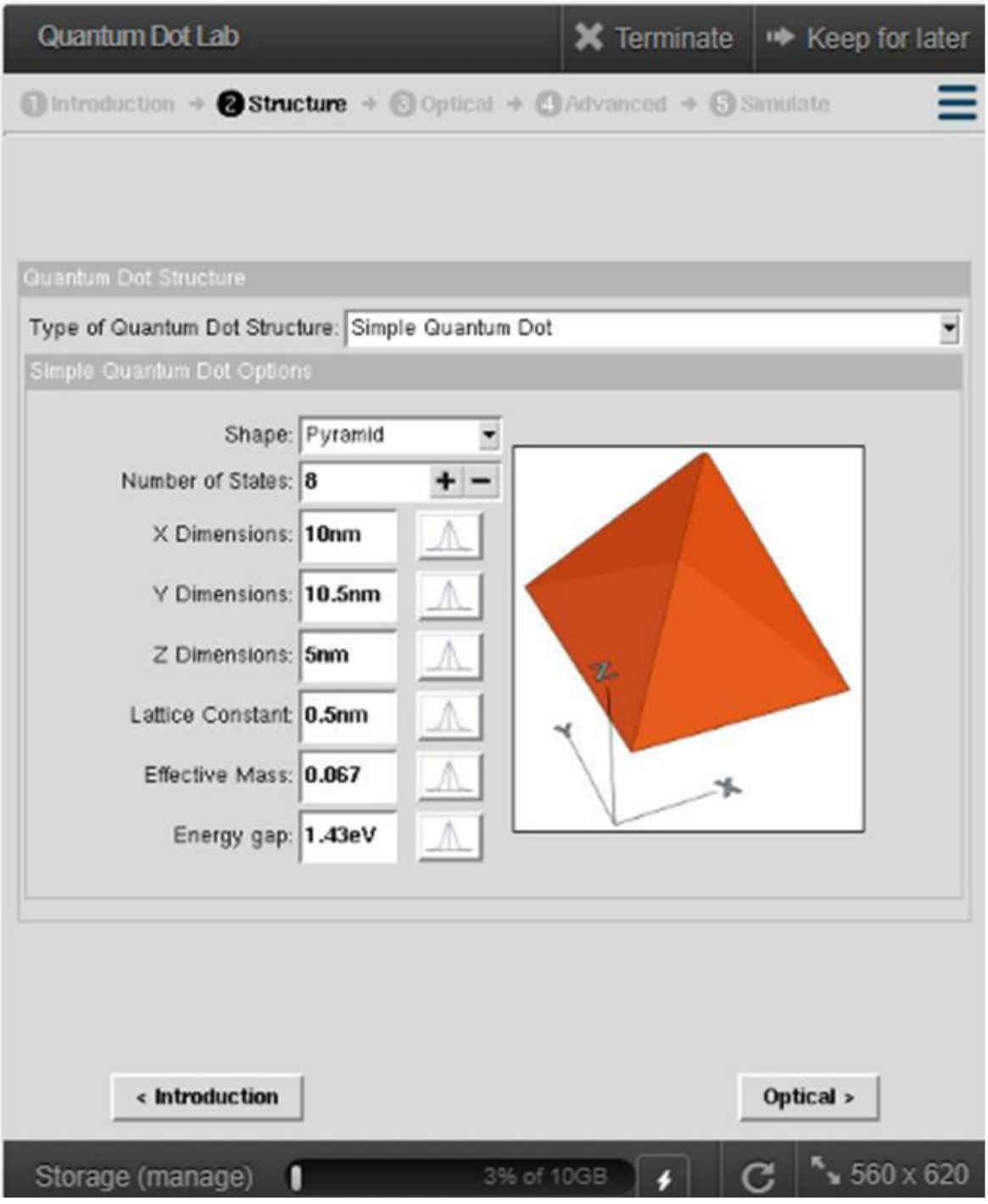
Theory:

Quantum dots (QD) are very small semiconductor particles, only several nanometres in size, so small that their optical and electronic properties differ from those of larger particles. They are a central theme in nanotechnology. Many types of quantum dot will emit light of specific frequencies if electricity or light is applied to them, and these frequencies can be precisely tuned by changing the dots' size, shape and material, giving rise to many applications.Quantum dots exhibit properties that are intermediate between those of bulk semiconductors and those of discrete molecules. Their optoelectronic properties change as a function of both size and shape. Larger QDs (diameter of 5–6 nm, for example) emit longer wavelengths resulting in emission colors such as orange or red. Smaller QDs (diameter of 2–3 nm, for example) emit shorter wavelengths resulting in colors like blue and green, although the specific colors and sizes vary depending on the exact composition of the QD.Because of their highly tunable properties, QDs are of wide interest. Potential applications include transistors, solar cells, LEDs, diode lasers and second-harmonic generation, quantum computing, and medical imaging.

Procedure:

- Open nanoHUB.
- Goto Resources -> Tools.
- Select Quantum Dot in Tags.
- Select Quantum dot -> Quantum dot Lab
- Press Launch tool in info section.
- Select the type of Quantum dot Structure
- Select Energy Bandgap and enter the value
- Press simulate to view results.
- Select Result - > Energy States
- Repeat the above steps for simple and multi-layer quantum dot
- Compare the results of various materials.

Output:


Quantum Dot Lab
Terminate
Keep for later
1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure:
Simple Quantum Dot
Simple Quantum Dot Options
Shape:
Pyramid
Number of States:
8
X Dimensions:
10nm
Y Dimensions:
10.5nm
Z Dimensions:
5nm
Lattice Constant:
0.5nm
Effective Mass:
0.067
Energy gap:
1.43eV
Z
Y
X
< Introduction
Optical >
Storage (manage)
3% of 10GB
560 x 620

Quantum Dot Lab | Terminate | Keep for later

1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate

Light Incident Angles

Angle Theta: 45deg

Angle Phi: 0deg

Sweep

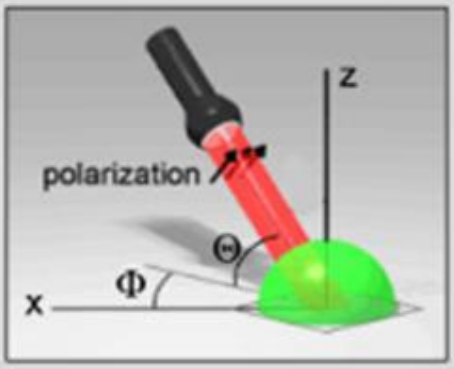
Sweep Parameter: Angle phi (deg)

Minimum: 0

Maximum: 90

Number of Points: 3 + −

Light Source



Absorption

Simple Q Dot - Absolute Fermi Level: no

Simple Q Dot - Rel. or Abs. Fermi Level: 0eV

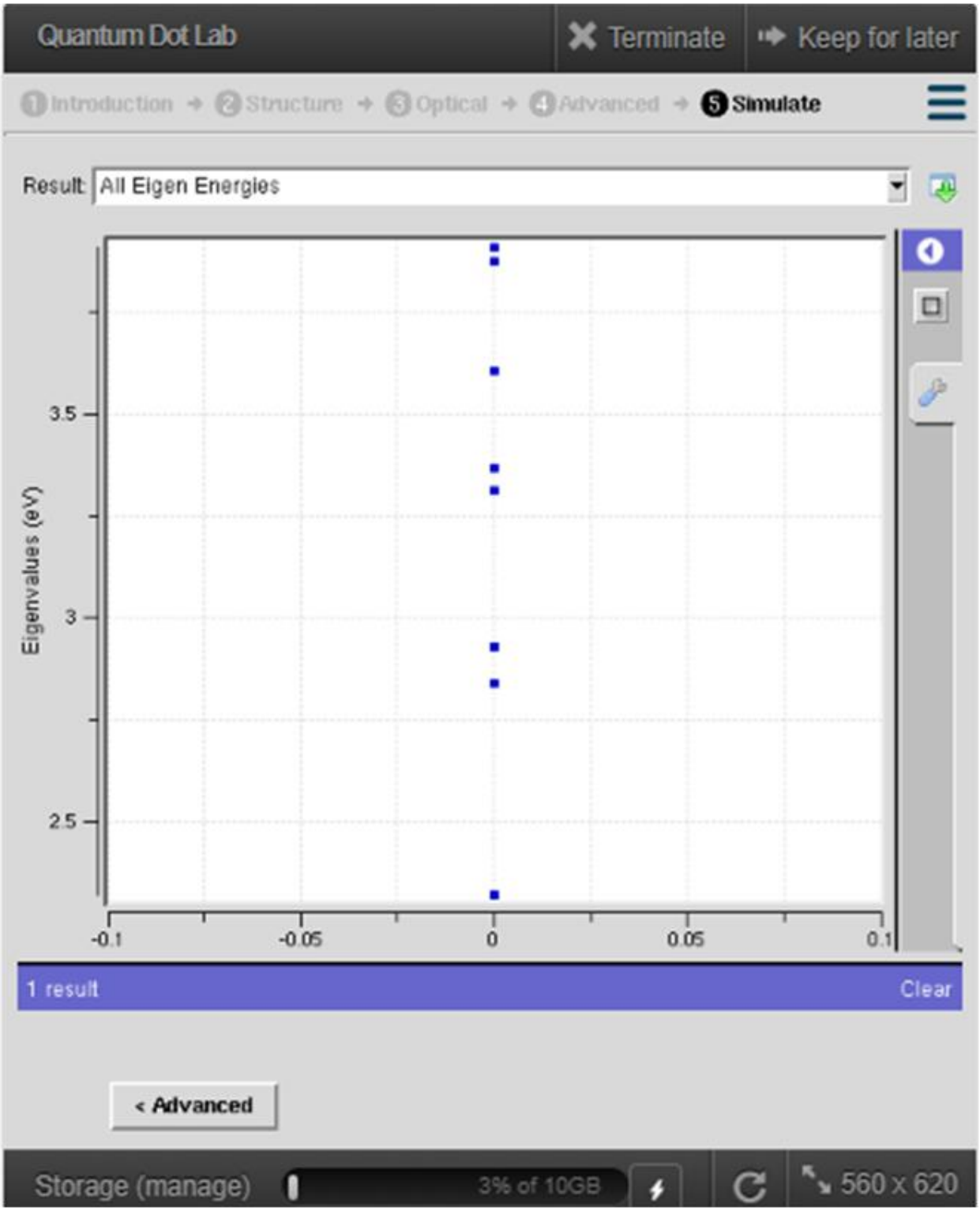
Multi-Layer - Fermi Level: 0.7eV

Temperature: 300K

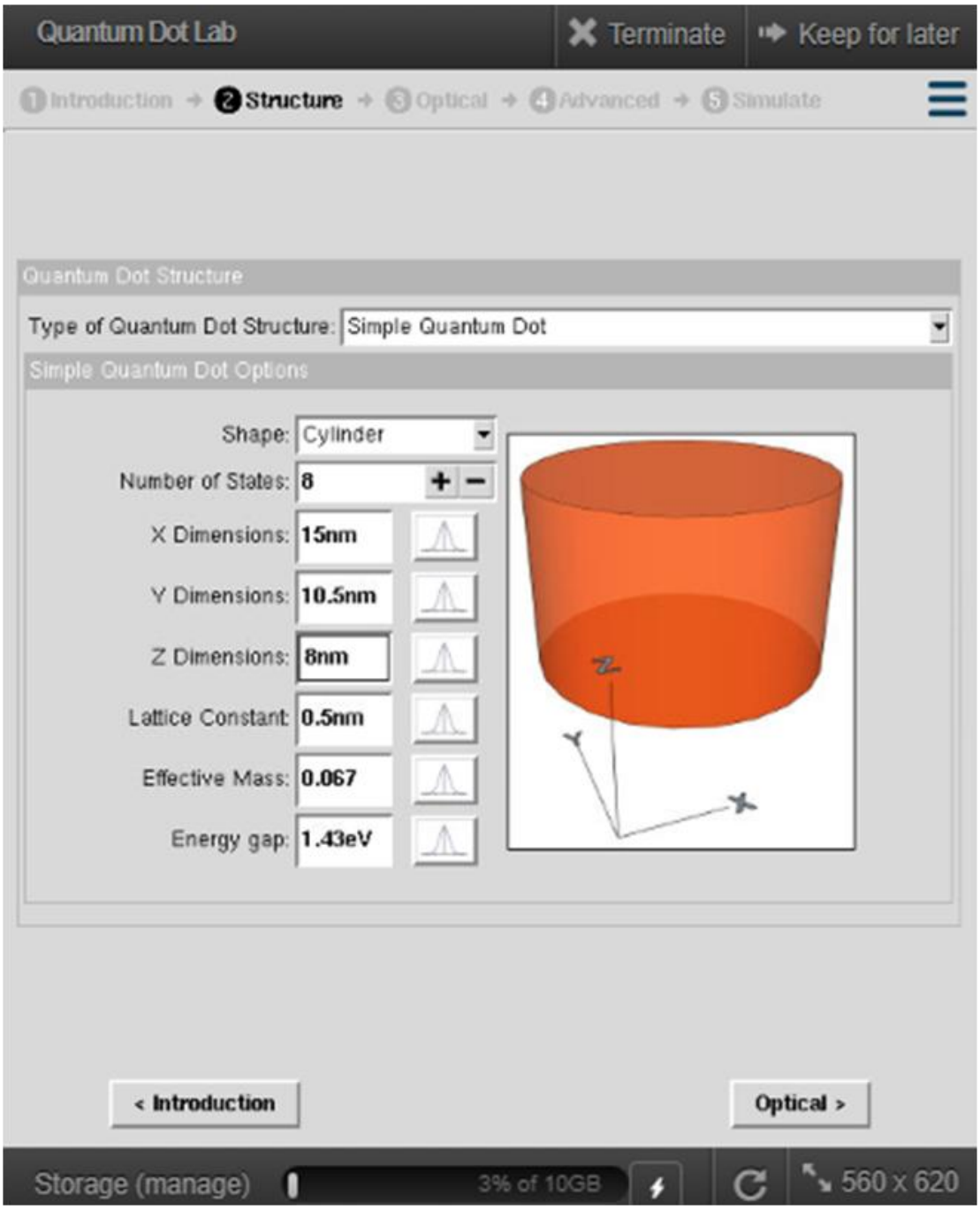
State Broadening: 0.001

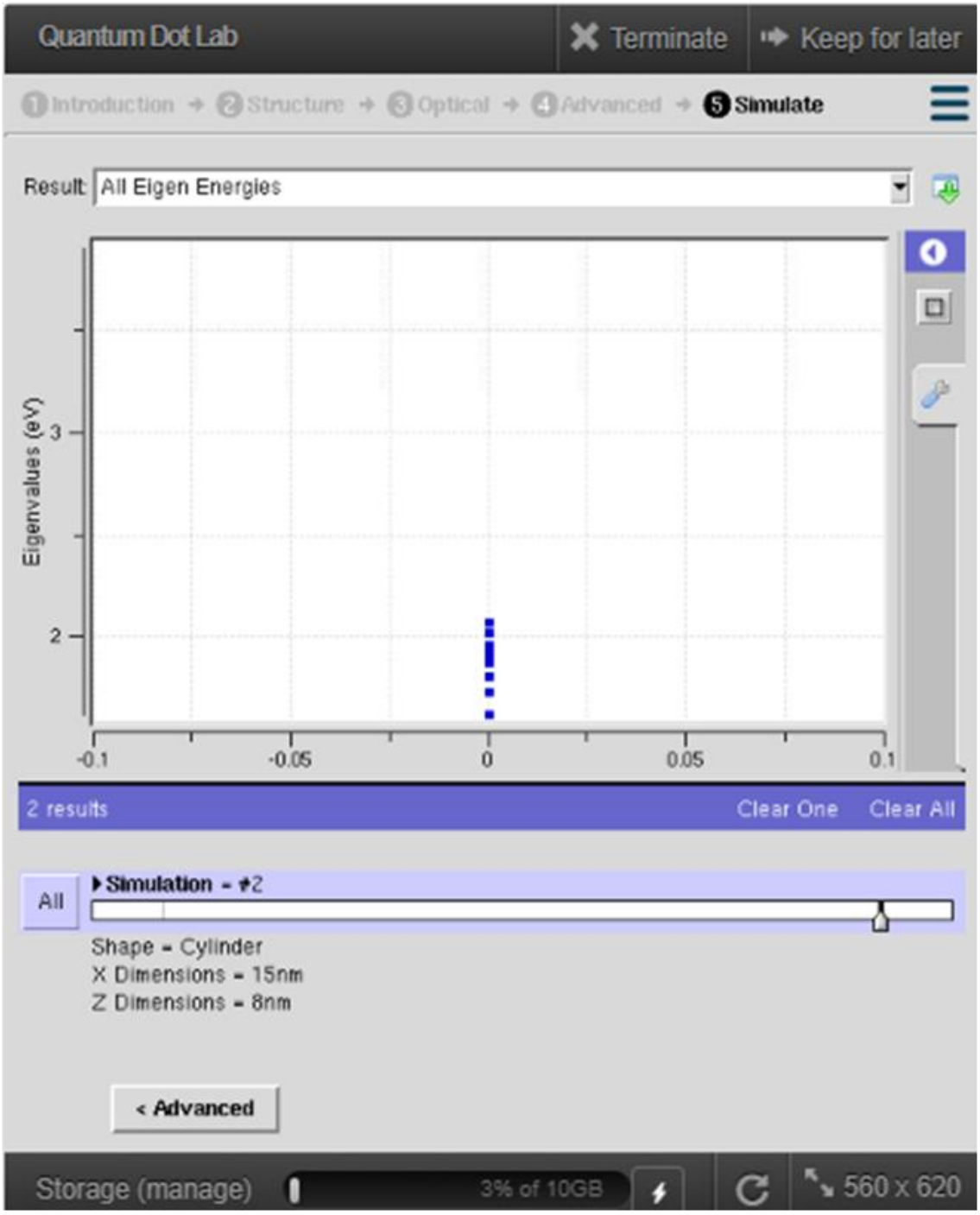
« Structure | Advanced »

Storage (manage) 3% of 10GB 560 x 620

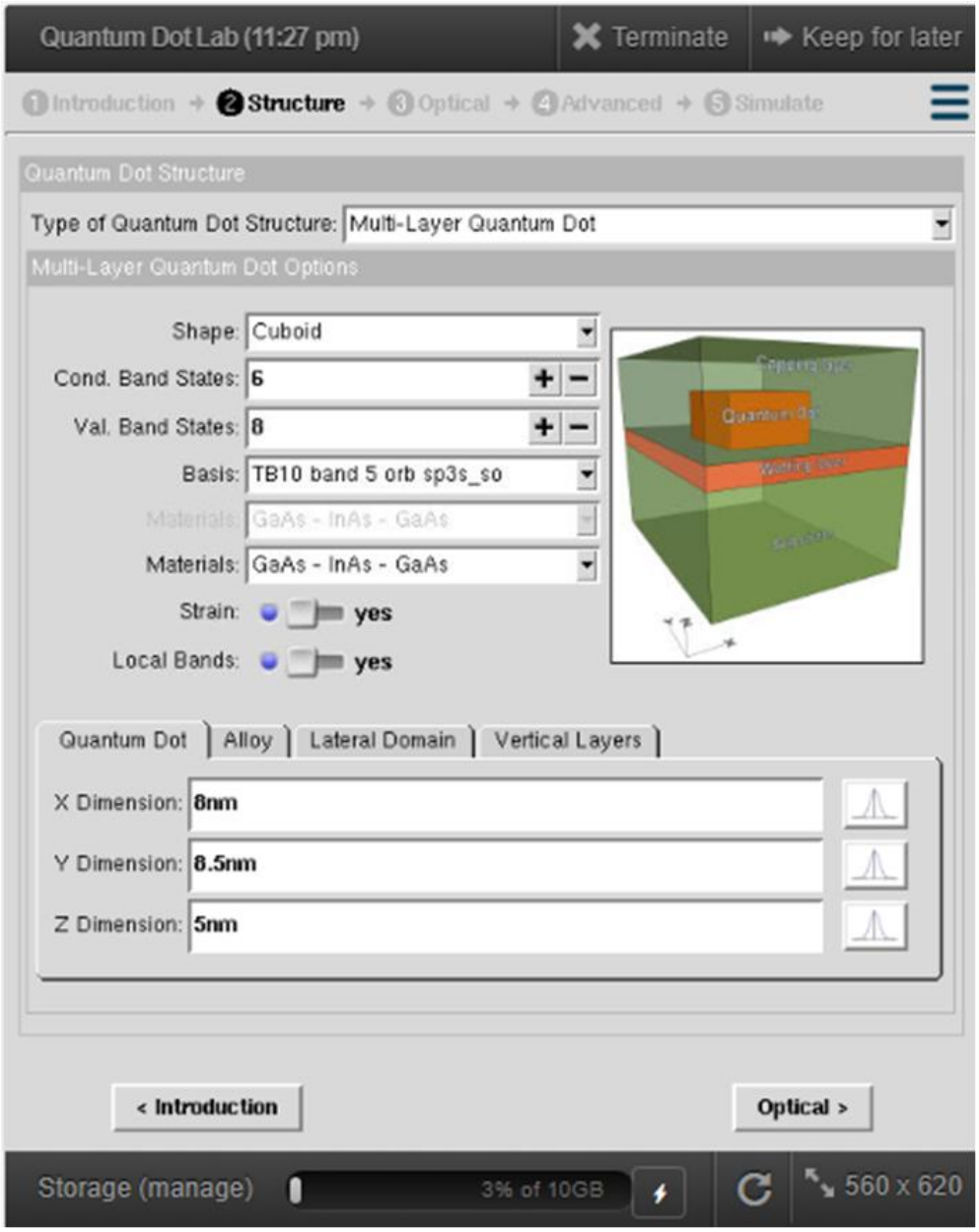

Quantum Dot Lab
Terminate
Keep for later
1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Result: All Eigen Energies
Eigenvalues (eV)
3.5
3
2.5
-0.1
-0.05
0
0.05
0.1
1 result
Clear
< Advanced
Storage (manage)
3% of 10GB
560 x 620

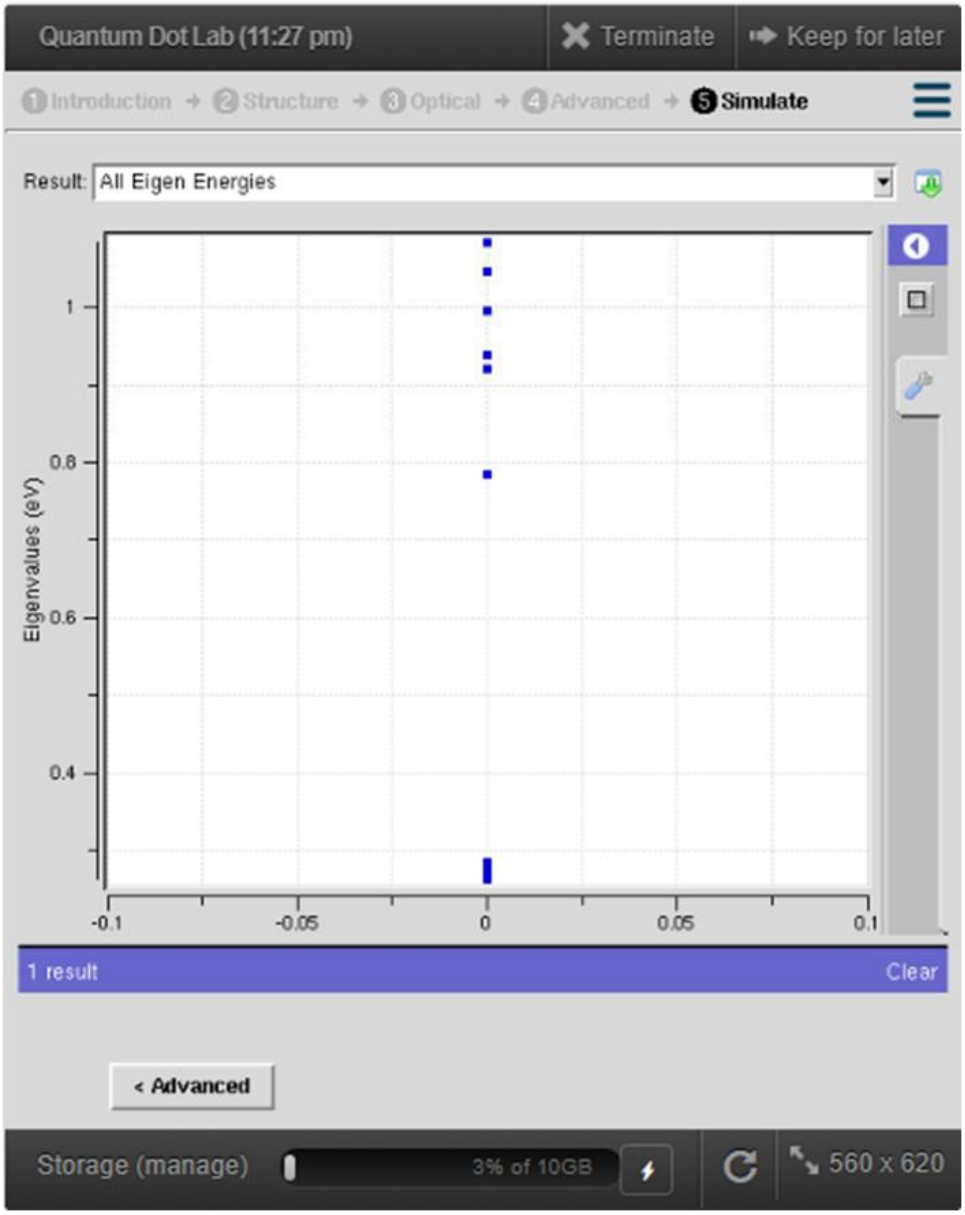
Case Study 1: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8 nm.





Case Study 2: Simulate the energy states of two layer cuboid quantum dots


Quantum Dot Lab (11:27 pm)
Terminate
Keep for later
1 Introduction
2 Structure
3 Optical
4 Advanced
5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure: Multi-Layer Quantum Dot
Multi-Layer Quantum Dot Options
Shape: Cuboid
Cond. Band States: 6
Val. Band States: 8
Basis: TB10 band 5 orb sp3s_so
Materials: GaAs - InAs - GaAs
Materials: GaAs - InAs - GaAs
Strain: yes
Local Bands: yes
Quantum Dot
Alloy
Lateral Domain
Vertical Layers
X Dimension: 8nm
Y Dimension: 8.5nm
Z Dimension: 5nm
< Introduction
Optical >
Storage (manage)
3% of 10GB
560 x 620



Analytical:

A thin film of SiO2 is deposited on a silicon substrate. The film has a thickness of 50 nm, and a refractive index of 1.5. Calculate the wavelength of light that experiences constructive interference in the film.

Given

- Thickness of the film, t=50 nm,
- Refractive index of the film, n=1.5.

The formula for constructive interference in a thin film with thickness t, refractive index n, and incident light of wavelength λ in a vacuum (or air) is given by:

$$2nt = m\lambda$$

where:

- t is the thickness of the film,
- n is the refractive index of the film,
- λ is the wavelength of the light in vacuum,
- m is an integer representing the order of the interference.

For the first-order constructive interference (where $m = 1$):

$$2nt = \lambda$$

Substituting the given values:

$$2 \cdot 1.5 \cdot 50\,\text{nm} = \lambda$$

$$\lambda = 150\,\text{nm}$$

Thus, the wavelength of light that experiences constructive interference in the film for the first order (m=1) is 150 nm. For higher orders (m=2, 3, etc.), the wavelength can be calculated similarly by multiplying 150 nm by the respective integer order:

- For $m = 2$: $\lambda = 2 \times 150\,\text{nm} = 300\,\text{nm}$
- For $m = 3$: $\lambda = 3 \times 150\,\text{nm} = 450\,\text{nm}$

Result:

The Simulation of energy states of quantum dots is accomplished.

Exp. 4. Kronig Penney Model

Aim:

To simulate Energy functional vs Energy graph using Kronig Penney model

Software: NanoHub Online Tool

Procedure:

- Open nanoHUB.
- Goto Resources -> Tools.
- Select Kronig Penney Model in Tags.
- Select Kronig Penney Model -> Periodic Potential Lab
- Press Launch tool in info section.
- Select Potential Type-> Step well
- Select Energy details -> Maximum Barrier Height ->1eV
- Press simulate to view results.
- Select Result - > Energy Functional Vs Energy

Theory:

The Kronig-Penney model is a simplified theoretical model used in solid-state physics to describe the behavior of electrons in a periodic potential, such as the potential experienced by electrons in a crystal lattice. This model was introduced by Ralph Kronig and William G. Penney in 1931 and is particularly useful for understanding electronic band structure in materials like semiconductors and insulators.

1. Basic Assumptions of the Model

- The model assumes that the electron in a one-dimensional system (simplified from the three-dimensional nature of real solids) moves in a periodic potential.
- This potential is typically modeled as a series of square wells (often called "delta function wells"), but the exact form can vary. The periodicity of the potential reflects the repeating structure of atoms in a crystal.
- The model assumes free electrons between these potential wells, meaning there are no interactions between the electrons themselves.

2. The Potential

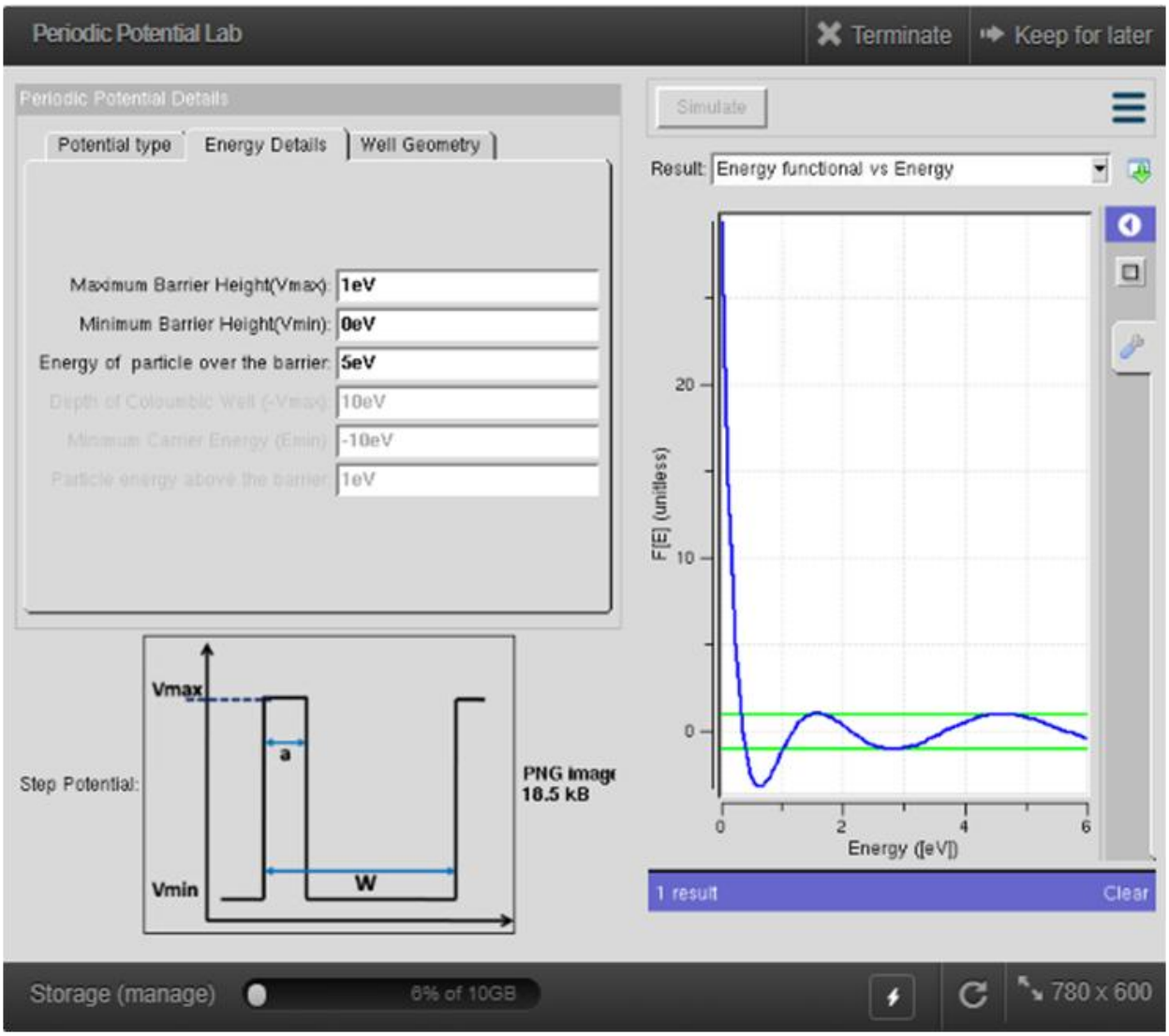
The potential in the Kronig-Penney model is periodic and is often represented as a series of delta functions or square wells in one dimension. A simple potential can be written as:

$$V(x) = \begin{cases} -V_0, & \text{for } |x| \leq a/2 \\ 0, & \text{for } |x| > a/2 \end{cases}$$

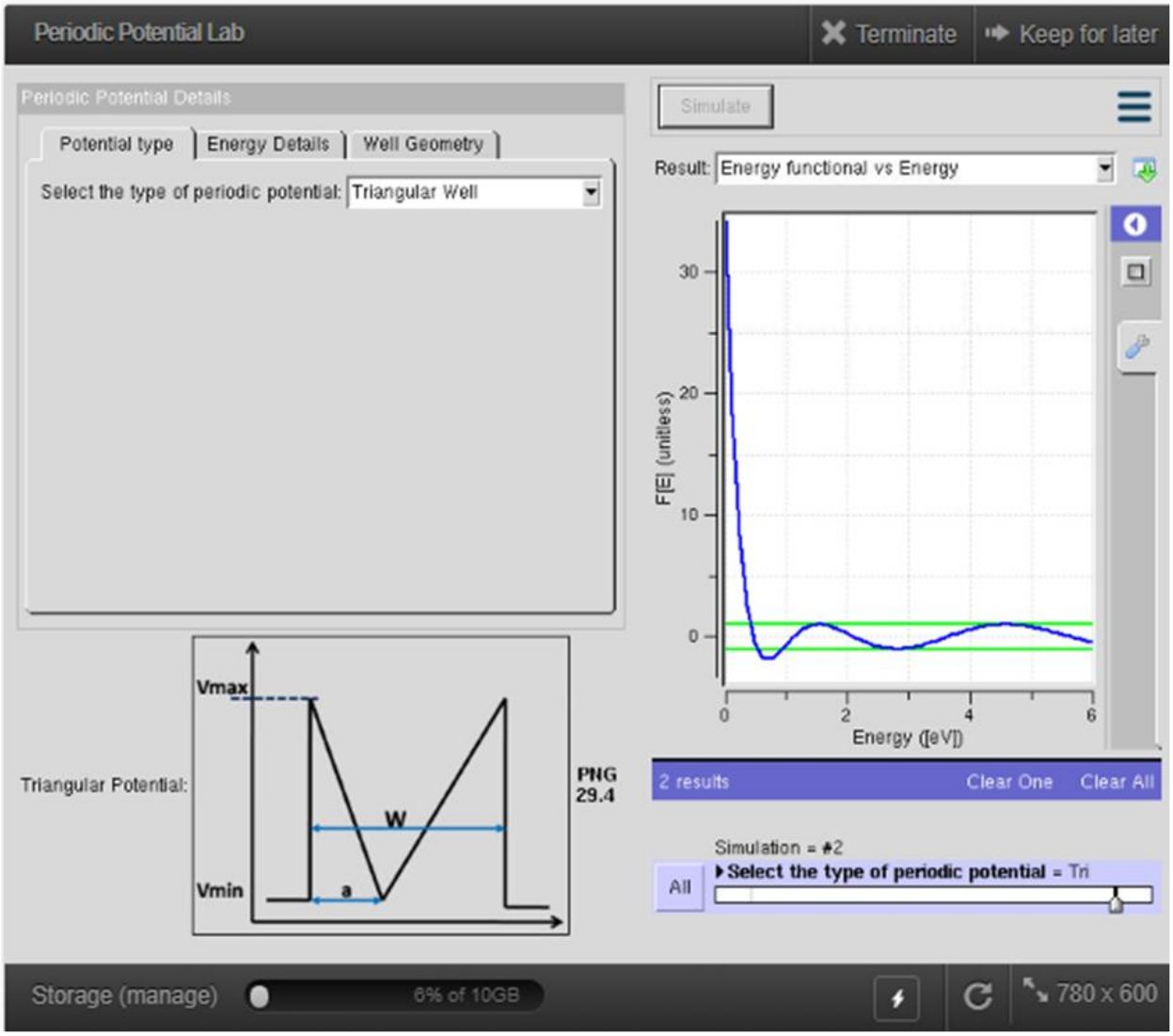
V0 is the strength of the potential well, a is the width of the potential well, and the potential is periodic with period b (i.e., the distance between two adjacent wells).

This means that the potential is a repeating square well every distance bbb, with each well having a depth V0 and a width a.

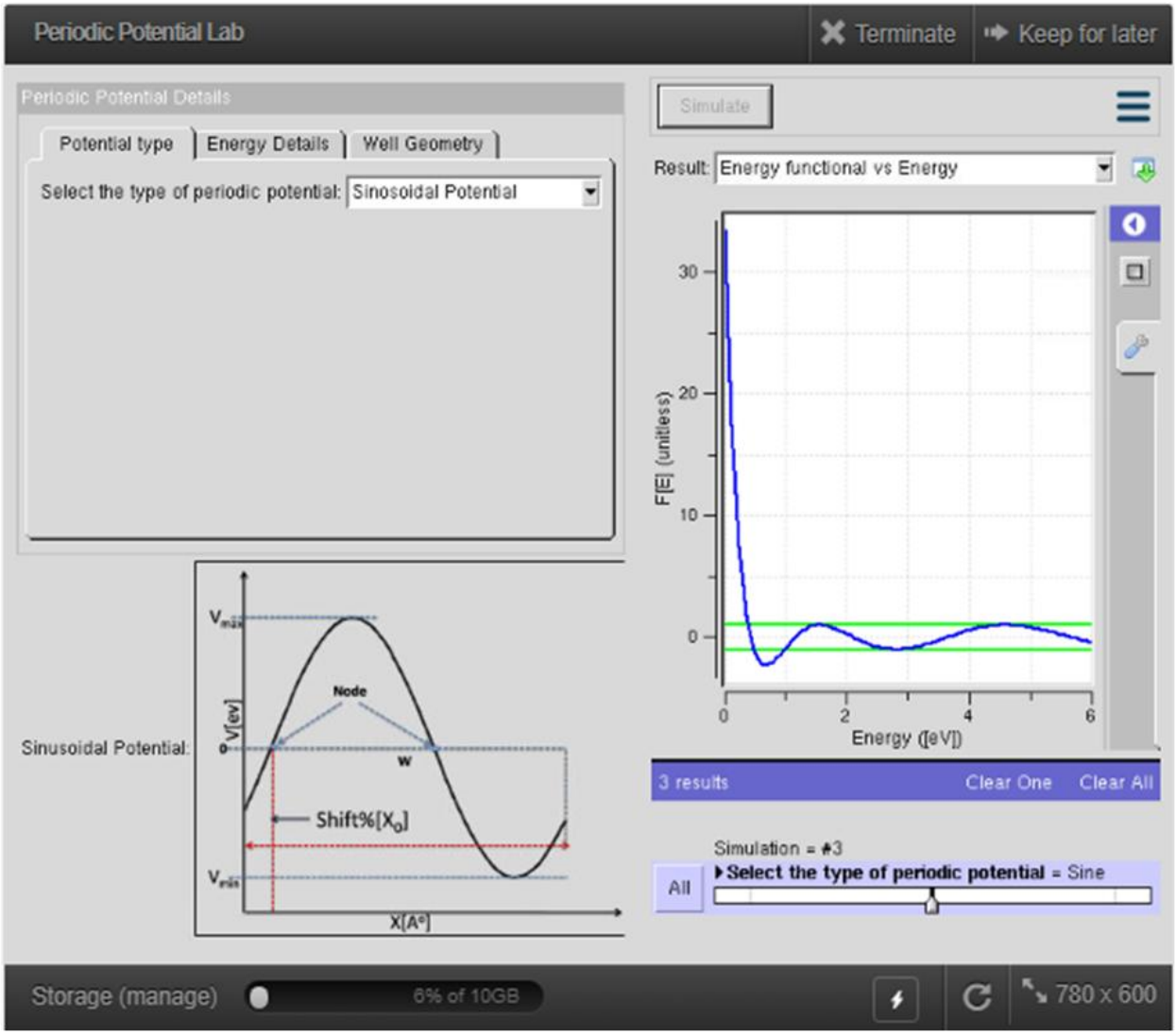
Output:



Case Study 1: Simulate the output for Triangular well



Case Study 2: Simulate the output for Sinusoidal Potential



Analytical:

Finding Energy Bands for a Simple Kronig-Penney Model

Given:

- Depth of potential well $V_0 = 1\,\text{eV}$
- Width of the potential well $a = 0.2\,\text{nm}$
- Periodicity of the potential $b = 1\,\text{nm}$
- Mass of the electron $m = 9.11 \times 10^{-31}\,\text{kg}$
- Planck's constant $\hbar = 1.055 \times 10^{-34}\,\text{J} \cdot \text{s}$

Given:

- $\hbar = 1.055 \times 10^{-34}\,\text{J} \cdot \text{s}$
- $m = 9.11 \times 10^{-31}\,\text{kg}$
- $b = 1\,\text{nm} = 1 \times 10^{-9}\,\text{m}$

We can calculate the first energy band E_1:

$$E_1 \approx \frac{(1.055 \times 10^{-34})^2 \pi^2}{2(9.11 \times 10^{-31})(1 \times 10^{-9})^2}$$

$$E_1 \approx 0.48\,\text{eV}$$

Thus, the first allowed energy band is approximately **0.48 eV**.

Result:

The energy functional vs Energy graph using the Kronig Penney model is simulated.

Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

Aim:

To simulate the band structure of nanowire for various materials.

Software: NanoHub Online Tool

Theory:

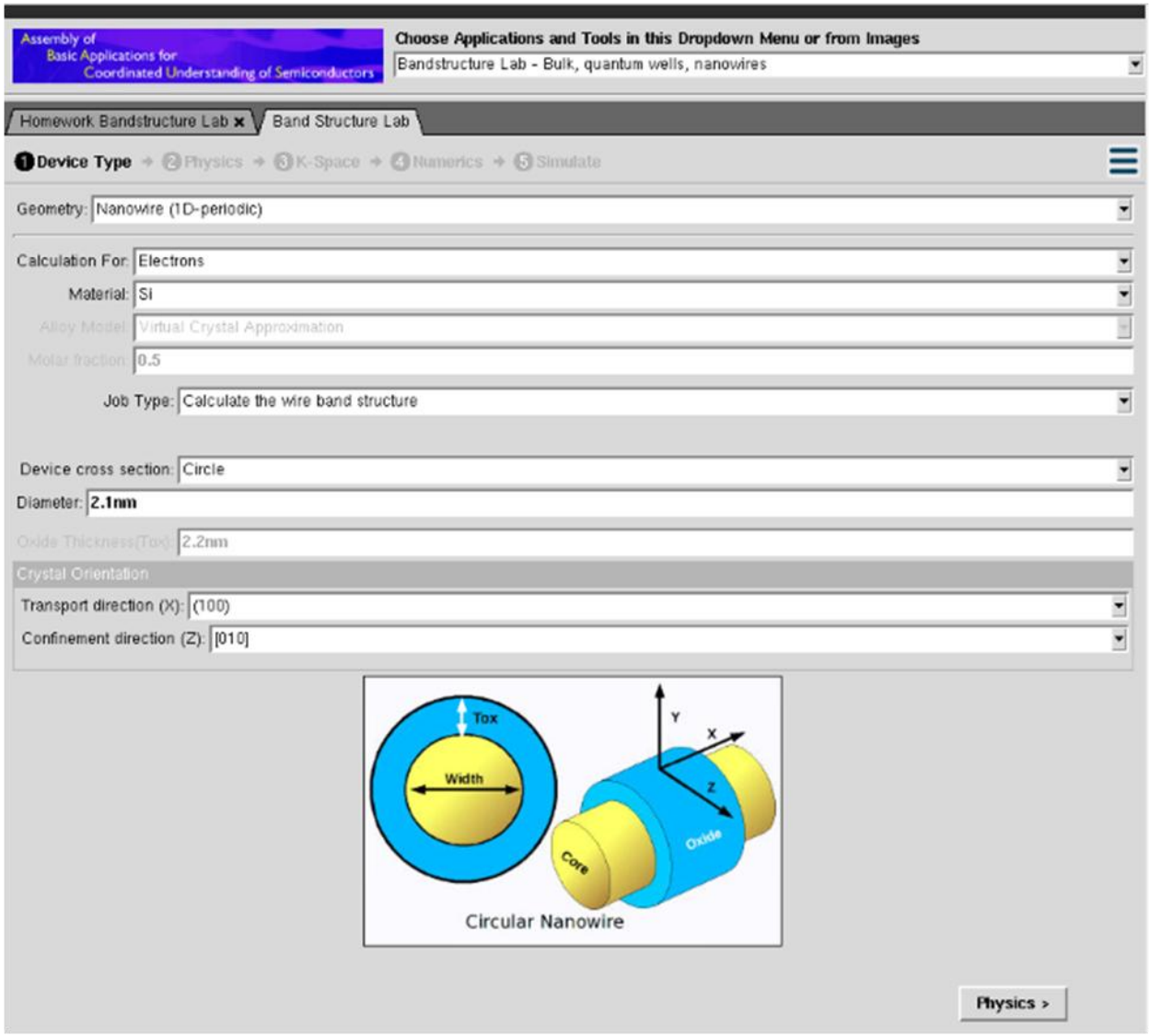
A quantum wire is an electrically conducting wire in which quantum effects influence the transport properties. Usually such effects appear in the dimension of nanometres.If the diameter of a wire is sufficiently small, electrons will experience quantum confinement in the transverse direction. As a result, their transverse energy will be limited to a series of discrete values. One consequence of this quantization is that the classical formula for calculating the electrical resistance of a wire

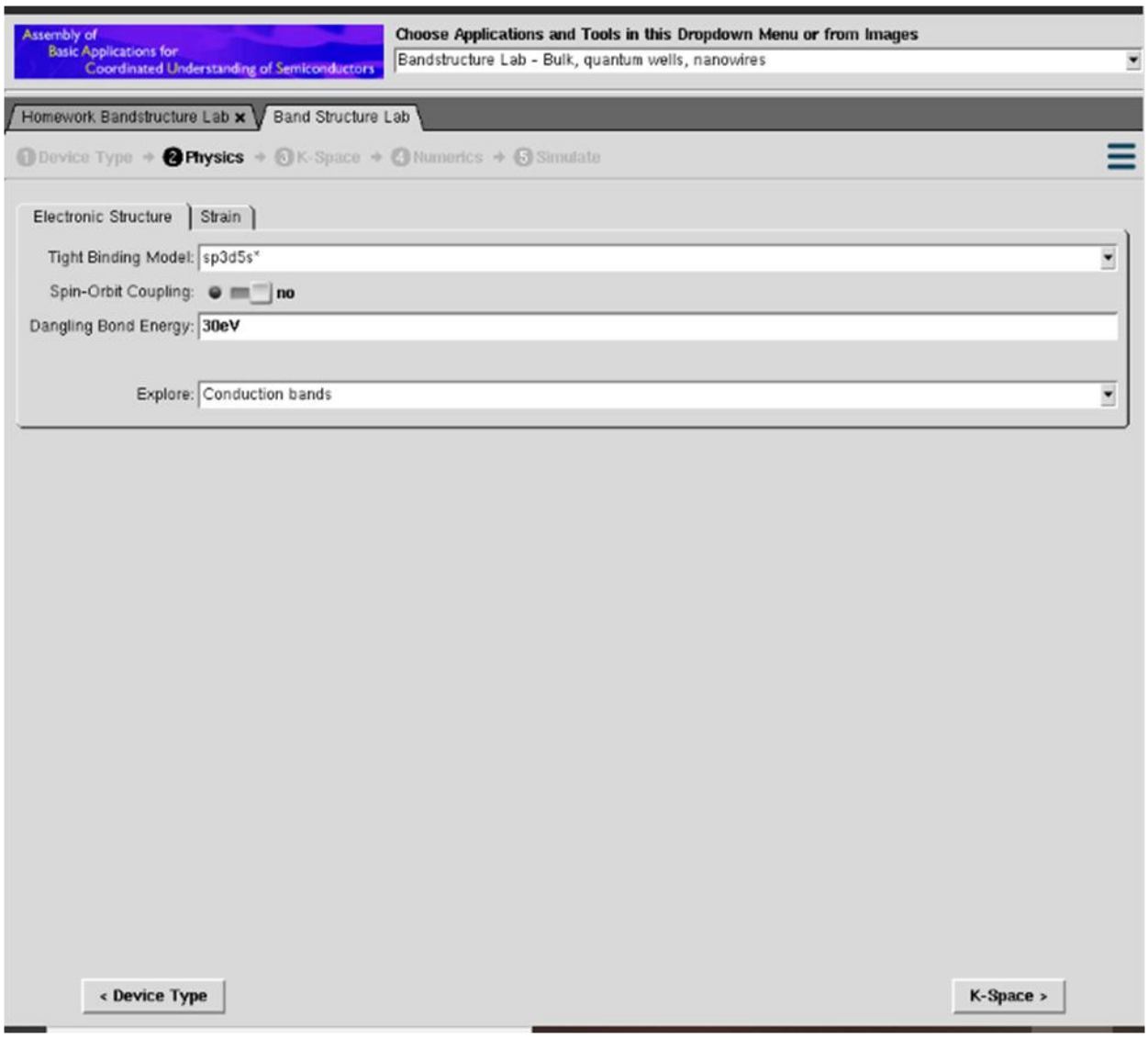
$$R = \rho \frac{l}{A}.$$

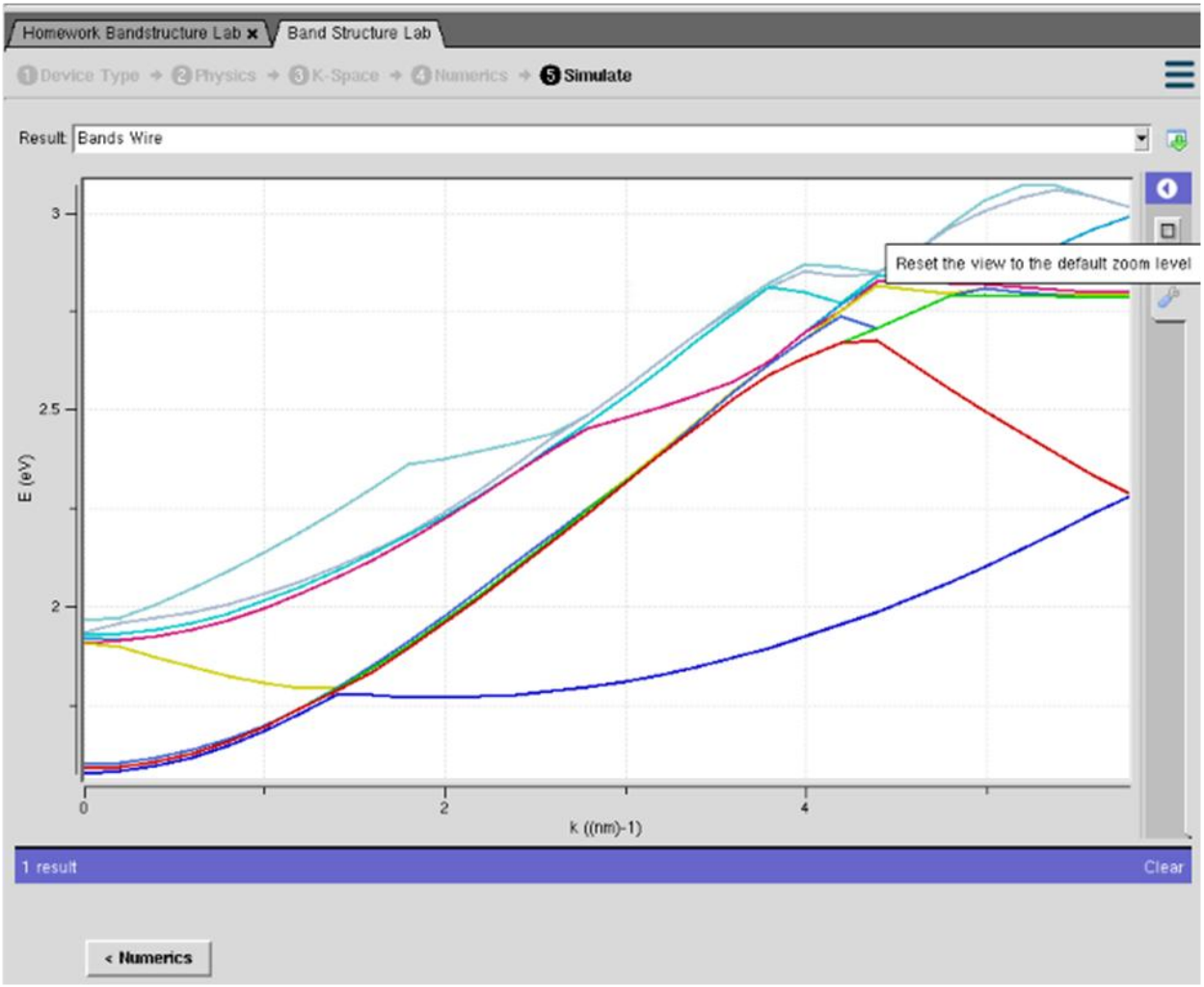
Procedure:

- Open nanoHUB.
- Goto Resources -> Tools.
- Select Nanowire in Tags.
- Select ABACUS - Assembly of Basic Applications for Coordinated Understanding of Semiconductors in Resources.
- Press Launch tool in the info section.
- Select the Applications -> Band structure Lab
- Select Geometry -> 1D Nanowire
- Device cross section -> Circular
- Select the Material Type
- Press simulate to view results.
- Compare the results of various materials.

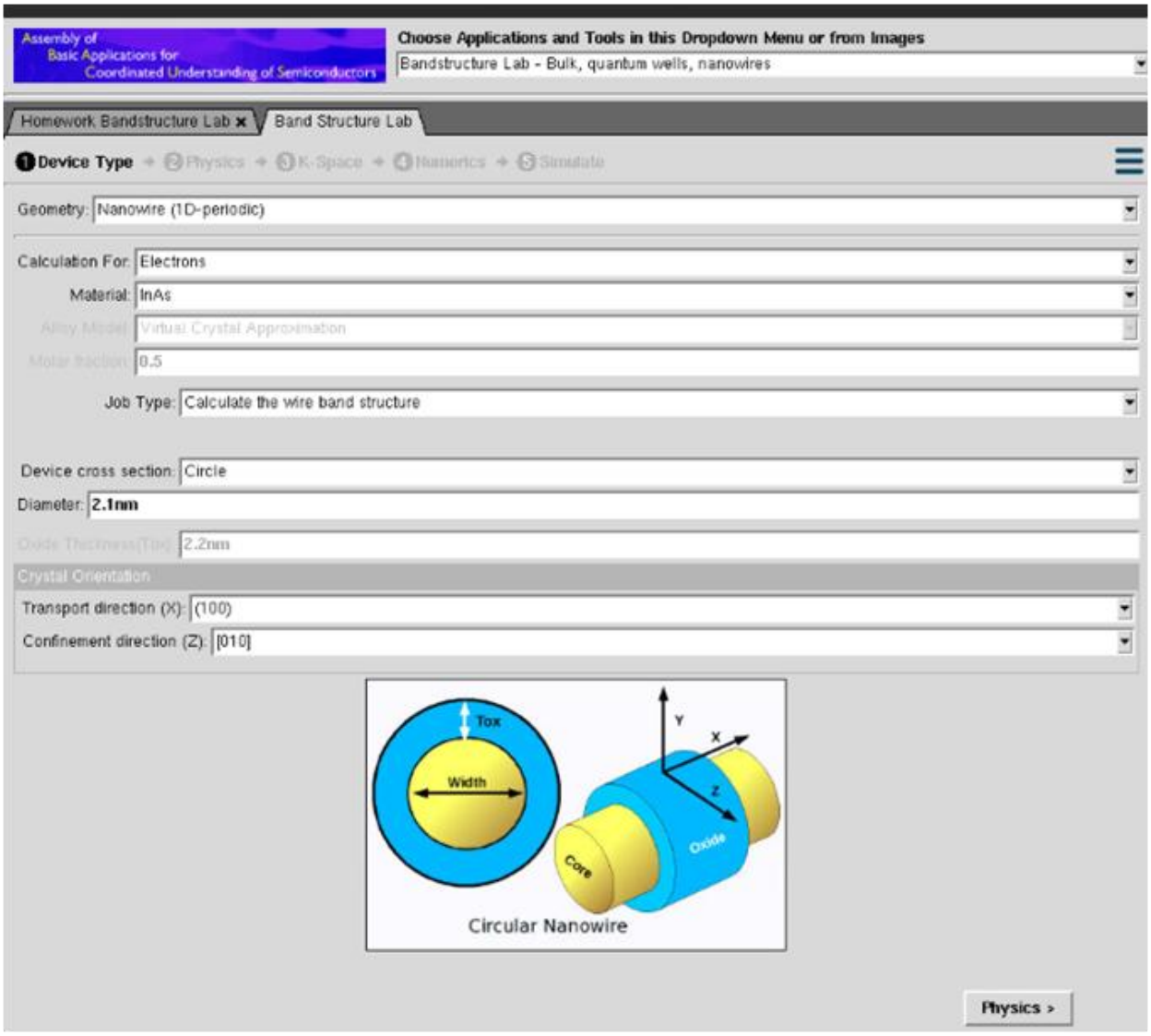
Output:

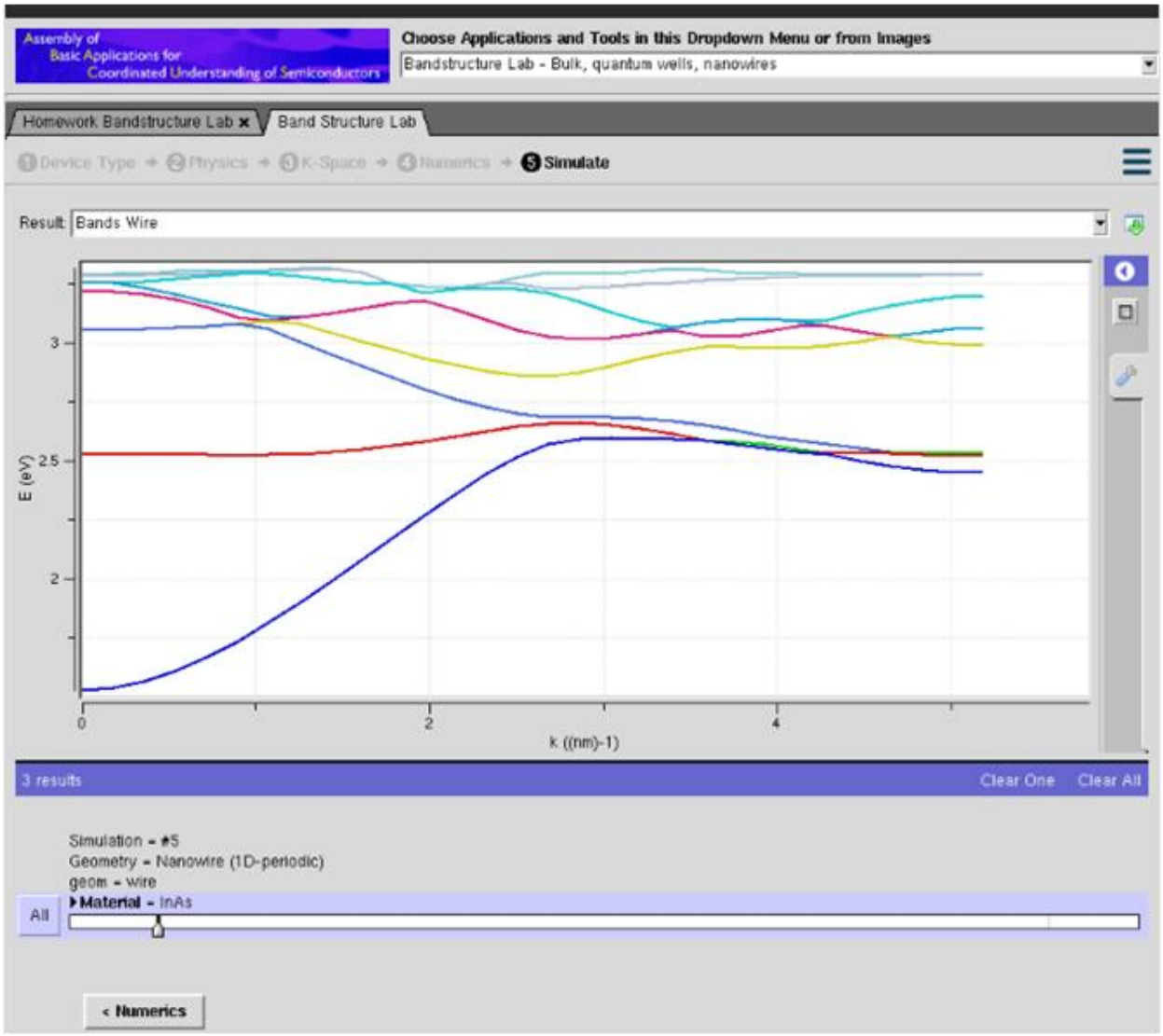

Assembly of
Basic Applications for
Coordinated Understanding of Semiconductors
Choose Applications and Tools in this Dropdown Menu or from Images
Bandstructure Lab - Bulk, quantum wells, nanowires
Homework Bandstructure Lab
Band Structure Lab
1 Device Type
2 Physics
3 K-Space
4 Numerics
5 Simulate
Geometry: Nanowire (1D-periodic)
Calculation For: Electrons
Material: Si
Alloy Model: Virtual Crystal Approximation
Molar fraction: 0.5
Job Type: Calculate the wire band structure
Device cross section: Circle
Diameter: 2.1nm
Oxide Thickness(Tox): 2.2nm
Crystal Orientation
Transport direction (X): (100)
Confinement direction (Z): [010]
Tox
Width
Y
X
Z
Core
Oxide
Circular Nanowire
Physics >


Assembly of
Basic Applications for
Coordinated Understanding of Semiconductors
Choose Applications and Tools in this Dropdown Menu or from Images
Bandstructure Lab - Bulk, quantum wells, nanowires
Homework Bandstructure Lab
Band Structure Lab
1 Device Type
2 Physics
3 K-Space
4 Numerics
5 Simulate
Electronic Structure
Strain
Tight Binding Model:
sp3d5s*
Spin-Orbit Coupling:
no
Dangling Bond Energy:
30eV
Explore:
Conduction bands
< Device Type
K-Space >

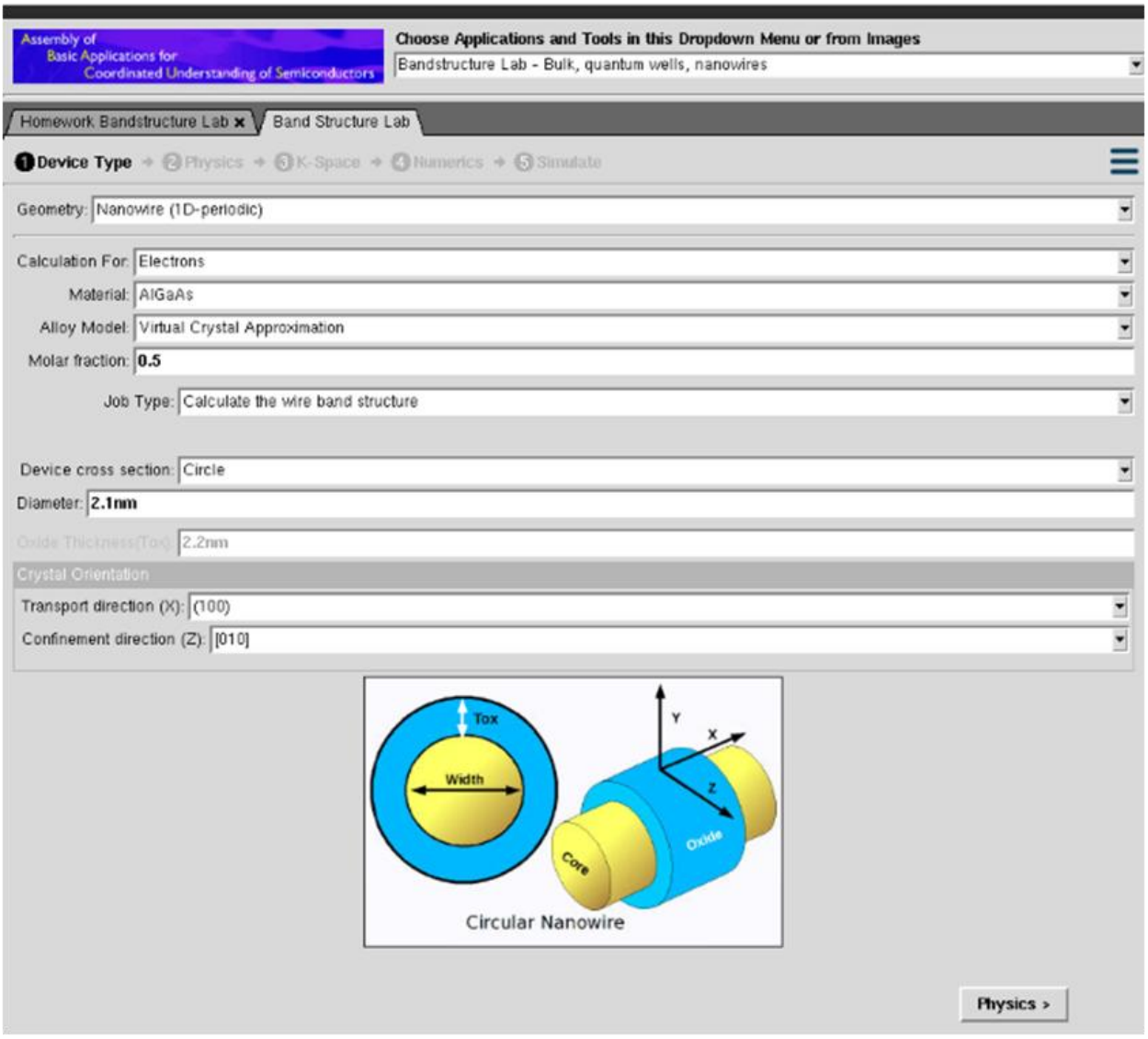

Homework Bandstructure Lab
Band Structure Lab
1 Device Type
2 Physics
3 K-Space
4 Numerics
5 Simulate
Result: Bands Wire
Reset the view to the default zoom level
E (eV)
3
2.5
2
0
2
4
k ((nm)-1)
1 result
Clear
< Numerics

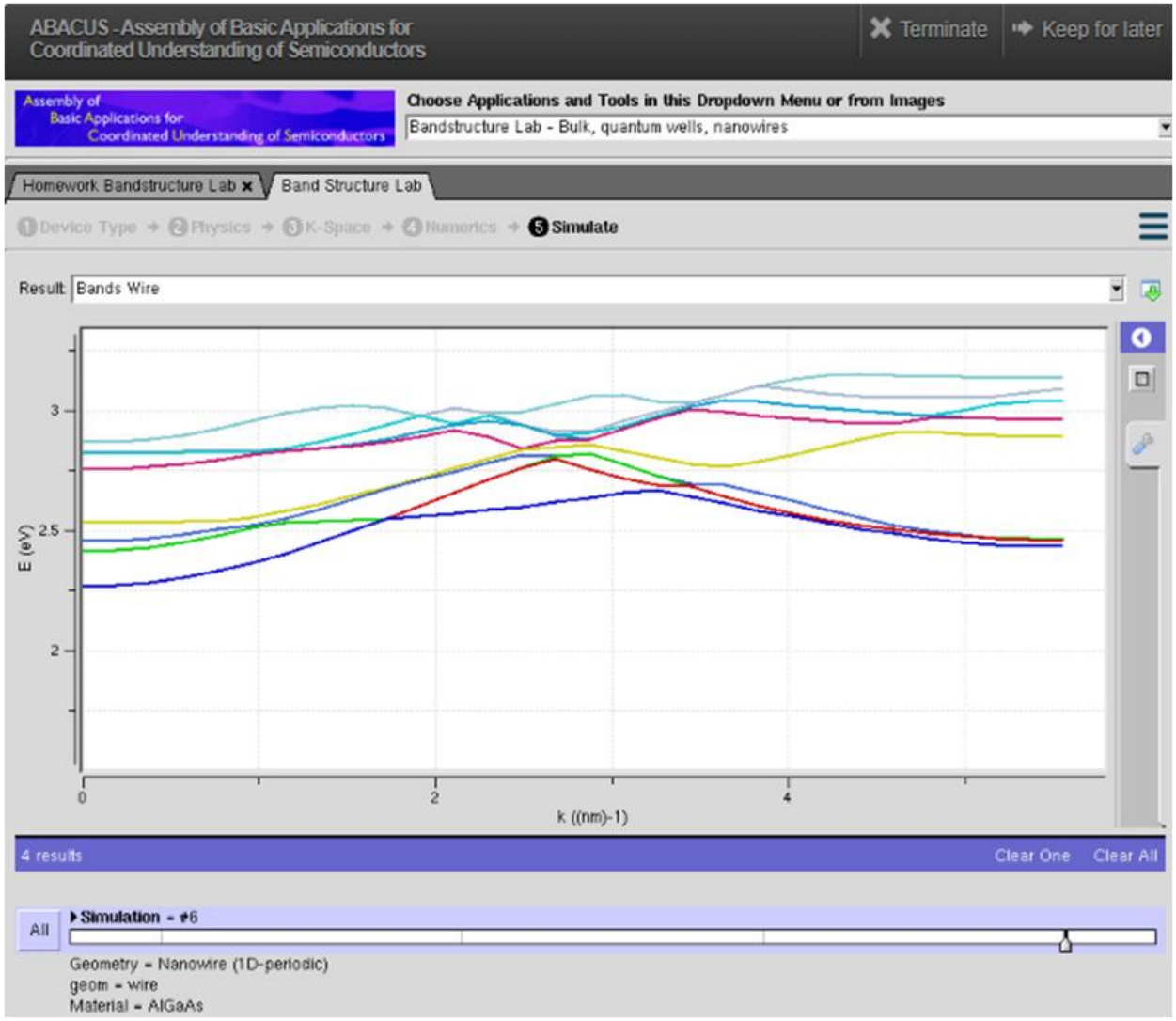
Case Study 1: Simulate the band-structure of InAs circular nanowire





Case Study 2: Simulate the band-structure of AlGaAs circular nanowire


Assembly of
Basic Applications for
Coordinated Understanding of Semiconductors
Choose Applications and Tools in this Dropdown Menu or from Images
Bandstructure Lab - Bulk, quantum wells, nanowires
Homework Bandstructure Lab
Band Structure Lab
1 Device Type
2 Physics
3 K-Space
4 Numerics
5 Simulate
Geometry: Nanowire (1D-periodic)
Calculation For: Electrons
Material: AlGaAs
Alloy Model: Virtual Crystal Approximation
Molar fraction: 0.5
Job Type: Calculate the wire band structure
Device cross section: Circle
Diameter: 2.1nm
Oxide Thickness(Tox): 2.2nm
Crystal Orientation
Transport direction (X): (100)
Confinement direction (Z): [010]
Tox
Width
Y
X
Z
Core
Oxide
Circular Nanowire
Physics »



Result:

Simulation of band structure of nanowire for various materials is accomplished.

Exp. 6. Demonstration of Quantum Tunneling

Aim

To learn the demonstration of Quantum Tunneling is analysed and the key learning points are observed using nano hub simulation tool

Abstract

Quantum tunneling, a phenomenon where particles penetrate energy barriers they classically cannot overcome, is pivotal in the realm of nanoelectronics. This paper demonstrates the principles of quantum tunneling and its applications in nanoscale devices, where quantum mechanical effects dominate. We explore the tunneling current behavior in nanoscale junctions and its dependence on barrier height, width, and material properties, employing advanced modeling and experimental data.

Key applications such as tunnel diodes, single-electron transistors, and flash memory cells are examined, highlighting their reliance on tunneling mechanisms for enhanced functionality. By leveraging tunneling effects, these devices achieve unprecedented efficiency, speed, and miniaturization, underscoring the transformative potential of quantum mechanics in modern electronics. This work also discusses challenges such as leakage currents and energy dissipation, offering insights into optimizing nanoelectronic device performance through material engineering and design innovations.

Tool name: Piece-Wise Constant Potential Barriers Sim2L

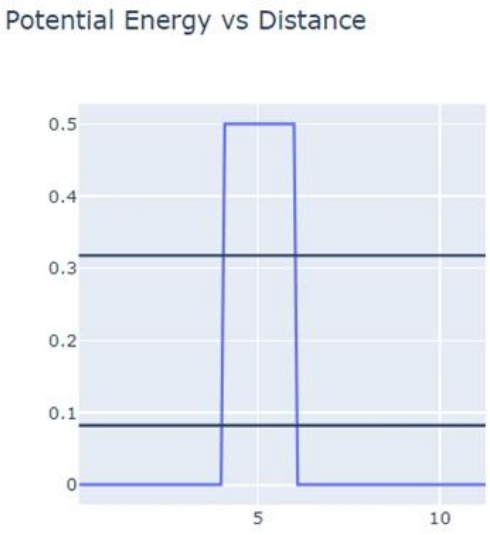
This tool simulates quantum mechanical transmission and reflection coefficients for potential barriers. It models structures with multiple segments of constant potential, useful for analyzing quantum tunneling, resonant tunneling diodes, and band formation in periodic potentials. The tool supports different approaches, including the transfer matrix and tight-binding methods, and allows for detailed analysis of barrier structures, effective mass variations, and resonance peaks. Researchers can apply this to explore quantum reflections and energy band behaviors in nanodevices.

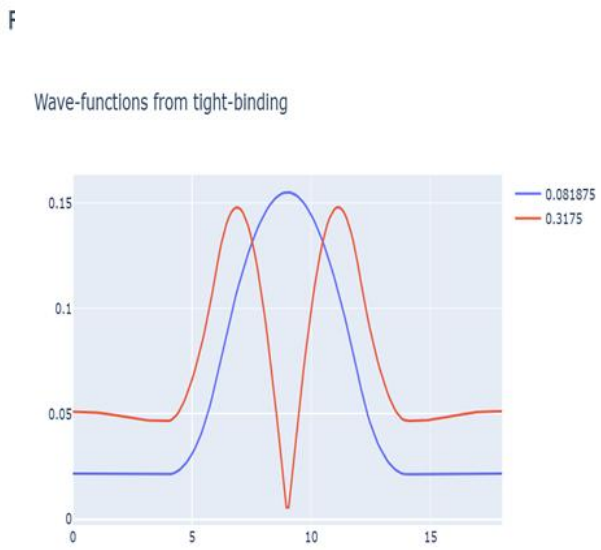
Procedure:

- Open nanoHUB.
- Goto Explore
- Goto Resources
- Select Tools- **Piece-Wise Constant Potential Barriers Sim2L**.
- Press Launch tool in info section.
- Enter into app mode tab in jupyter environment.

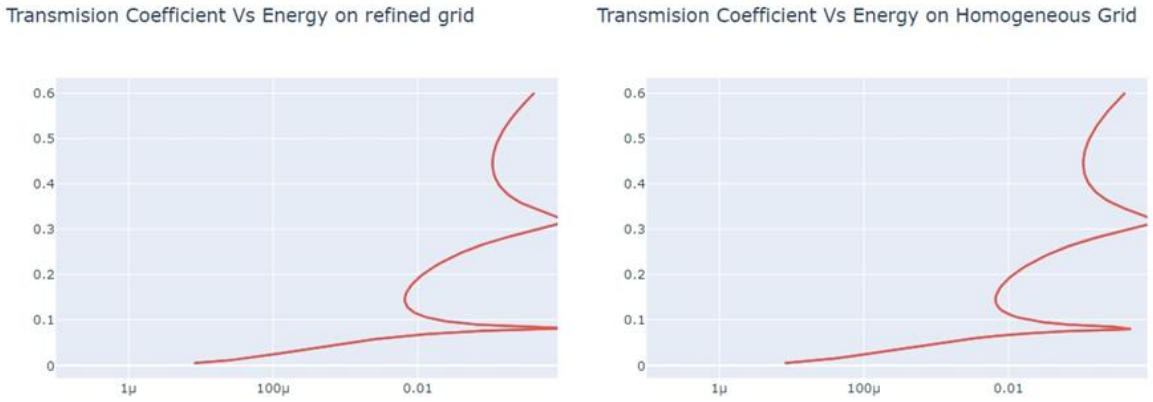
- Press simulate to view results.
- Compare the results.

Outputs

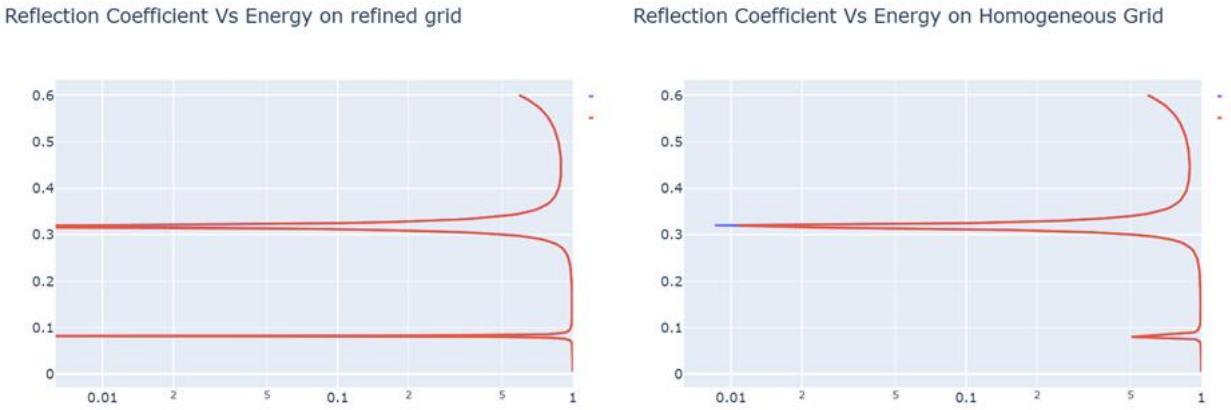




Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid



Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid



Analytical Problem

Analytical Problem: Demonstration of Quantum Tunneling

A particle with mass $m = 9.1 \times 10^{-31}\,\text{kg}$ (electron mass) and energy $E = 0.5\,\text{eV}$ is incident on a potential barrier of height $V_0 = 1.0\,\text{eV}$ and width $a = 1.0\,\text{nm}$.

Tasks:

1. **Transmission Probability:**
   - Derive an expression for the transmission probability T using the approximate solution for a rectangular barrier.
   - Calculate the numerical value of T for the given parameters.
     (Hint: Use the approximation $T \approx e^{-2\kappa a}$, where $\kappa = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$).
2. **Impact of Barrier Width:**
   - If the barrier width a is reduced to $0.8\,\text{nm}$, calculate the new transmission probability T'.
   - Determine the ratio T'/T to evaluate the exponential sensitivity of tunneling to the barrier width.
3. **Tunneling Current:**
   - Consider a tunneling junction where the tunneling current density J is proportional to T. If $J_0 = 1.0\,\text{nA/m}^2$ for $a = 1.0\,\text{nm}$, calculate the tunneling current density for $a = 0.8\,\text{nm}$.
4. **Discussion:**
   - Explain qualitatively how quantum tunneling imposes a limit on miniaturization in nanoelectronics. Relate your explanation to the scaling of gate oxide thickness in MOSFETs.

Key Equations:

1. **Tunneling Coefficient κ:**

$$\kappa = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

2. **Transmission Probability T:**

$$T \approx e^{-2\kappa a}$$

3. **Tunneling Current Density:**

$$J \propto T$$

By solving this problem, one can understand the exponential dependence of quantum tunneling on the barrier properties and its implications for nanoelectronic device design

Result: Thus the Demonstration of Quantum Tunneling is analysed and the key learning points are observed using nano hub simulation tool

Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

Aim:

To study the current-voltage characteristics and transconductance efficiency of a single-gate MOSFET using the NanoHUB FetToy Simulator.

Tools Required:

1. Computer with an internet connection.
2. FetToy Simulator on the NanoHUB platform.

Theory:

A single-gate MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) is a fundamental semiconductor device used for amplification and switching in modern electronics. It operates by modulating the flow of charge carriers (electrons or holes) between the source and drain terminals, controlled by the gate voltage (V_G).

Key characteristics of a MOSFET include:

1. **Drain Current (I_D)**: This represents the current flowing from the drain to the source. It is influenced by both the gate voltage (V_G) and drain voltage (V_D).
2. **Transconductance Efficiency (g_m/I_D)**: This is a measure of how effectively the MOSFET converts gate voltage modulation into a change in drain current. Higher gm/ID values indicate better efficiency.
3. **Saturation Region**: In this region, ID becomes relatively constant despite increasing VD . This occurs when the MOSFET channel is fully pinched off.

The experiment uses the **NanoHUB FetToy Simulator** to explore the MOSFET's electrical properties by varying V_G and V_D . By analyzing the I_D vs. V_D and I_D vs. V_G curves, important parameters such as the saturation region and transconductance efficiency can be quantified. This enables a deeper understanding of MOSFET performance in electronic circuits.

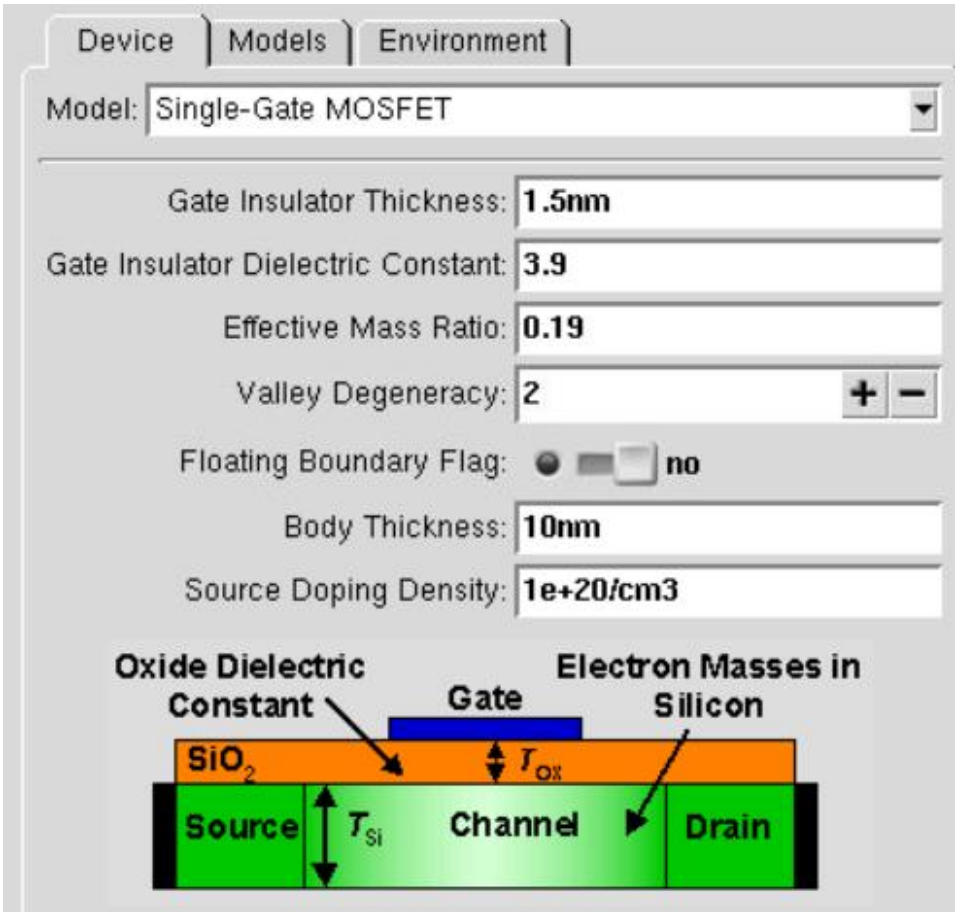
Procedure:

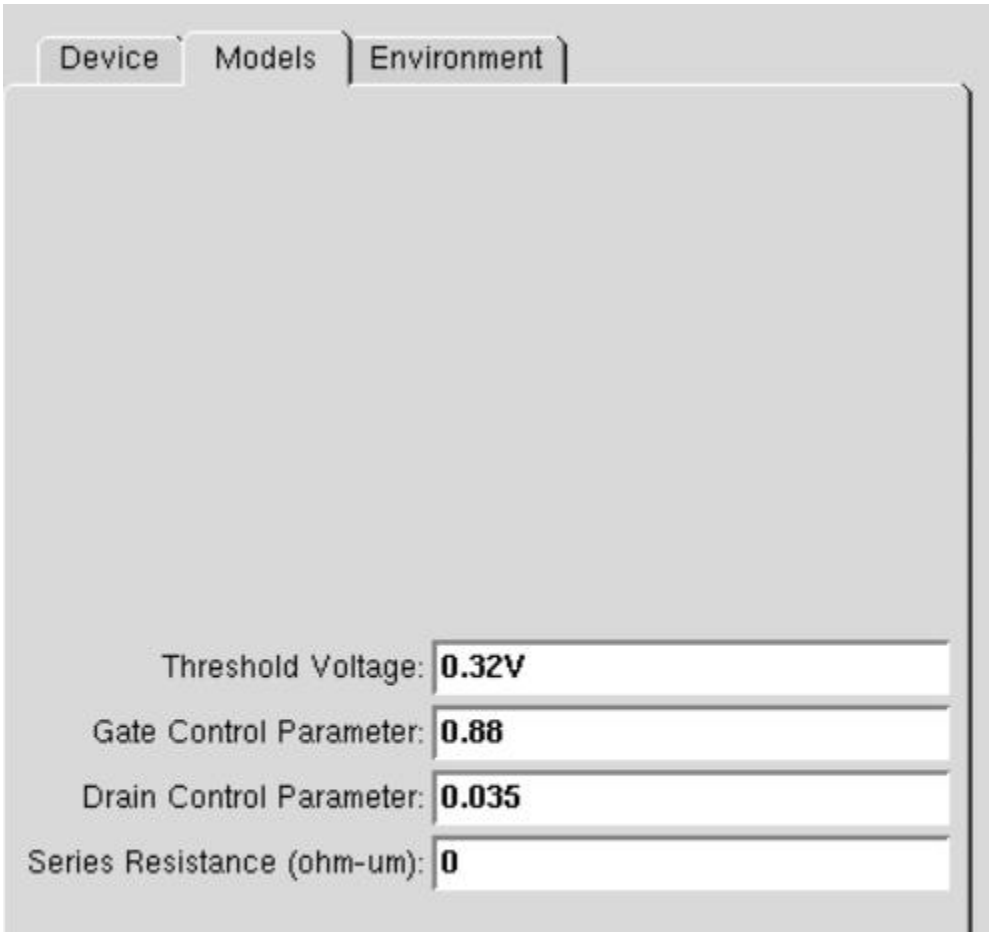
1. **Access the FetToy Simulator**:
    - Log in to the NanoHUB platform.
    - Navigate to the **FetToy Simulator** tool.
2. **Set Up the Device Parameters**:
    - Select **Device** tab:
        - Choose Model: **Single-Gate MOSFET**.
        - Enter the following parameters:

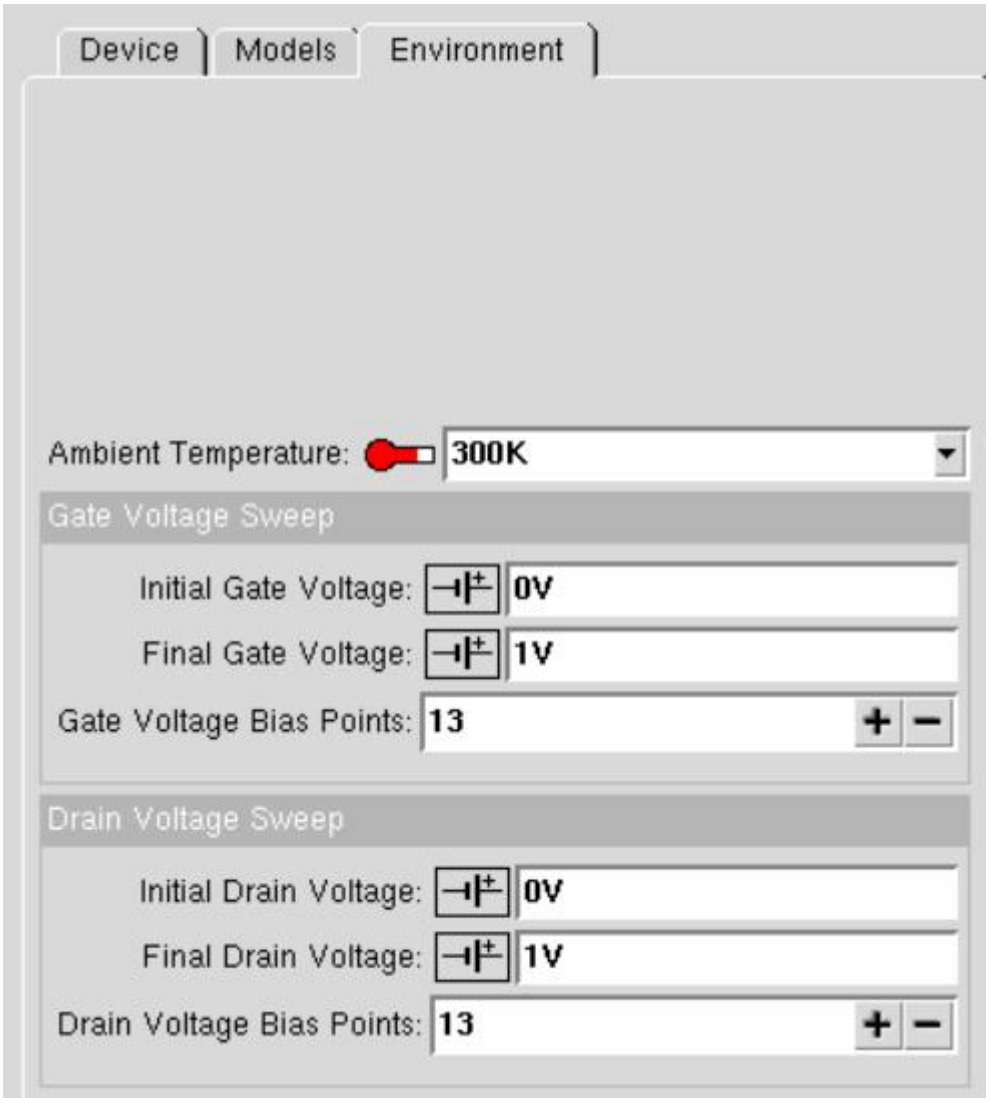
        - Gate Insulator Thickness: 1.5 nm.
        - Gate Insulator Dielectric Constant: 3.9.
        - Effective Mass Ratio: 0.19.
        - Valley Degeneracy: 2.
        - Body Thickness: 10 nm
        - Source Doping Density: 1 x 10^{20} cm^{-3}.
3. **Set Up the Electrical Parameters**:
    - Go to the **Models** tab:
        - Threshold Voltage: 0.32 V.
        - Gate Control Parameter: 0.88.
        - Drain Control Parameter: 0.035.
        - Series Resistance: 0 Ω μm.
4. **Configure the Environment**:
5. Go to the **Environment** tab:
    - Ambient Temperature: 300K (default).
    - Set up voltage sweeps:
        - **Gate Voltage Sweep**:
            - Initial Gate Voltage: 0 V.
            - Final Gate Voltage: 1 V.
            - Gate Voltage Bias Points: 13.
        - **Drain Voltage Sweep**:
            - Initial Drain Voltage: 0 V.
            - Final Drain Voltage: 1 V.
            - Drain Voltage Bias Points: 13.
6. **Run the Simulation**:
    - Click the **Simulate** button to generate the results.
7. **Analyze the Output**:
    - Navigate to the results section:
        - View **Drain Current vs. Gate Voltage** plot.
        - View **Drain Current vs. Drain Voltage** plot.
        - View **gm/Id vs. Gate Voltage** plot.
8. **Extract Data**:
    - Export the simulation results for further analysis. Data points for gate voltage (VGd), drain voltage (VD), and current (ID) can be downloaded.

9. Perform Numerical Analysis:

- Use the extracted data to calculate the transconductance efficiency (gm/ID) or determine the saturation region manually if not directly available.


Device
Models
Environment
Model: Single-Gate MOSFET
Gate Insulator Thickness: 1.5nm
Gate Insulator Dielectric Constant: 3.9
Effective Mass Ratio: 0.19
Valley Degeneracy: 2
Floating Boundary Flag: no
Body Thickness: 10nm
Source Doping Density: 1e+20/cm3
Oxide Dielectric Constant
Gate
Electron Masses in Silicon
SiO2
Tox
Source
Tsi
Channel
Drain


Device
Models
Environment
Threshold Voltage:
0.32V
Gate Control Parameter:
0.88
Drain Control Parameter:
0.035
Series Resistance (ohm-um):
0



Formulae Used:

1. **Transconductance Efficiency**:

$$\frac{g_m}{I_D} = \frac{\Delta ID}{\Delta VG \cdot ID}$$

2. **Drain Current Change:**

$$\Delta ID = I_D\ (V_G = x) - I_D\ (V_G = y)$$

Case Analysis:

Case 1: Increase in V_G :

- **Observation**: As V_G increases, the drain current (ID) also increases exponentially due to the MOSFET entering the saturation region earlier.

Case 2: Variation in V_D :

- **Observation**: Drain current (I_D) saturates for higher V_D at a fixed V_G .

Tabulation for Results:

Gate Voltage (V_G)	**Drain Voltage (V_D)**	**Drain Current (I_D)**	**Transconductance Efficiency (g_m/I_D)**
V	**V**	**µA/µm**	**1/V**
0.25	**1.0**	**24.4**	**25**
0.5	**1.0**	**517**	**5.8**
0.75	**1.0**	**1400**	**2.94**

Input Values:

Gate Voltage (VG): 0–1 V in 13 steps.

Drain Voltage (VD): 0–1 V in 13 steps.

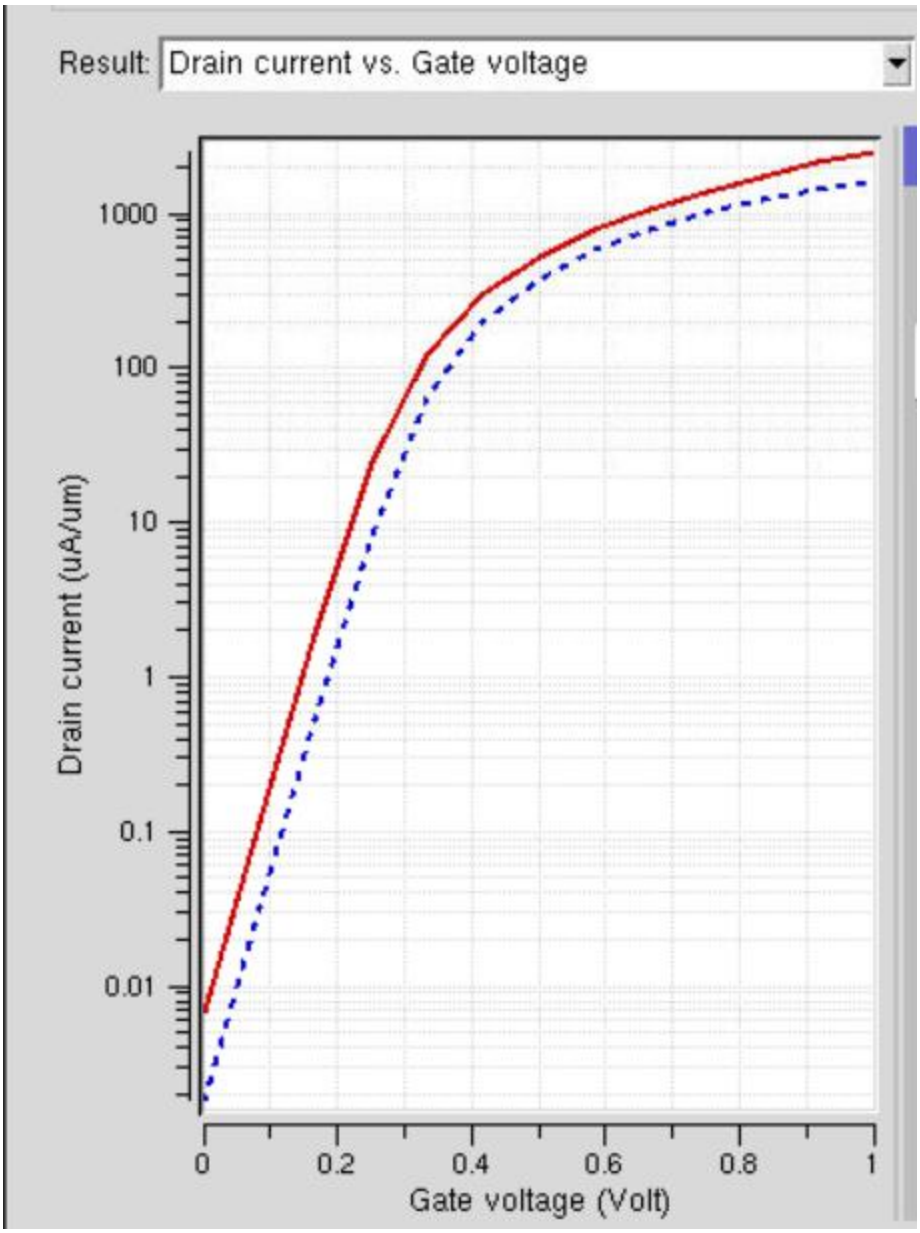
Device parameters as listed above.

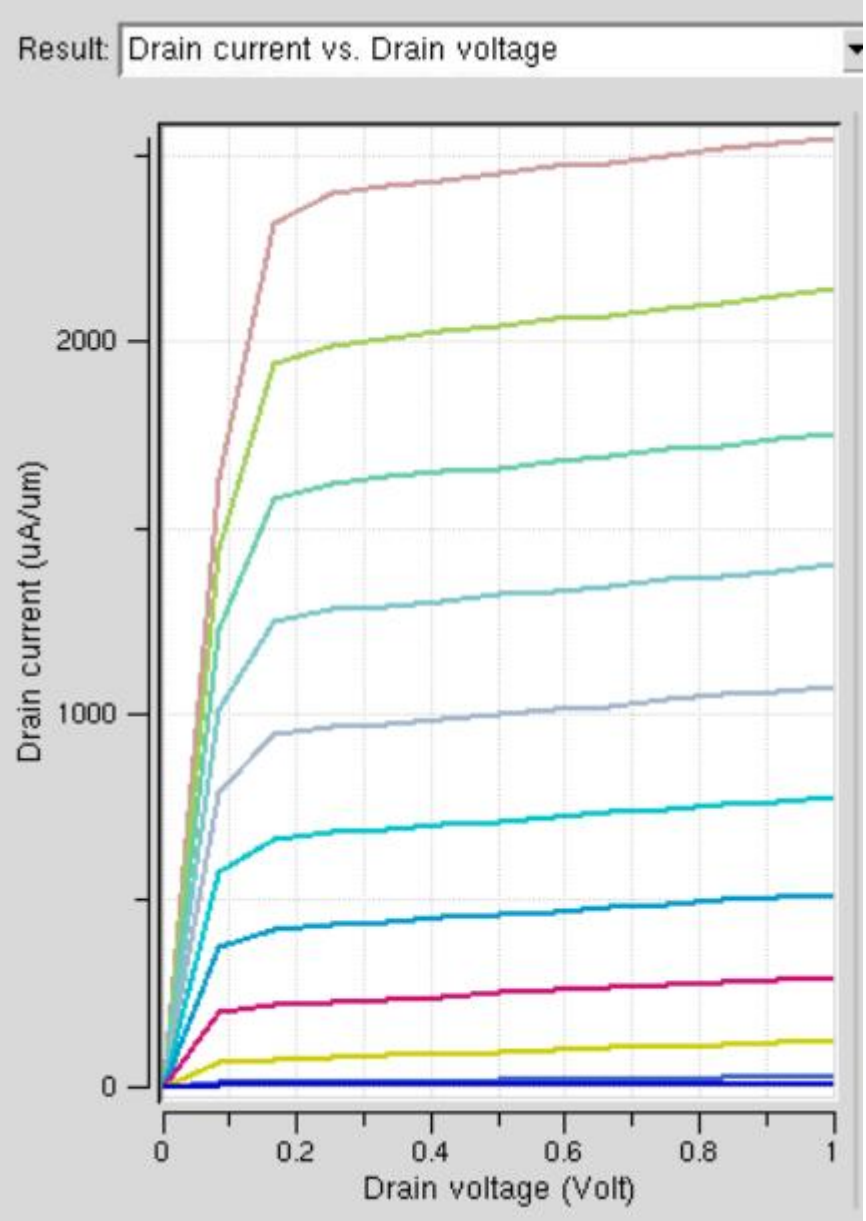
Sample Calculation:

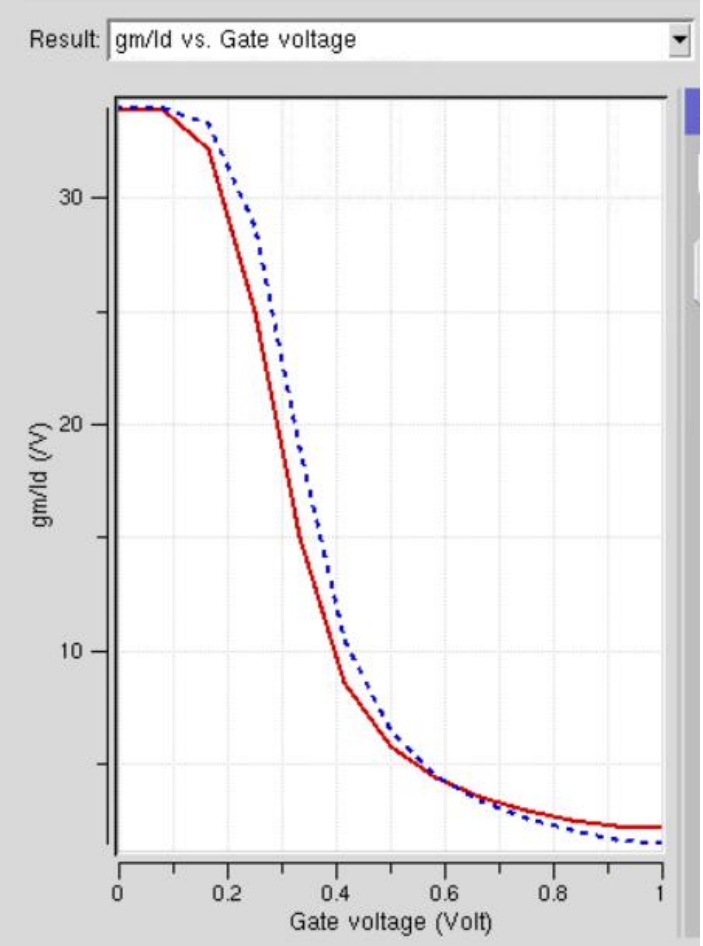
1. For VG=0.5 V, VD=1.0: ID=517 µA/µm.

2. Change in ID for VG=0.5 V to VG=0.75 V ΔID=1400−517=883 µA/µm.

3. Transconductance Efficiency for VG=0.25 V: gm/ID=25 V^{-1}.


Result: Drain current vs. Gate voltage
1000
100
10
1
0.1
0.01
Drain current (uA/um)
0
0.2
0.4
0.6
0.8
1
Gate voltage (Volt)


Result: Drain current vs. Drain voltage
Drain current (uA/um)
2000
1000
0
0
0.2
0.4
0.6
0.8
1
Drain voltage (Volt)



Results:

1. The MOSFET exhibits saturation at higher VD for VG=0.5 V.
2. As gm/ID decreases with increasing VG , indicating reduced efficiency at higher gate voltages.

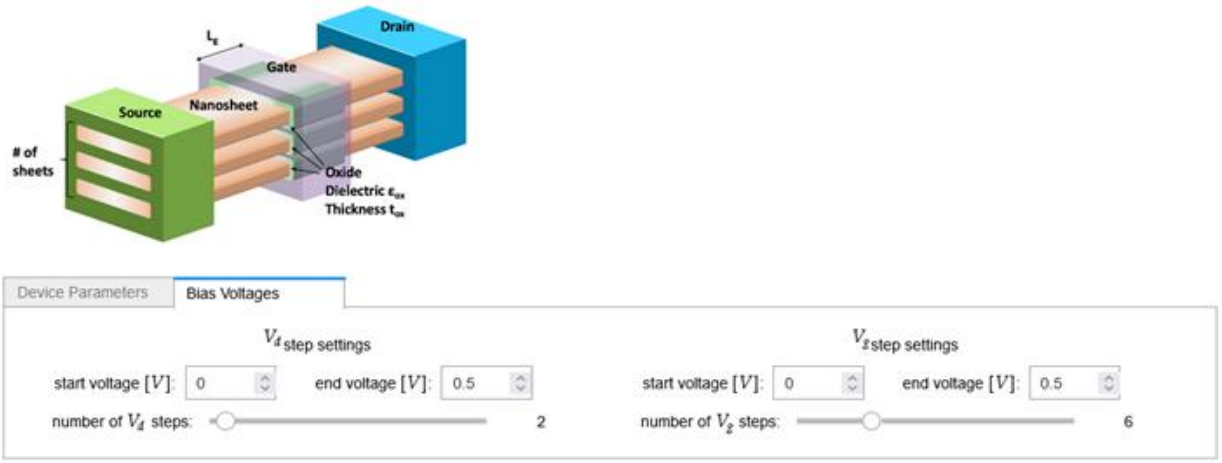
Exp. 8. Simulation of VI Characteristics of FINFET, CNTFET and GFET using NanoHub

(a) FINFET

Aim:

To Obtain the VI Characteristics of FinFET by varying the Gate Channel length and oxide thickness

Software: NanoHub Online Tube



Background Concepts:

FinFET technology takes its name from the fact that the FET structure used looks like a set of fins when viewed.

The main characteristic of the FinFET is that it has a conducting channel wrapped by a thin silicon "fin" from which it gains its name. The thickness of the fin determines the effective channel length of the device.

In terms of its structure, it typically has a vertical fin on a substrate which runs between a larger drain and source area. This protrudes vertically above the substrate as a fin.

The gate orientation is at right angles to the vertical fin. And to traverse from one side of the fin to the other it wraps over the fin, enabling it to interface with three side of the fin or channel.

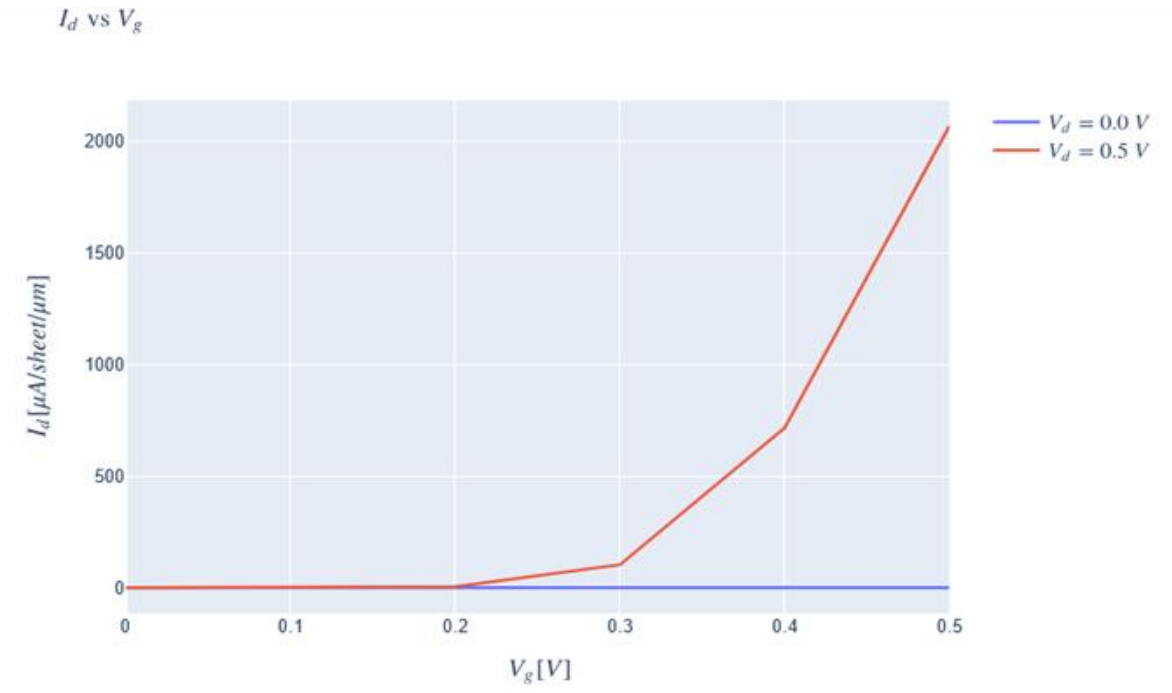
This form of gate structure provides improved electrical control over the channel conduction and it helps reduce leakage current levels and overcomes some other short-channel effects.

The term FinFET is used to describe any fin-based, multigate transistor architecture regardless of number of gates.

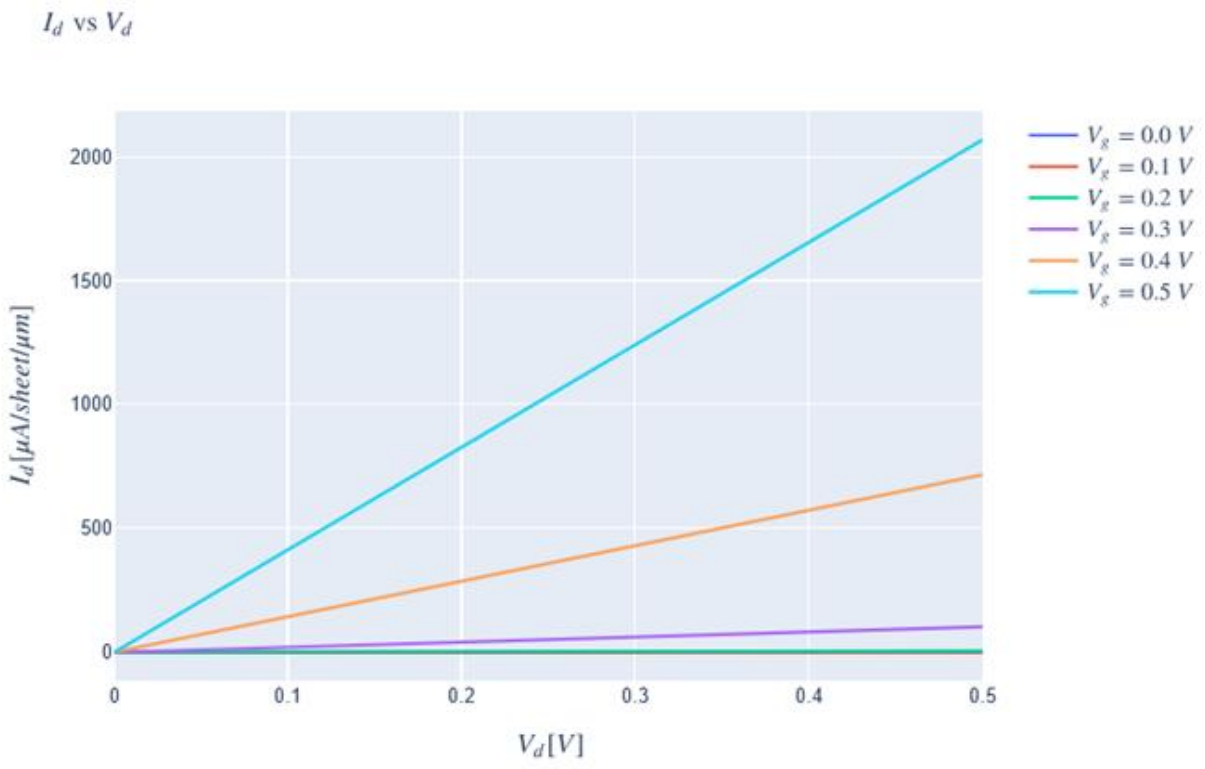
Procedure:

- Open nanoHUB.
- Goto Resources -> Tools.
- Select FinFET
- Press Launch tool in info section.
- Select the appropriate device and bias parameters-> Simulate
- Select -> the appropriate simulated output-> VI Characteristics
- Repeat the above steps for the given case study problem statement

Output:

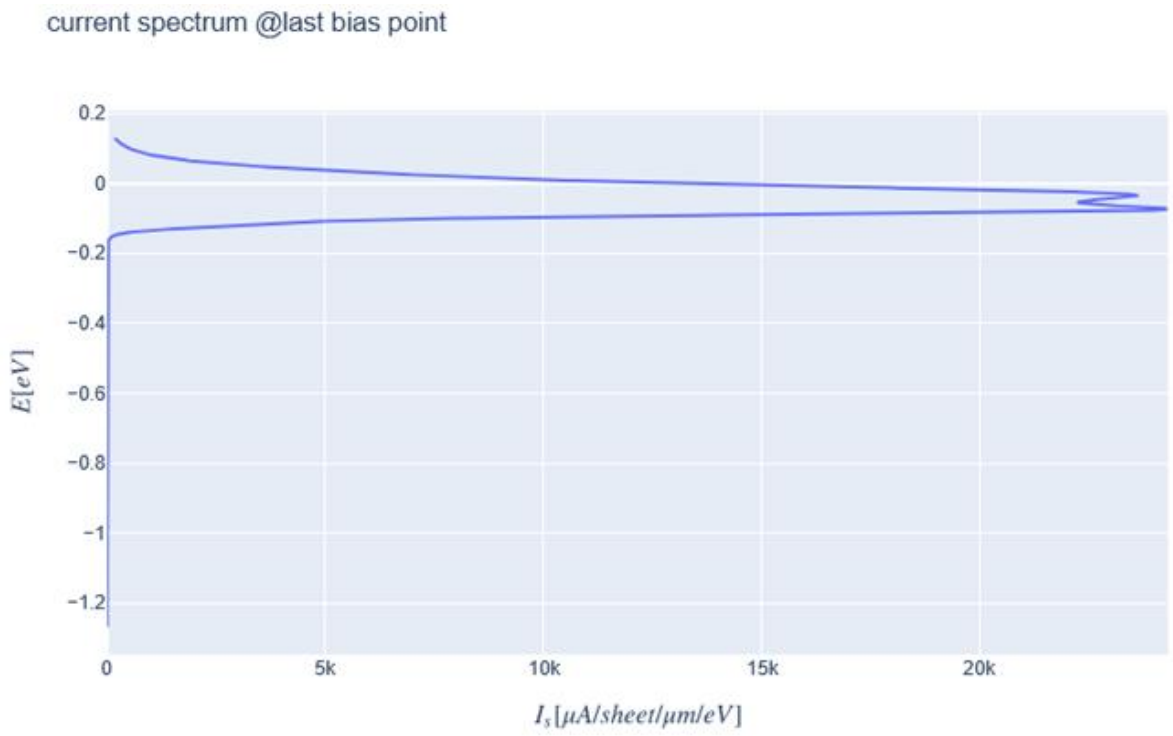


The output Drain current Performance Vs Gate supply voltage of FinFET device



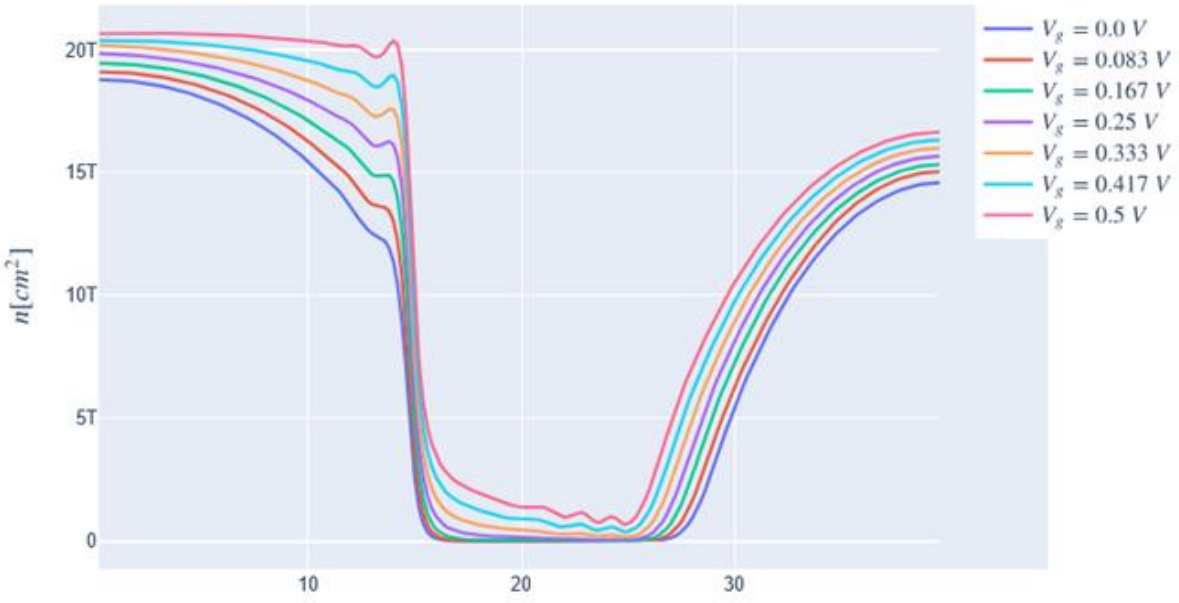
The output Drain current Performance Vs Drain voltage of FinFET device

Case Study 1: Simulate & obtain the current spectrum characteristics for the last bias point of FinFET. Analyze the performance for different Oxide layer thicknesses



Case Study 2: Simulate & obtain the electron density per sheet characteristics of FinFET. Analyze its performance with different Gate region length

electron density per sheet



Problem Statement:

A FinFET has the following parameters:

$H_{\text{fin}} = 30\,\text{nm}$, $W_{\text{fin}} = 10\,\text{nm}$, $L_{\text{ch}} = 20\,\text{nm}$, $C_{\text{ox}} = 2\,\mu\text{F/cm}^2$, $\mu = 200\,\text{cm}^2/\text{V\cdotps}$, $V_{th} = 0.4\,\text{V}$.

- Find the drain current in linear and saturation regions for $V_{GS} = 0.8\,\text{V}$ and $V_D = 0.3\,\text{V}$ (linear) and $V_D = 1.0\,\text{V}$ (saturation).

The calculated drain currents for the given FinFET parameters are:

1. **Linear Region** ($V_D = 0.3\,\text{V}$): $I_D = 10.5\,\mu\text{A}$
2. **Saturation Region** ($V_D = 1.0\,\text{V}$):

$$I_D = 11.2\,\mu\text{A}$$

Step-by-Step Analysis

1. Understanding the Device Geometry and Parameters

The effective width of the FinFET (W_{eff}) is determined by the fin height and width:

$$W_{\text{eff}} = 2 \cdot H_{\text{fin}} + W_{\text{fin}}$$ Substituting the given values:

$$W_{\text{eff}} = 2 \cdot 30\,\text{nm} + 10\,\text{nm} = 70\,\text{nm} = 70 \times 10^{-9}\,\text{m}.$$

2. Drain Current in the Linear Region

In the linear region, the drain current equation is:

$$I_D = \mu \cdot C_{\text{ox}} \cdot \frac{W_{\text{eff}}}{L_{\text{ch}}} \cdot [(V_{GS} - V_{th})V_D - \frac{V_D^2}{2}].$$

Substituting the values:

$$\mu = 200 \times 10^{-4}\,\text{m}^2/\text{V}\backslash\text{cdotps},\ C_{\text{ox}} = 2\,\mu\text{F/cm}^2 = 2 \times 10^{-6}\,\text{F/m}^2,$$

$$W_{\text{eff}} = 70 \times 10^{-9}\,\text{m},\ L_{\text{ch}} = 20 \times 10^{-9}\,\text{m},\ V_{GS} = 0.8\,\text{V},\ V_{th} = 0.4\,\text{V},\ V_D = 0.3\,\text{V}.$$

$$I_D = (200 \times 10^{-4}) \cdot (2 \times 10^{-6}) \cdot \frac{70 \times 10^{-9}}{20 \times 10^{-9}} \cdot [(0.8 - 0.4) \cdot 0.3 - \frac{0.3^2}{2}].$$

Simplifying:

$$I_D = 10.5\,\mu\text{A}.$$

3. **Drain Current in the Saturation Region**

In the saturation region, the drain current equation is:

$$I_D = \frac{\mu \cdot C_{ox} \cdot W_{eff}}{2 \cdot L_{ch}} \cdot (V_{GS} - V_{th})^2.$$

Substituting the values:

$$I_D = \frac{(200 \times 10^{-4}) \cdot (2 \times 10^{-6}) \cdot 70 \times 10^{-9}}{2 \cdot 20 \times 10^{-9}} \cdot (0.8 - 0.4)^2.$$

Simplifying:

$$I_D = 11.2\,\mu\text{A}.$$

Observations

1. **Linear Region:**
   - The drain current depends on both V_D and V_{GS}, following a parabolic relationship.
   - For small V_D, I_D increases approximately linearly with V_D.
2. **Saturation Region:**
   - The drain current depends only on V_{GS} and V_{th}.
   - Once $V_D > V_{GS} - V_{th}$, the channel becomes "pinched off," and further increases in V_D have little effect on I_D.

OUTCOME:

Thus, the simulation of VI characteristics of FINFET by varying the gate channel length and oxide thickness is analyzed and key learning points are observed

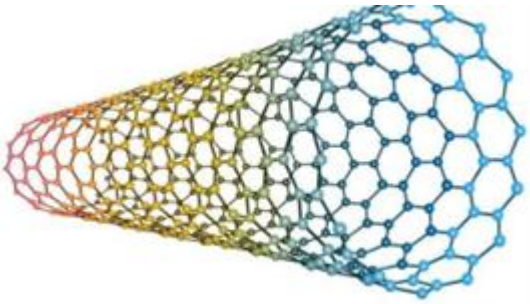
(b) CNTFET

Aim:

To obtain the density of states and conductance of CNTFET by varying the diameter and length.

Software: NanoHub Online Tube

Theory:



Carbon nanotubes (CNTs) are allotropes of carbon with a cylindrical nanostructure. These cylindrical carbon molecules have unusual properties, which are valuable for nanotechnology, electronics, optics and other fields of materials science and technology. Owing to the material's exceptional strength and stiffness, nanotubes have been constructed with length-to-diameter ratio of up to 132,000,000:1. For a given (n,m) nanotube, if n = m, the nanotube is metallic; if n − m is a multiple of 3 and n ≠ m and nm ≠ 0, then the nanotube is quasi-metallic with a very small band gap, otherwise the nanotube is a moderate semiconductor. Thus all armchair (n = m) nanotubes are metallic, and nanotubes (6,4), (9,1), etc. are semiconducting. The diameter of CNT can be given as:

$$d = \frac{a}{\pi}\sqrt{(n^2 + nm + m^2)}$$

where a = 0.246 nm.

The angle of CNT is given by

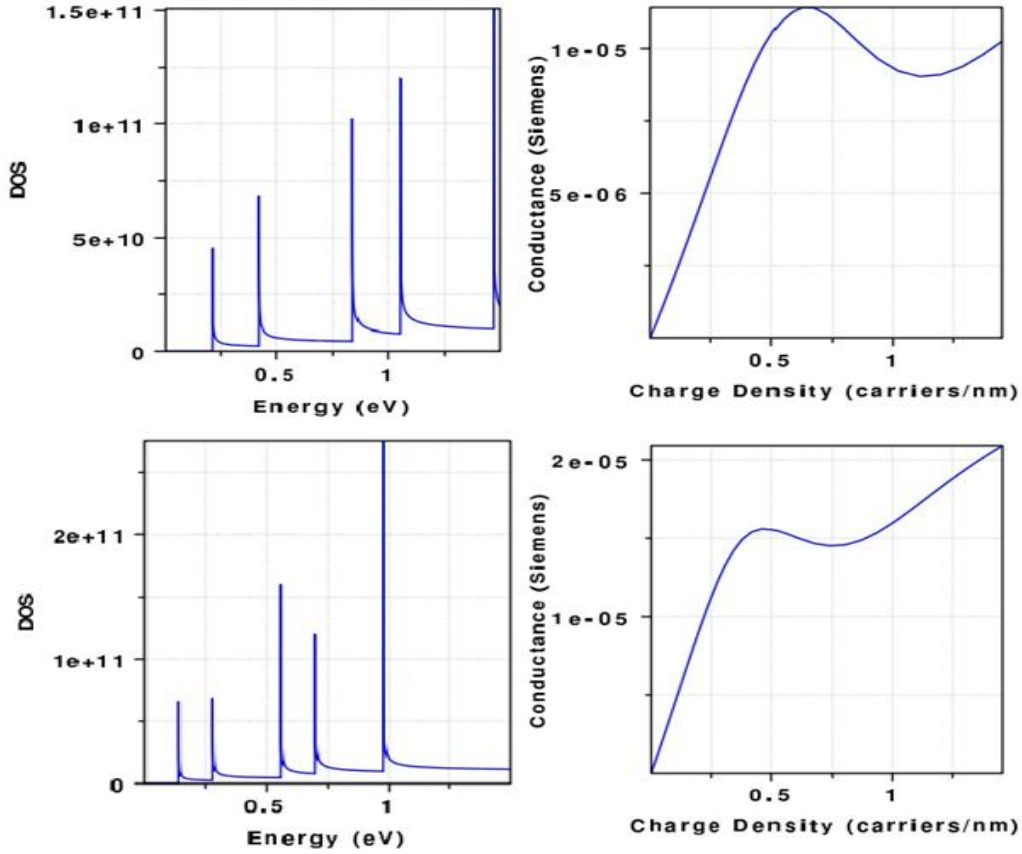
$$\tan\theta = \frac{\sqrt{3}m}{2n + m}$$

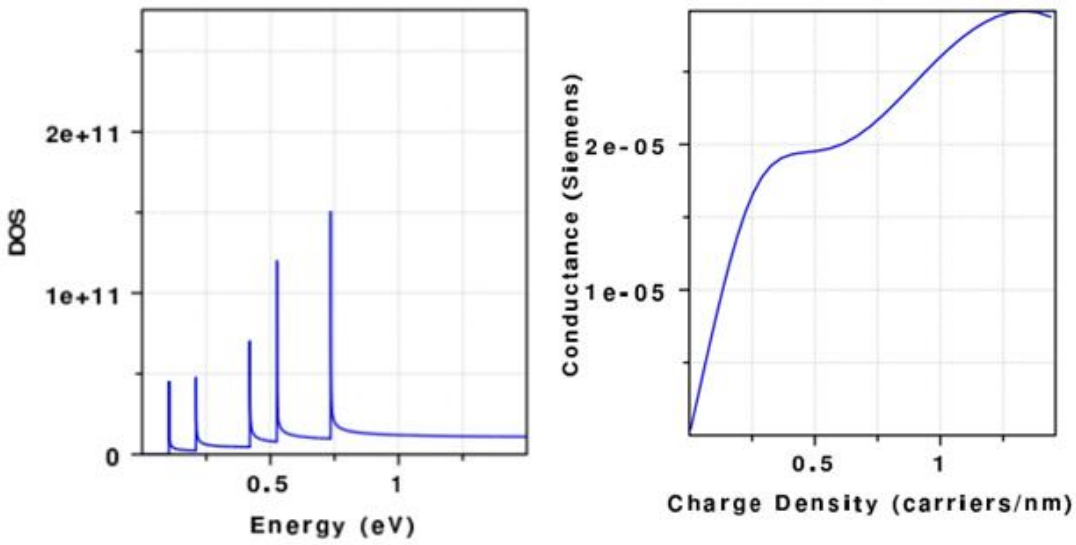
Procedure:

- **Open nanoHUB.**
- **Goto Resources -> Tools.**

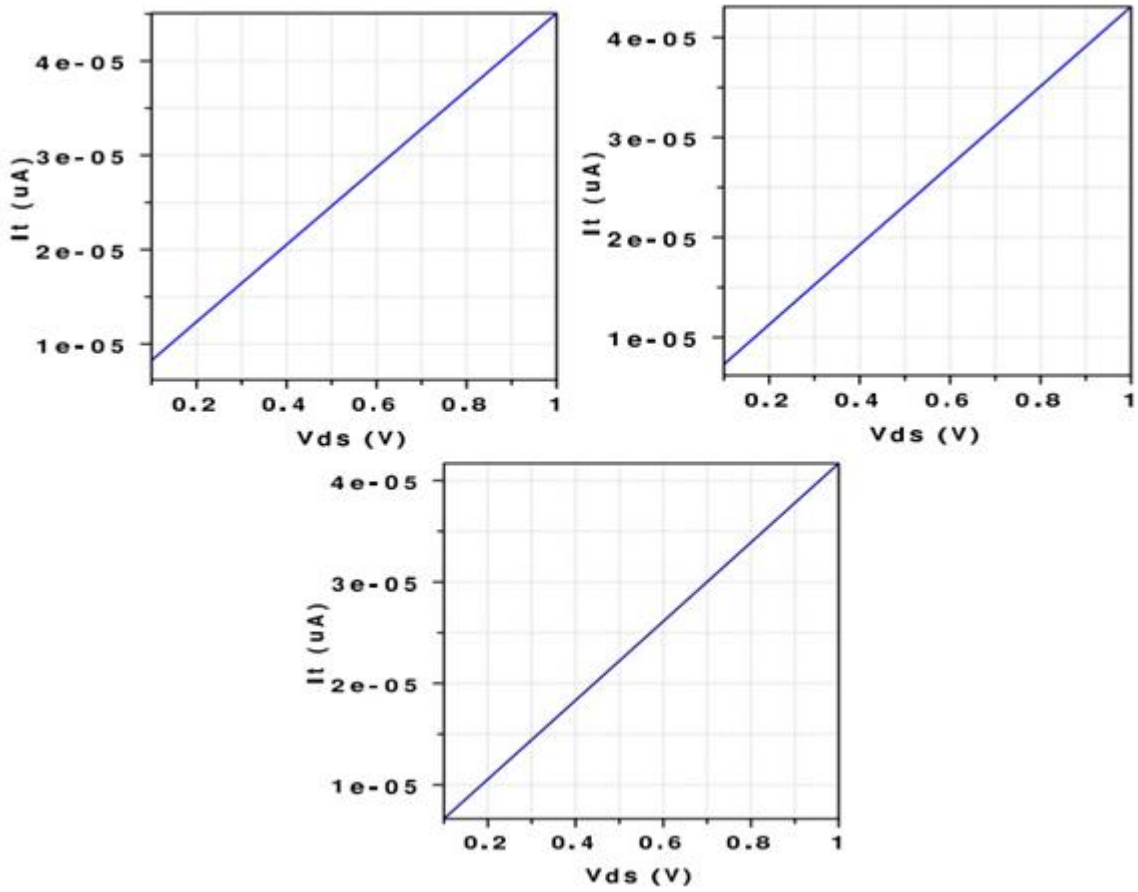
- **Select CNTFET ->CNT Mobility**
- Press Launch tool in info section.
- Select the -> diameter of CNT
- Select -> Simulate
- Select ->Result -> conductance
- Select ->Result -> Density of states
- Repeat the above for various diameter
- Select the -> Length of CNT
- Select -> Simulate
- Select ->Result -> conductance
- Select ->Result -> Density of states

Output:

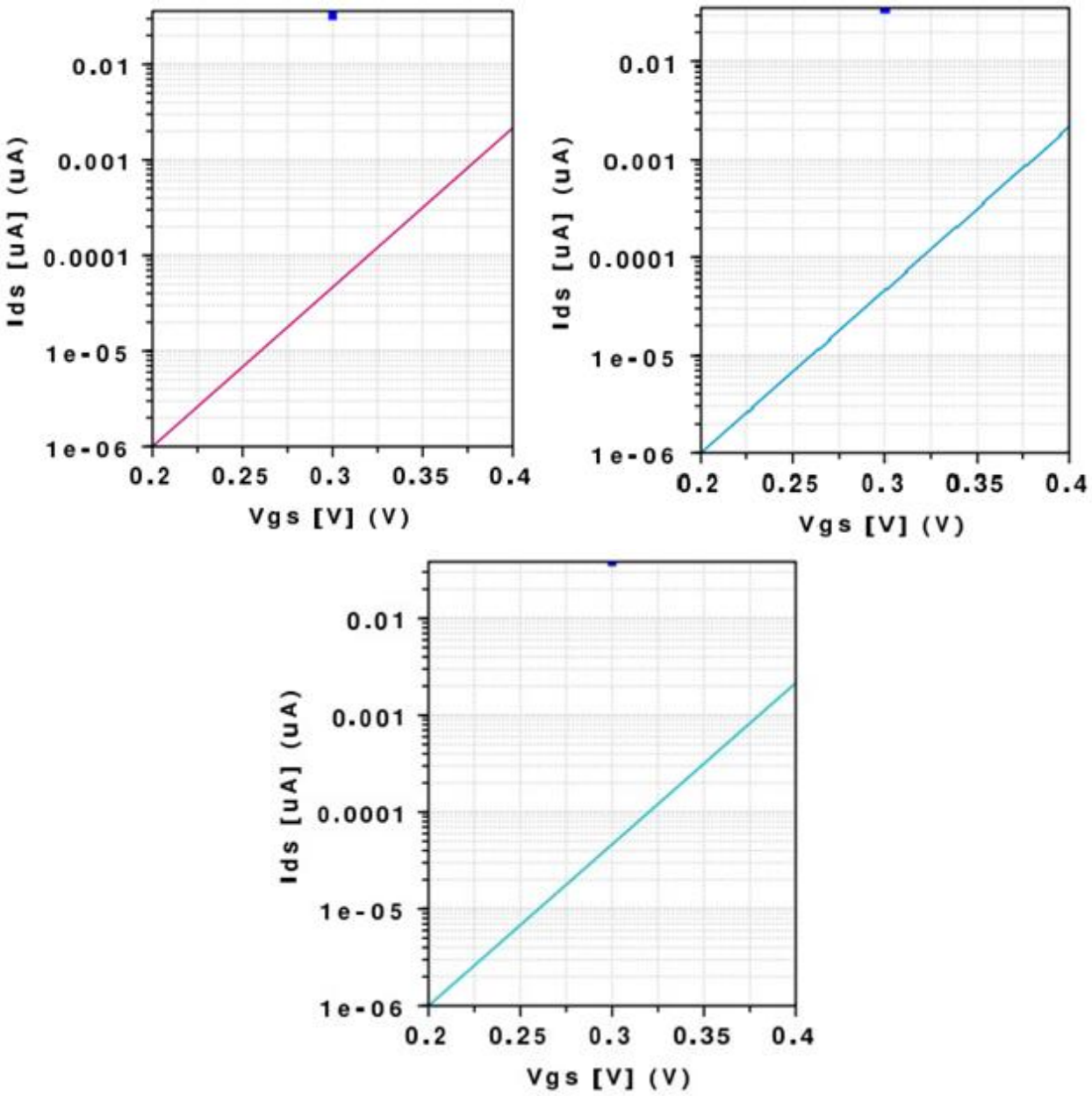




Case study 1: Simulate & Obtain the VI characteristics of CNT-FET by varying Schottky barrier and oxide thickness



Case study 2: Simulate & Obtain the VI characteristics of CNT-FET by varying CNT diameter



Outcome:

Thus the simulation of density of states and conductance of CNTFET by varying the diameter and length is analyzed and key learning points are observed.

(c) GFET

Aim:

To simulate the Voltage-Current (VI) characteristics of a Graphene Field Effect Transistor (GFET) using appropriate simulation tools, analyze its performance under varying gate voltages and drain-source voltages, and understand its behavior as a high-mobility semiconductor device.

Software: NanoHub Online Tube

Theory:

Graphene is a semimetal with small overlap between the valence and the conduction bands (zero bandgap material). It is an allotrope (form) of carbon consisting of a single layer of carbon atoms arranged in a hexagonal lattice.It is the basic structural element of many other allotropes of carbon, such as graphite, diamond, charcoal, carbon nanotubes and fullerenes.

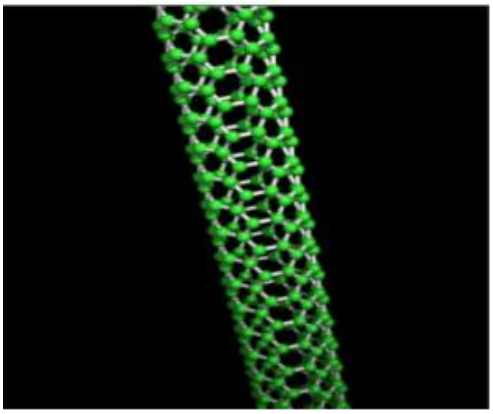


Fig 1. Graphene

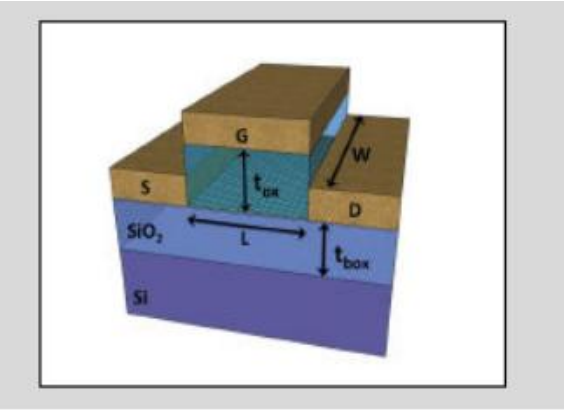


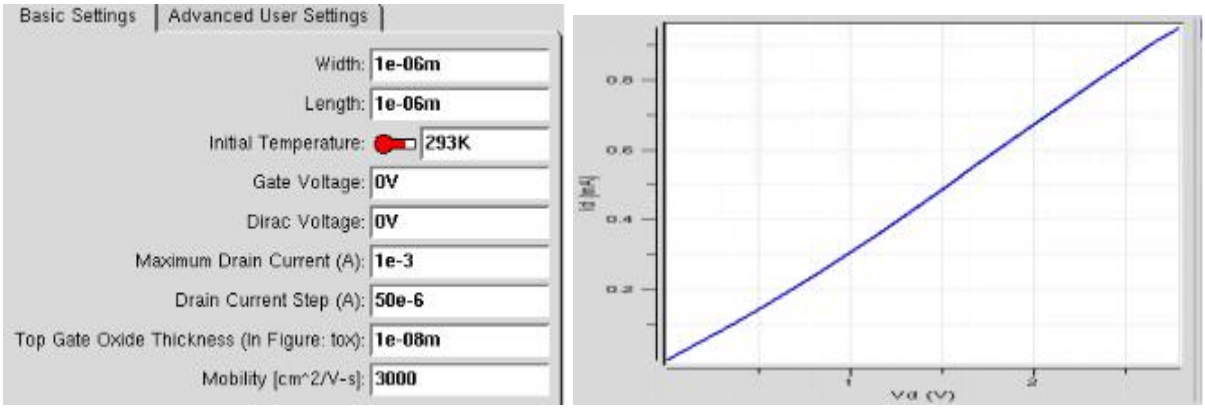
Fig 2. Graphene

Graphene is a zero-gap semiconductor, because its conduction and valence bands meet at the Dirac points, which are six locations in momentum space, on the edge of the Brillouin zone, divided into two non- equivalent sets of three points. The two sets are labeled K and K'. The sets give graphene a valley degeneracy of gv = 2. By contrast, for traditional semiconductors the primary point of interest is generally Γ, where momentum is zero. However, if the in-plane direction is confined, in which case it is referred to as a nanoribbon, its electronic structure is different. If it is "zig-zag", the bandgap is zero. If it is "armchair", the bandgap is non-zero. The corresponding resistivity of graphene sheets would be 10−6 Ω· cm. This is less than the resistivity of silver, the lowest otherwise known at room temperature.

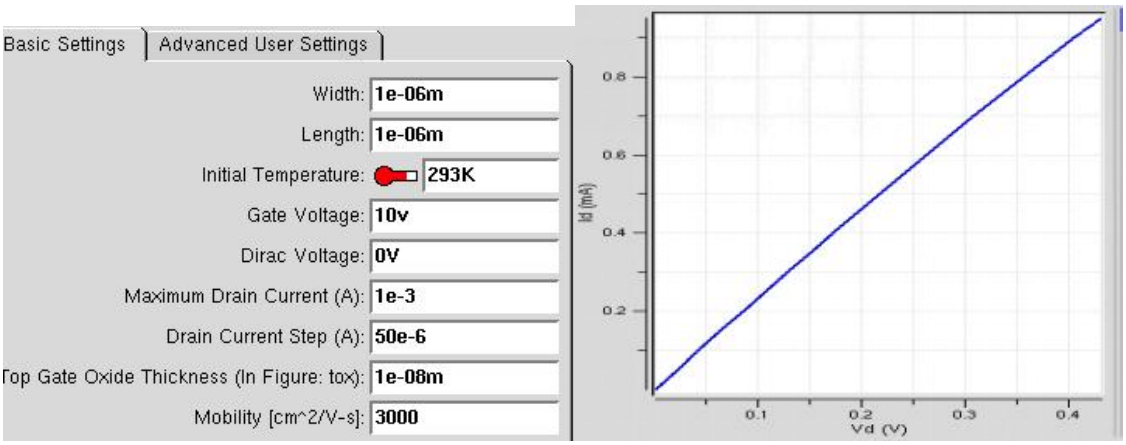
Procedure:

- Open nanoHUB.
- Goto Resources -> Tools.
- Select Graphene -> GFET
- Press Launch tool in info section.
- Select the -> Width
- Click on Simulation
- Select ->Result -> Ids vs Vds
- Repeat the above steps for various width
- Select -> Length
- Select ->Result -> Ids vs Vds
- Repeat the above steps for various width

Output:



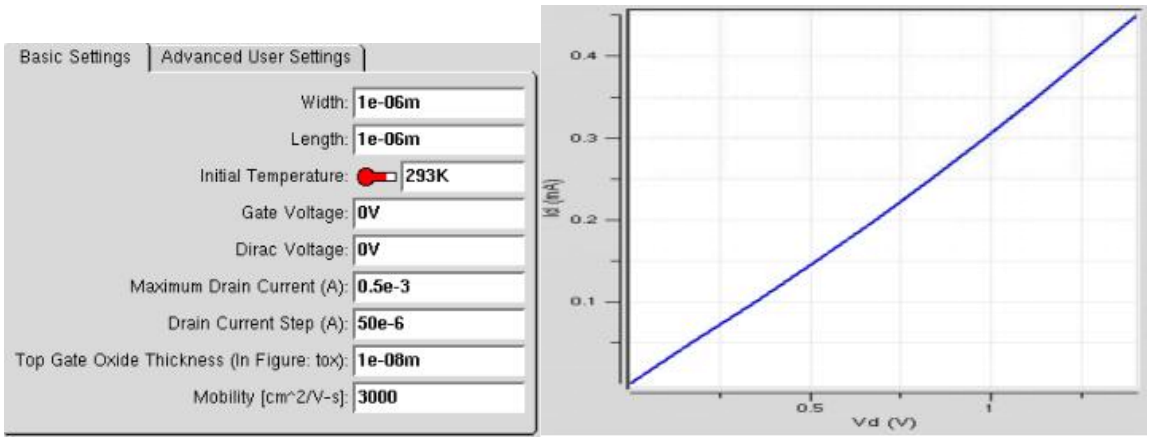
Case Study 1:How does varying the gate voltage (Vgs) affect the drain current (Ids) in a GFET?



Inference:For Vgs>0V (Positive Gate Voltage):

- Vgs attracts electrons to the graphene channel, increasing the electron carrier density.
- The drain current (Ids) increases linearly in the ohmic region and saturates in the saturation region as Vgs increases further.
- Varying Vgs directly modulates Ids in a GFET, demonstrating graphene's ambipolar nature. This behavior distinguishes GFETs from conventional FETs and enables their use in high-frequency electronics and analog circuits.

Case Study 2: What factors determine the maximum drain current (Ids,max) in a Graphene Field Effect Transistor (GFET)?

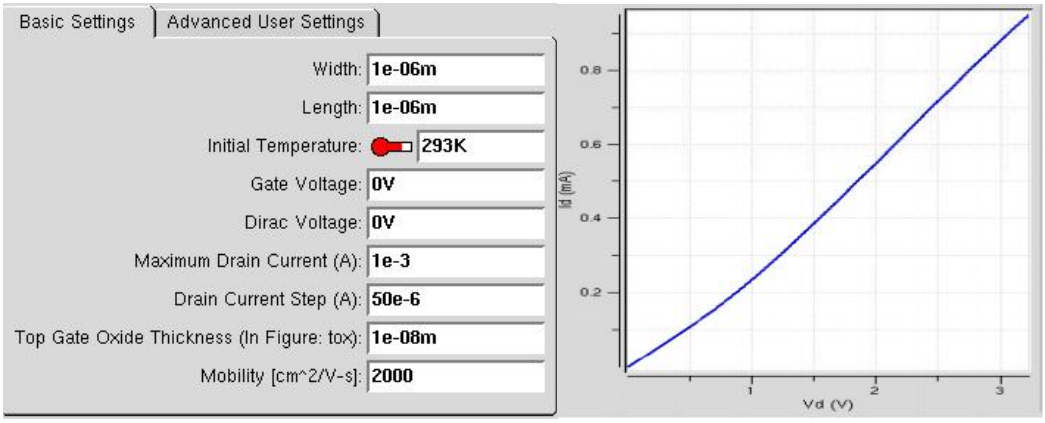


Inference:· The VI curve exhibits a symmetric behavior around the Dirac point (Vgs=0V), where the drain current is at its minimum.

· For positive Vgs, the current is dominated by electrons, while for negative Vgs, holes dominate the conduction.

Ids,max was observed at a specific Vgs and Vds combination due to graphene's high mobility and low intrinsic resistance. The maximum drain current of 0.5×10−30.5 times A was achieved under specific gate and drain voltage conditions, confirming the simulation's accuracy.

Case Study 3: To investigate how the mobility of charge carriers in a Graphene Field-Effect Transistor (GFET) with a mobility of 2000 cm²/V·s influences its current-voltage (VI) characteristics.



Inference:

· The simulated VI curves show a relatively high drain current (Ids) for a given gate-source voltage (Vgs), demonstrating efficient carrier transport in the graphene channel. High carrier mobility (2000 cm2/Vs) minimizes scattering effects, enabling rapid carrier movement and improved current conduction.

Result:

The VI characteristics of the Graphene Field Effect Transistor (GFET) were successfully simulated. The device demonstrated:

- · A sharp increase in current with increasing Vgs due to high carrier mobility, with the drain current reaching a maximum of 0.5 mA at Vgs=5V.
- · A low threshold voltage of approximately 0.5 V, indicating efficient operation at low gate voltages.

· The linear and saturation regions of operation were clearly observed, showing typical GFET behavior. The simulation confirms the potential of GFETs for high-speed and low-power applications in future electronic devices.

Exp. 9. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using NanoHUB

Aim:

To obtain the Tunnelling Magnetoresistance (TMR) by varying barrier height. Software: NanoHub Online Tube

Theory:

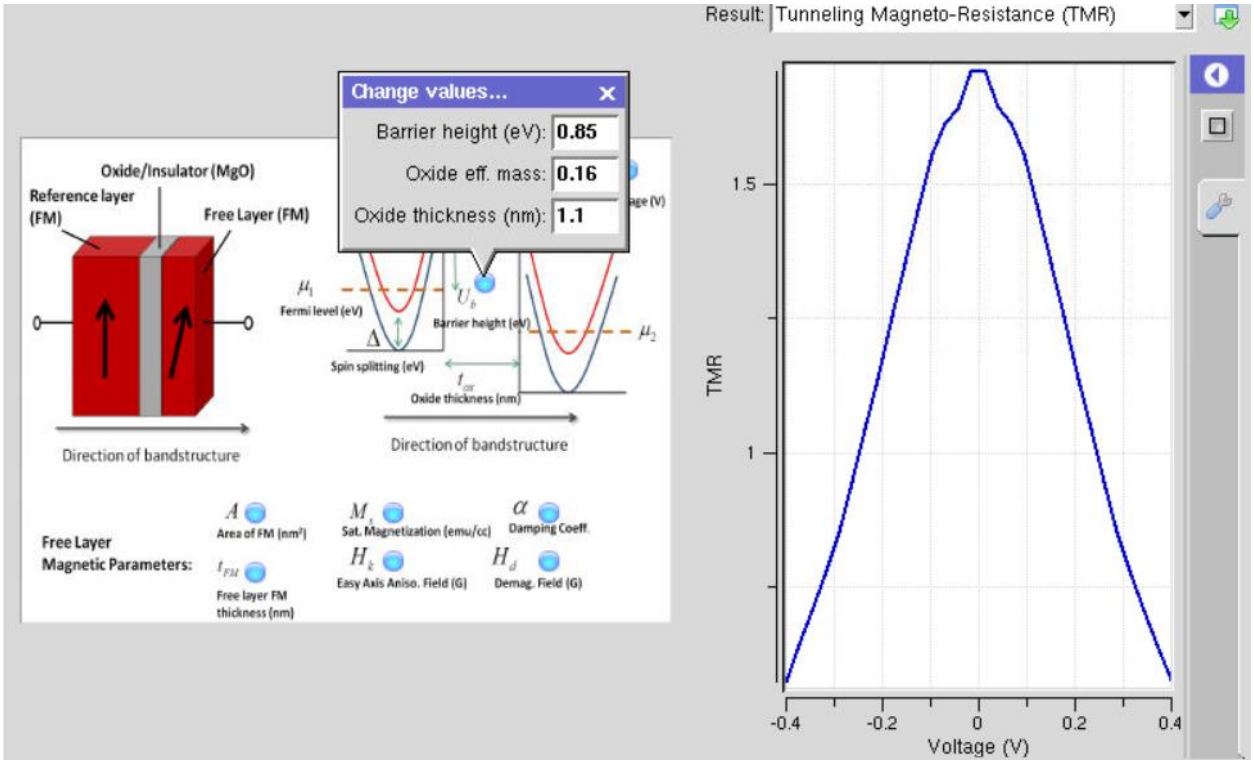
Tunnel magnetoresistance (TMR) is a magneto resistive effect that occurs in a magnetic tunnel junction (MTJ), which is a component consisting of two ferromagnets separated by a thin insulator. If the insulating layer is thin enough (typically a few nanometres), electrons can tunnel from one ferromagnet into the other. Since this process is forbidden in classical physics, the tunnel magnetoresistance is a strictly quantum mechanical phenomenon. Magnetic tunnel junctions are manufactured in thin film technology. On an industrial scale the film deposition is done by magnetron sputter deposition; on a laboratory scale molecular beam epitaxy, pulsed laser deposition and electron beam physical vapor deposition are also utilized. The junctions are prepared by photolithography.

Procedure:

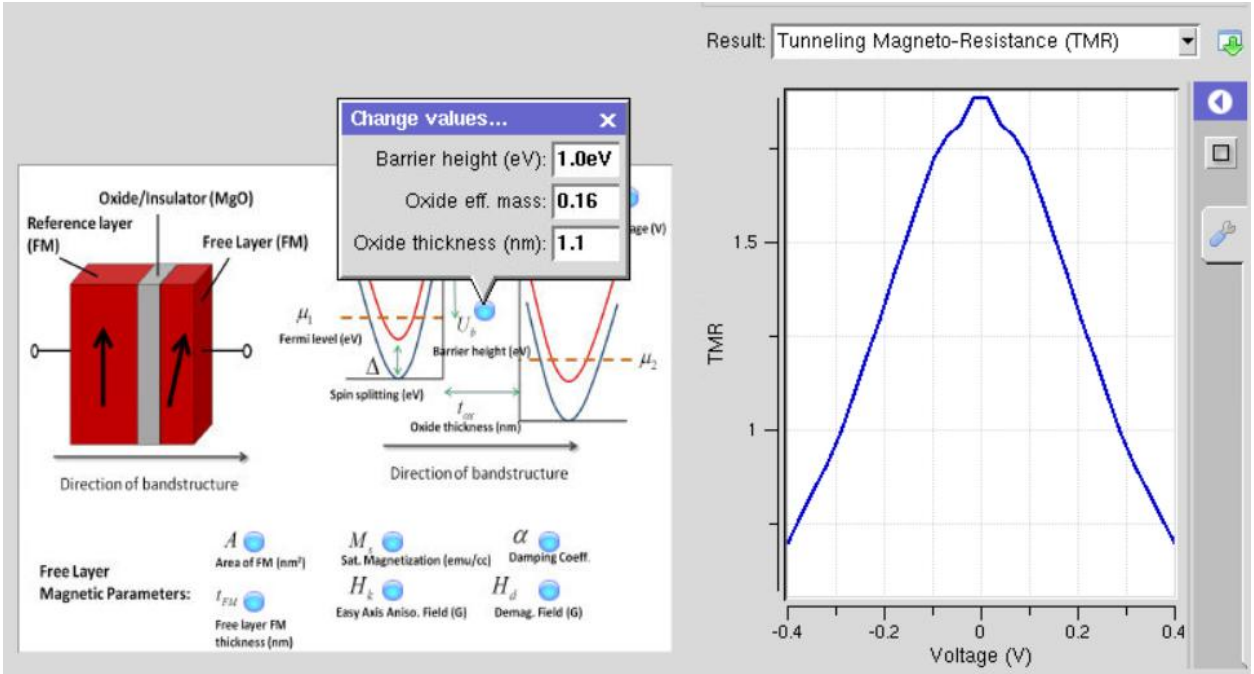
- Open nanoHUB.
- Goto Resources -> Tools.
- Select Magnetic Tunnel Junction Lab
- Press Launch tool in info section.
- Select the -> barrier height
- Click on Simulation
- Select ->Result -> Tunnel magnetoresistance (TMR)
- Repeat the above steps for various barrier height

Output:

Case Study 1: Characteristics of TMR with barrier height of 0.85eV



Case Study 2: Characteristics of TMR with barrier height of 1.0eV



Result: SIMULATION OF TUNNELLING MAGNETORESISTANCE (TMR) BY VARYING BARRIER HEIGHT IS ACCOMPLISHED.

Exp. 10. Biosensor and Bio-FET characteristics using NanoHub

Aim:

1. To simulate the threshold voltage of a BioFET-based biosensor as a function of electrolyte pH (4 to 8).
2. To analyze the transfer characteristics (IdsI_{ds}Ids vs. VfgV_{fg}Vfg) of the BioFET for varying pH levels and interpret the sensor's performance.

Tools Required:

1. NanoHUB BioSensor Lab (Version 2): Used for simulating threshold voltage and transfer characteristics.
2. Excel/MATLAB: For plotting and analyzing simulation results against theoretical predictions.

Theory:

1. pH Sensitivity:
    - The potential at the BioFET sensor surface is modulated by the pH of the surrounding electrolyte. This is governed by the Nernst equation: E=E0−(2.303 RT)/F pH where the theoretical sensitivity is 59 mV/pH at 25∘C.
2. Transfer Characteristics:
    - The transfer characteristics (Ids vs. Vfg) describe the variation of drain current (Ids) with the gate voltage (Vfg), influenced by the electrolyte pH. The characteristics shift as pH changes, indicating effective sensing.
3. Key Parameters:
    - Threshold Voltage (Vt): Reflects the potential at which the device turns on and shifts linearly with pH.
    - Current Sensitivity (Ids): Demonstrates changes in the channel current with pH variation.

Simulation Parameters:

Device Parameters of Sensor:

- Sensor Type: EGFET (Extended-Gate Field Effect Transistor).
- Biological Parameters:
    - Analyte: DNA.
    - Receptor Density: 1×10^12.
- Ambient Conditions:
    - Incubation Time: 60 minutes.

- Temperature: 300 K (25°C).

pH Variation:

- Simulated for pH values ranging from 4 to 8.

Procedure:

Step 1: Threshold Voltage Simulation

1. Open the BioSensor Lab in NanoHUB.
2. Set the device parameters as specified above.
3. Vary the pH of the electrolyte from 4 to 8 in steps of 1.
4. Record the threshold voltage (Vt) for each pH.

Step 2: Transfer Characteristics Simulation

1. Using the same tool, select EGFET as the sensor type.
2. Fix the electrolyte pH and vary the gate voltage (Vfg) between 0.4 V and 1 V.
3. Record the drain current (Ids) for each pH value.

Step 3: Data Analysis

1. Plot the threshold voltage (Vt) as a function of pH.
2. Overlay the transfer characteristics (Ids vs. Vfg) for different pH values.
3. Compare the simulated results with theoretical predictions.

Results and Observations:

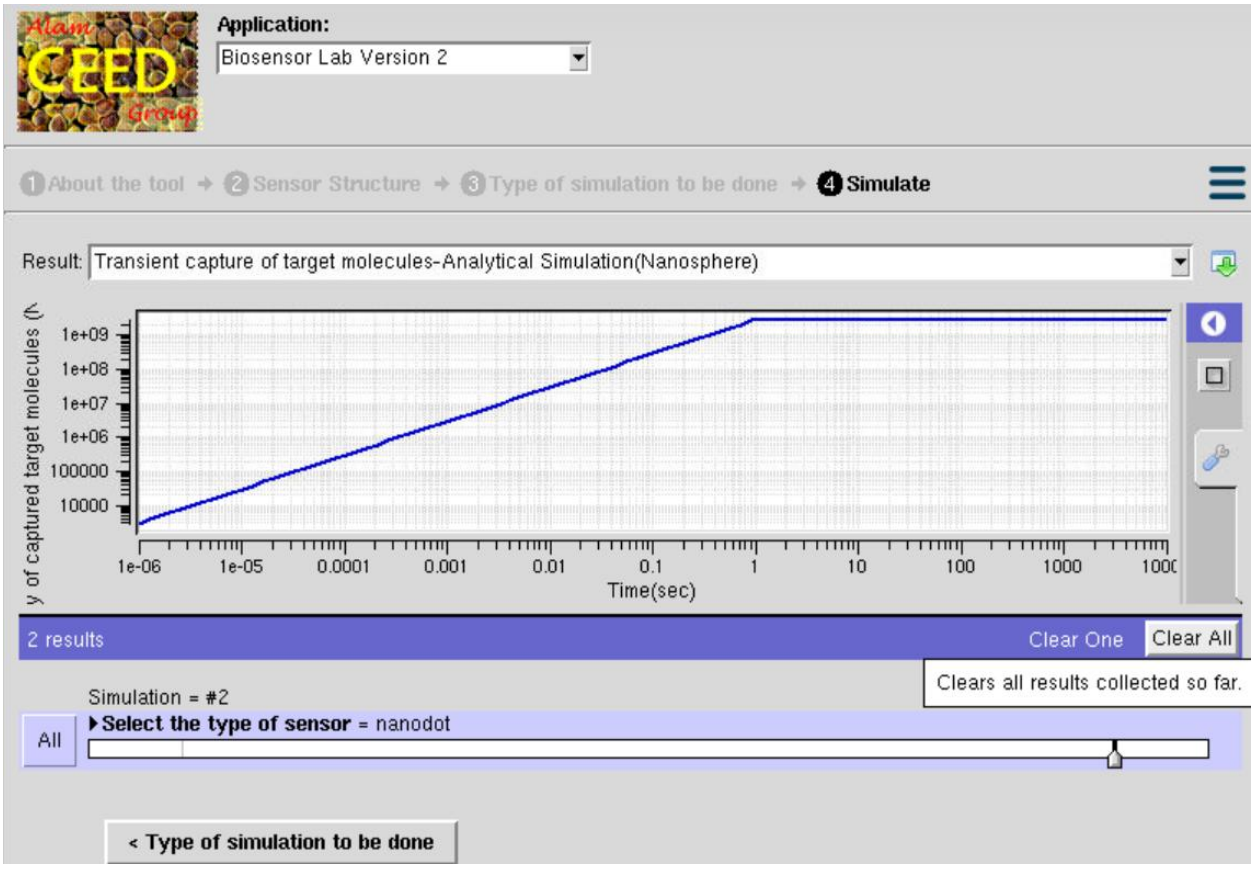
1. Threshold Voltage as a Function of pH:

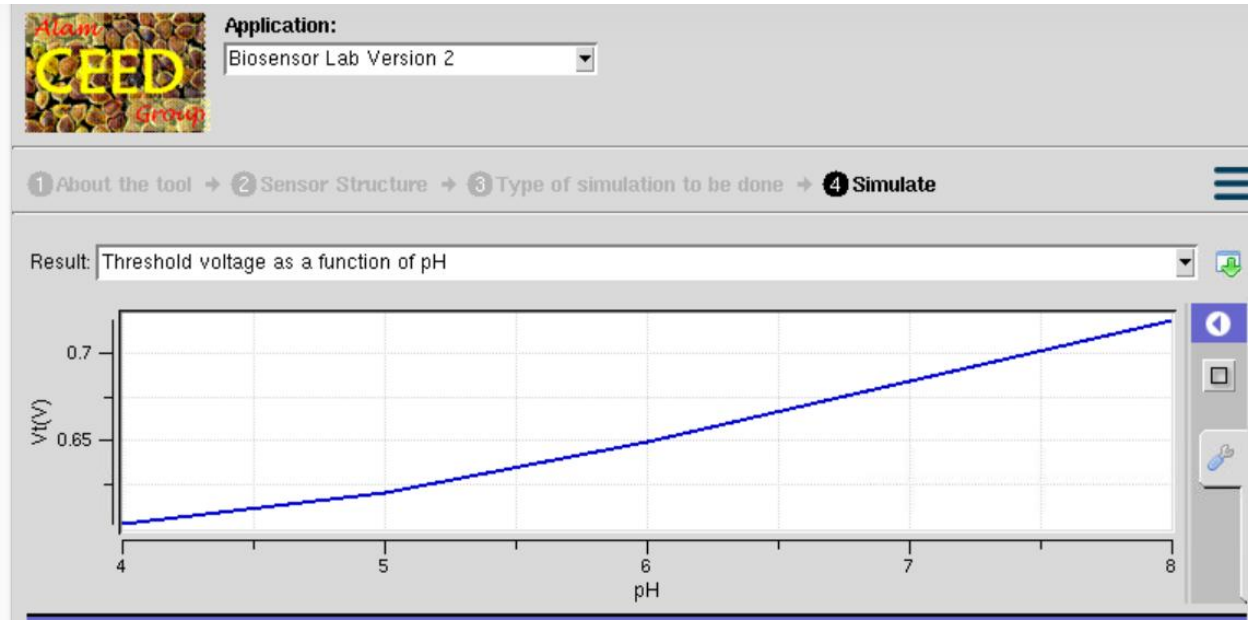
- Simulation Result:
  - The threshold voltage increases linearly with pH, as shown in the plot below:
  - For pH 4 to pH 8, the threshold voltage shifts between 0.65 V and 0.75 V.
- Observation:
  - The linear relationship validates the Nernstian behavior of the BioFET with an approximate sensitivity of 59 mV/pH59 \, \text{mV/pH}59mV/pH, matching theoretical expectations.

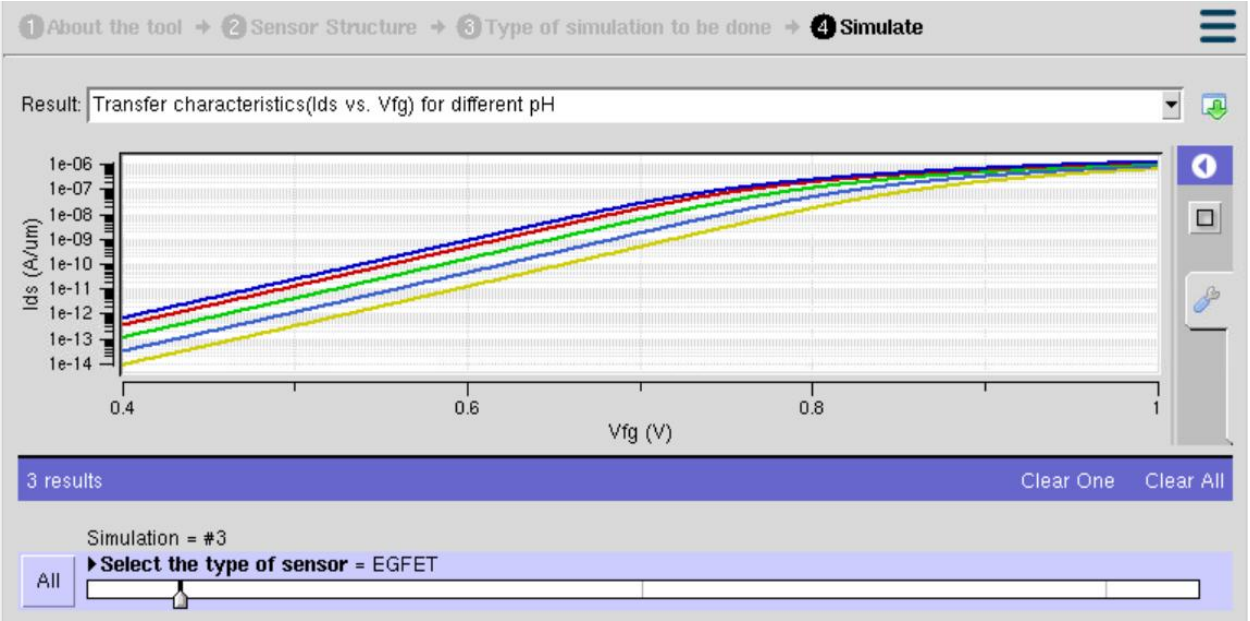
2. Transfer Characteristics (Ids vs. Vfg) for Different pH:

- Simulation Result:
  - The transfer characteristics exhibit a shift in IdsI_{ds}Ids with varying pH levels, as shown below:

    - Higher pH results in an increased threshold voltage, shifting the curve to the right.
- Observation:
    - The results demonstrate effective detection of pH changes, with the BioFET showing higher IdsI_{ds}Ids for lower VfgV_{fg}Vfg at lower pH levels.







Conclusion:

1. The BioFET sensor demonstrates linear sensitivity to pH changes, with a near-theoretical response of 59 mV/pH59 \, \text{mV/pH}59mV/pH.
2. The transfer characteristics (Ids vs. Vfg) effectively shift with varying pH, confirming the sensor's ability to detect electrolyte variations.
3. The EGFET configuration is validated as a reliable model for BioFET simulations, offering insights into potential optimizations for practical applications.

Future Scope:

1. Extend the analysis to wider pH ranges (e.g., 2 to 12) to identify saturation effects.
2. Study the influence of different insulator materials (e.g., SiO_2, HfO_2) on sensitivity and noise performance.
3. Simulate the BioFET with different analytes (e.g., proteins or ions) to generalize the model for various biosensing applications.

Exp. 11. Calculate the Potential of a pH SENSOR using NanoHub

Aim:

To obtain the Potential of a sensor by varying the pH value of an electrolyte and

insulator thickness

Software: NanoHub Online Tube

Theory:

An Electrolyte–insulator–semiconductor (EIS) sensor is a sensor that is made of these three components:

an electrolyte with the chemical that should be measured

an insulator that allows field-effect interaction, without leak currents between the two other components

a semiconductor to register the chemical changes

The EIS sensor can be used in combination with other structures, for example to construct a light-addressable potentiometric sensor (LAPS).

Procedure:

- Open nanoHUB.
- Goto Resources -> Tools.
- Select -> ENBIOS 1D
- Press Launch tool in info section.
- Select the -> semiconductor/insulator/electrolyte
- Click on Simulation
- Select ->Result -> Potential

- Select -> electrolyte -> change pH value
- Click on simulation
- Select -> Result -> Potential
- Repeat the above steps for different pH values
- Select -> Insulator-> change thickness and length
- Click on simulation
- Select -> Result -> Potential
- Repeat the above steps for different thickness and length

Parameters:

Model Output: Result:

Exp. 12. Rietveld refinement & lattice parameter determination (single-phase material) using FullProf software

Aim:

To perform Rietveld refinement of X-ray diffraction (XRD) data and determine the lattice parameters of crystalline materials using FullProf software.

Software Used:

FullProf Suite (Version 7 or later)

Procedure:

- Data Preparation: Obtain the experimental XRD pattern for a single-phase material in .dat or .xyd format.
- Note the radiation source (e.g., Cu Kα, λ = 1.5406 Å).
- Project Setup in FullProf:
- Open the FullProf software and load the diffraction data file.
- Enter initial structural parameters such as the space group (e.g., Fd-3m for Si) and lattice constant obtained from JCPDS data.
- Parameter Initialization:
- Select the peak profile function (e.g., pseudo-Voigt for general fitting).
- Set instrumental parameters and choose a suitable background polynomial (typically of order 4).
- Initially, activate refinement for the scale factor and zero shift.

- Sequential Refinement:
- Run refinement for the scale factor and background parameters.
- Gradually activate refinement for lattice parameters, peak shape, and preferred orientation, if applicable.
- Continue the iterative refinement until the difference between the observed and calculated patterns is minimized.
- Result Extraction:
- After convergence, record the refined lattice parameter (a), Rwp, Rp, and χ^2 values.
- Compare the obtained results with standard literature data.
- Visualization:
- View and analyze the observed, calculated, and difference diffraction patterns displayed by FullProf.

Theory:

The Rietveld refinement method is a powerful computational technique used to extract precise structural information from X-ray diffraction (XRD) data. It works by fitting a calculated diffraction pattern, based on an initial structural model, to the experimentally observed pattern through least-squares minimization.
In this process, parameters such as lattice constants, atomic positions, thermal factors, peak shapes, and background are refined iteratively until the difference between the observed and calculated patterns is minimized.
For a single-phase material, the refinement mainly aims to:

- Accurately determine the lattice parameter (a) of the cubic structure.
- Evaluate the goodness of fit through parameters like Rwp (weighted profile R-factor), Rp, and χ^2.

The precision of the refinement depends on the correct selection of background function, peak profile, and initial lattice parameters (usually taken from JCPDS data).
In X-ray diffraction, each peak corresponds to the constructive interference from specific crystal planes. However, due to instrumental and sample-related effects (e.g., strain, crystallite size, and instrumental broadening), the peaks are not infinitely sharp but have finite width and shape.
FullProf models these diffraction peak shapes using various mathematical functions, primarily:

- Gaussian function: Represents instrumental broadening and is symmetric.
- Lorentzian function: Represents broadening due to small crystallite size or strain.

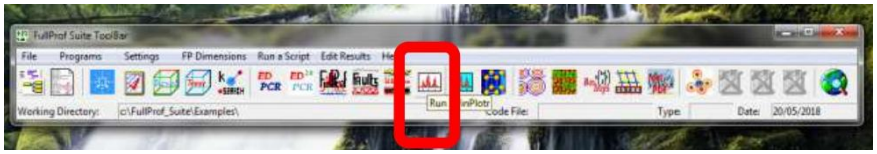
- Pseudo-Voigt function: A linear combination of Gaussian and Lorentzian, used in most real cases.

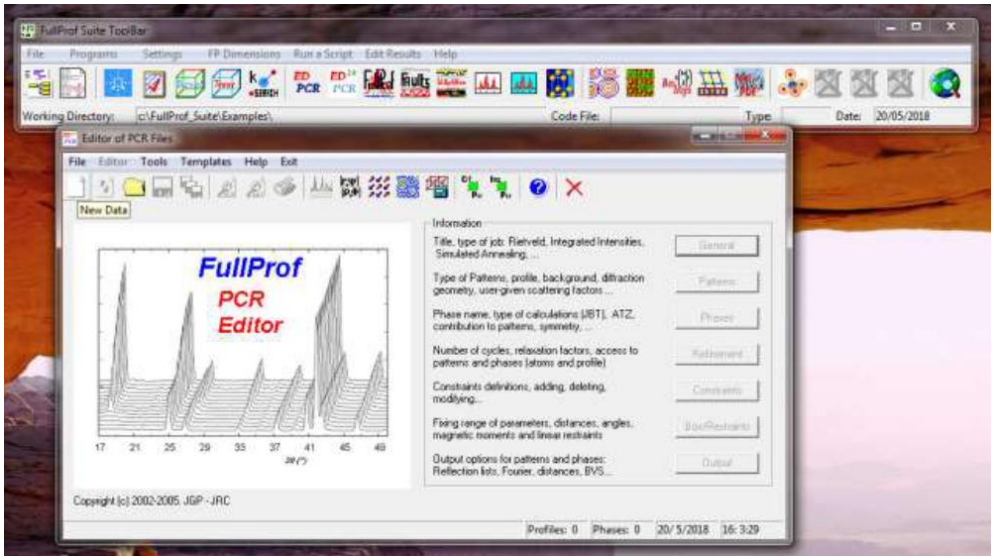
The choice of peak shape function influences the fit quality, refinement stability, and accuracy of lattice parameter determination. Hence, testing different peak functions helps identify the most suitable model for a given material and instrument setup.

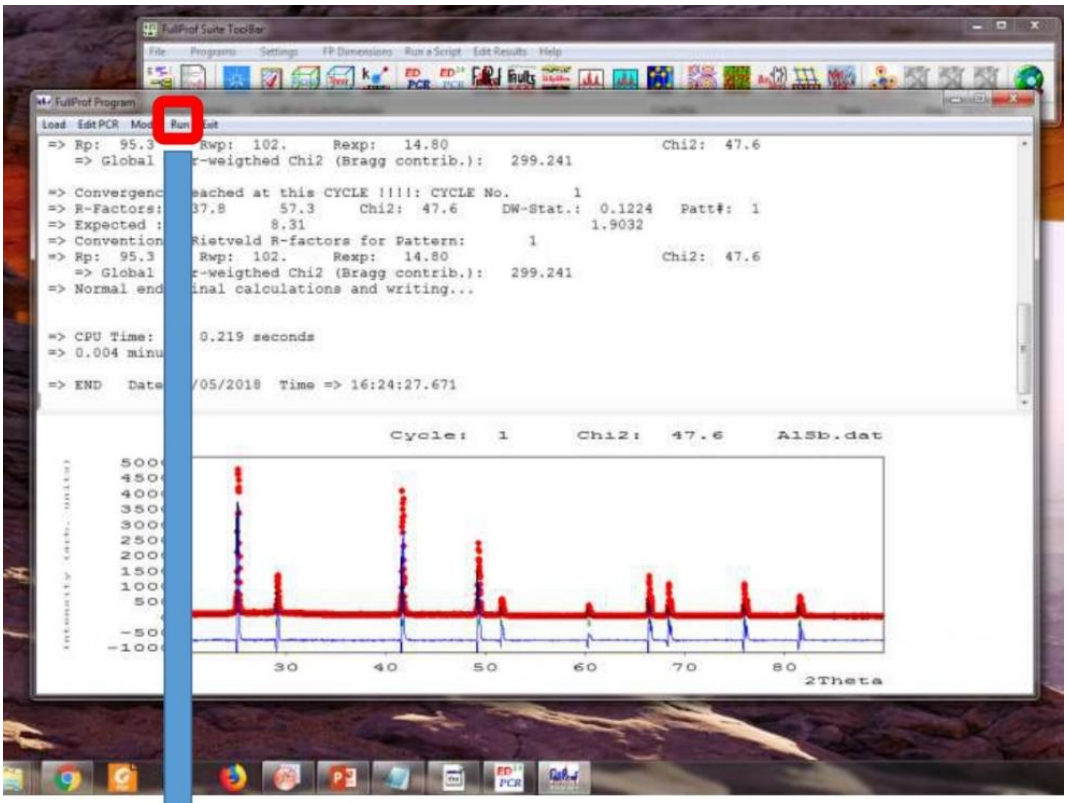
Output:

- Refined XRD pattern showing **observed, calculated, and difference** curves.
- Optimized **lattice parameters (a, b, c)**.
- Statistical indicators such as **Rp, Rwp, χ^2**, and residual errors.

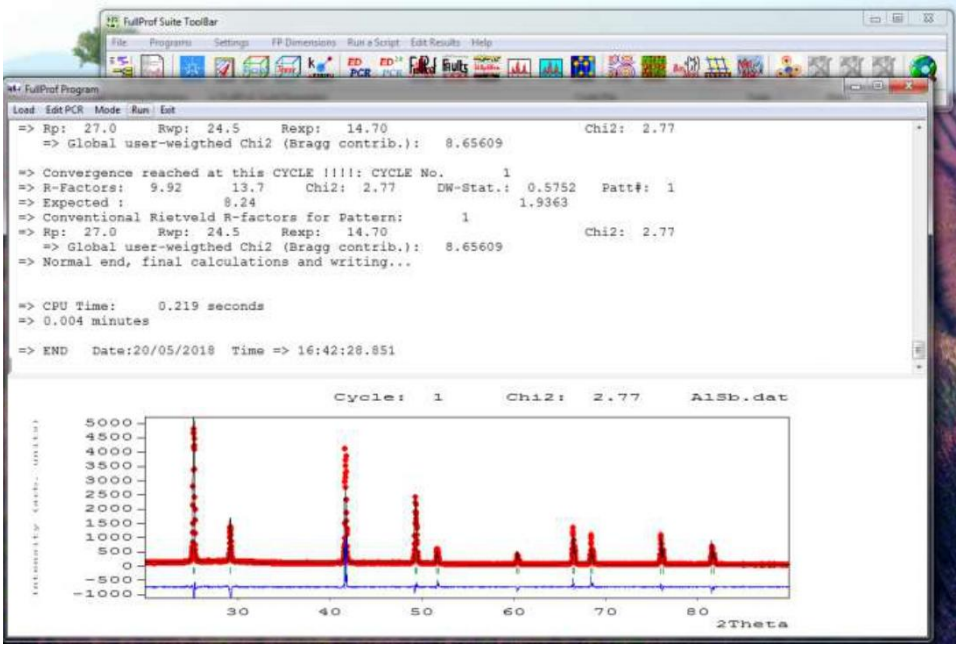
<u>Perform Rietveld refinement for a compound semiconductor (AlSb) and determine its lattice parameter. Compare with standard JCPDS values.</u>



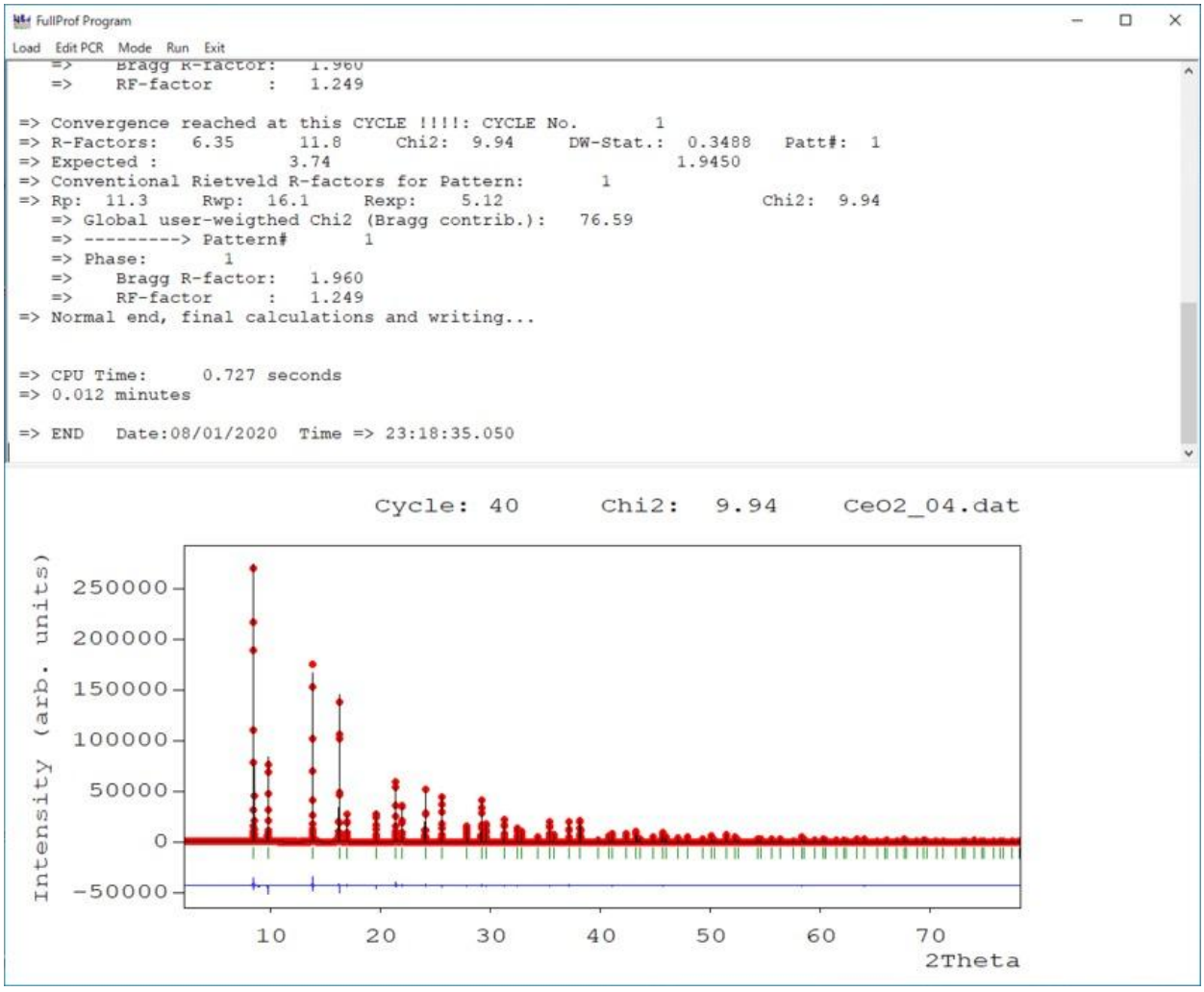




Keep clicking on Run till Chi2 value saturates. Then close.



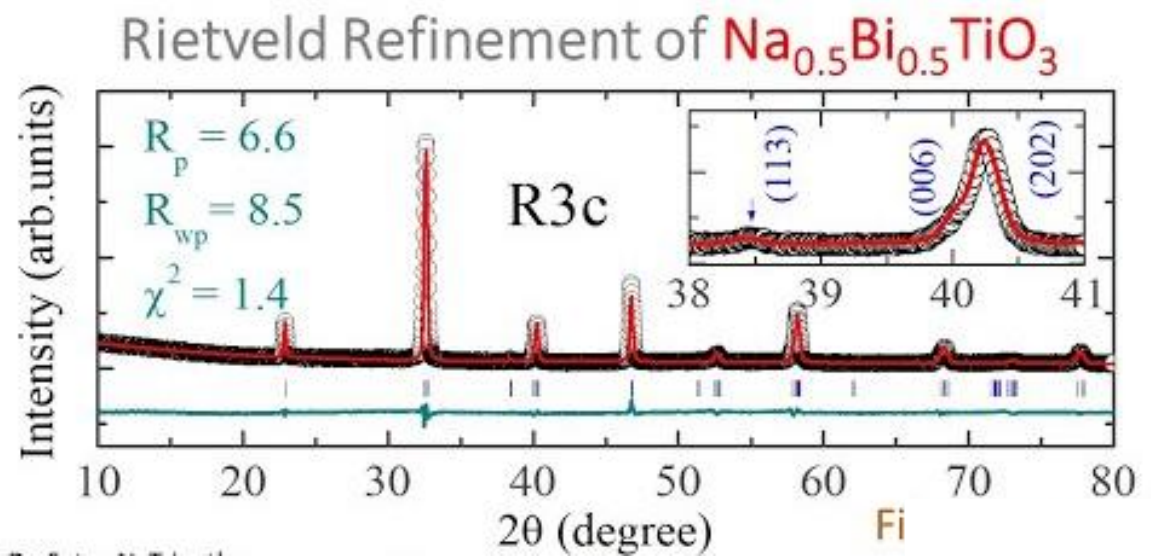
Case Study 1: Perform Rietveld refinement for a metal oxide material (CeO_2) using the pseudo-Voigt peak profile. Analyze the Rwp and χ^2 values.



Case Study 2: Determination of lattice parameters for metal oxide material (ZnO).



Case Study 3: Determination of lattice parameters for doped metal oxide material ($Na_{0.5}Bi_{0.5}TiO_3$).



Result:

The Rietveld refinement using FullProf successfully determined the lattice parameters of the tested materials. The low R_{wp} and χ^2 values confirm good agreement between observed and calculated patterns, validating the reliability of the refinement process.

Analytical:

Given the observed diffraction peak at 2θ=28.5° for CeO_2 using Cu Kα radiation (λ = 1.5406 Å):

$$d = \frac{\lambda}{2Sin\theta}$$

For cubic structure:

$$a = d\sqrt{h^2 + k^2 + l^2}$$

For (111) reflection, calculate the **lattice constant (a)** and compare it with the refined value from FullProf